Open19 Brick Mechanical Specifications

Linux Foundation Sustainable Scalable Infrastructure Alliance

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Chapter 1. Change log

Author	Date	Version	Change Details
My Truong	November 2023	2.0	V2 Spec release

Definitions

ASHRAE

American Society of Heating, Refrigerating and Air-Conditioning Engineers

ASHRAE W27

ASHRAE specification for water temperature range from 2 °C to 27 °C

OCP

Open Compute Project

SFP

SNIA SFF-8402 SFP+ 1X Pluggable Transceiver Solution

SNIA

Storage Networking Industry Association

QSFP

SNIA SFF-8679 QSFP+ 4X Pluggable Transceiver Solution

QSFP-DD

QSFP-DD MSA QSFP DOUBLE DENSITY 8X AND QSFP 4X PLUGGABLE TRANSCEIVERS

Chapter 2. Open19 Bricks

Four brick designs are the basis of Open19 V2. They are:

- SHHW: Single-High Half-Width, "Brick"
- DHHW: Double-High Half-Width
- SHDW: Single-High Double-Width, 1RU server equivalent
- DHDW: Double-High Double-Width, 2RU server equivalent

Chapter 3. Brick Mechanical Specification [SHHW]

3.1. SHHW Brick

The SHHW brick form-factor is a half-wide 1RU commonly used as the basis of 2U 4-node designs.

3.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

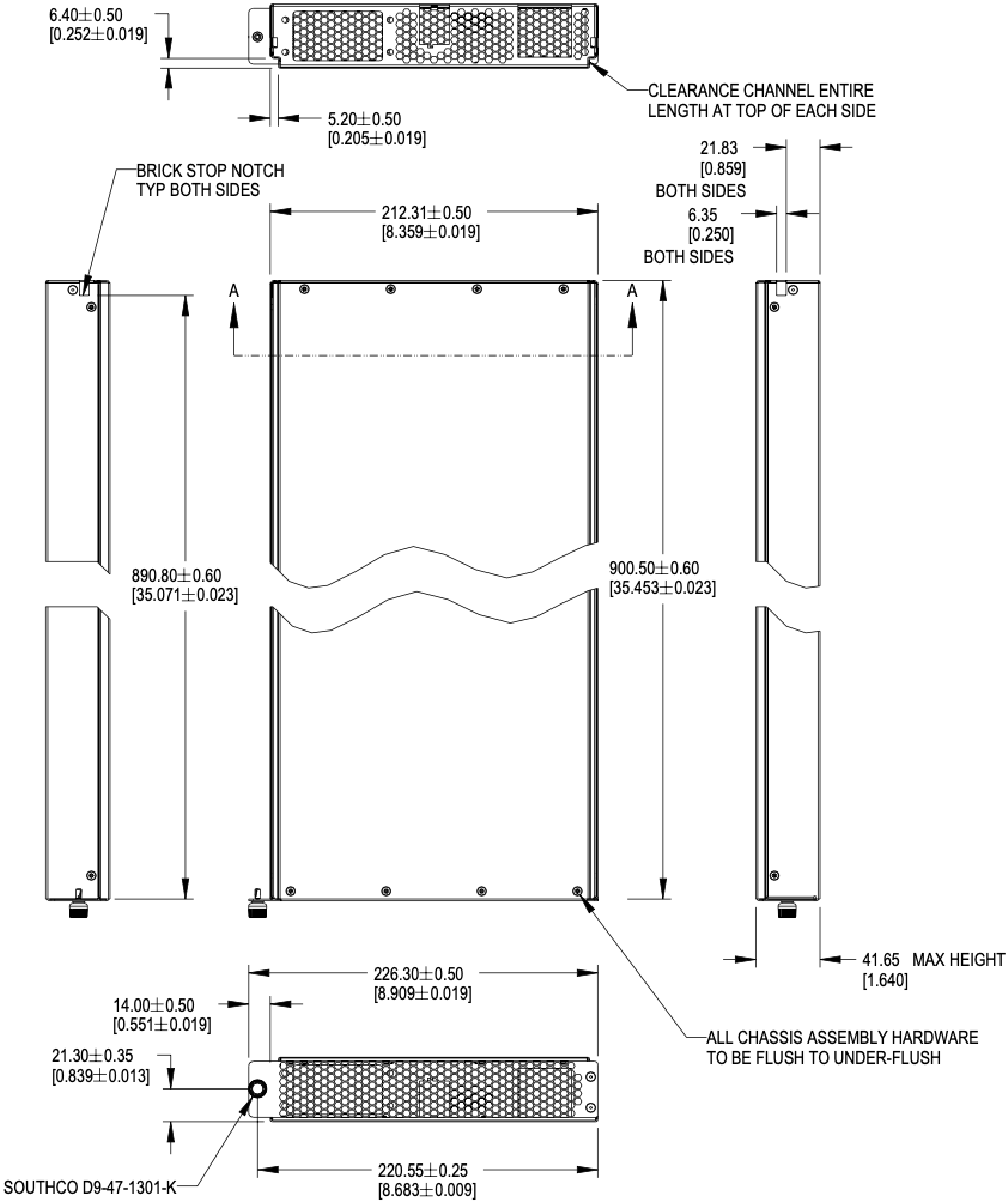


Figure 1. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

For half-width designs using the OCP DC-MHS M-DNO board specification, note the tolerance guidance in order to fit boards in a half-width chassis. This requires setting the nominal width at 209.55mm or smaller.

3.3. Mechanical features

3.3.1. Brick retention

The brick is secured into the cage with a single Southco D9-47-1302-K quarter-turn fastener, located on the left side of the faceplate. This is a change from V1. The change aligns the retention screw with high force areas of the optional liquid cooling interface.

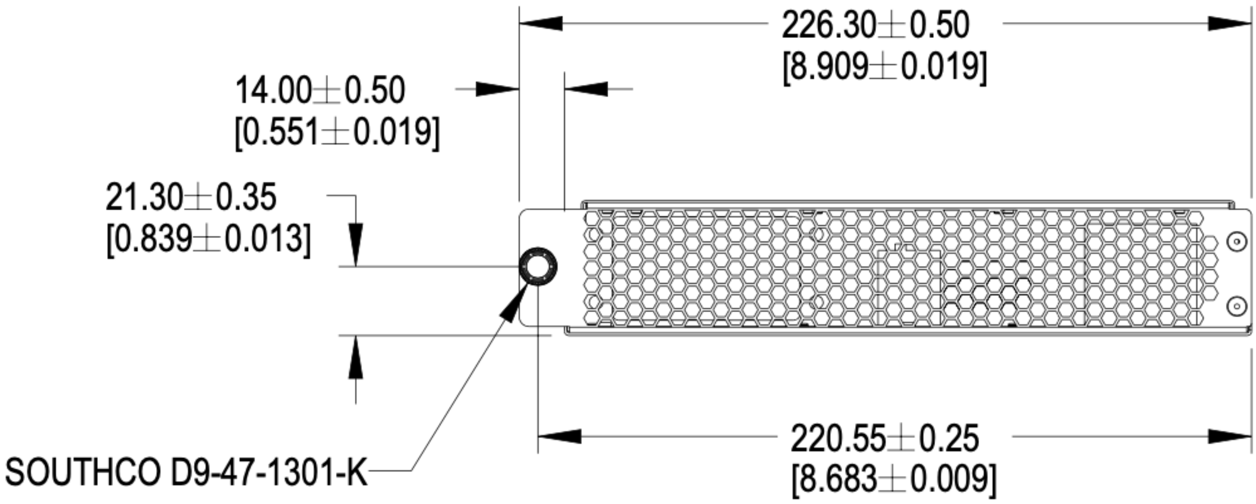


Figure 2. Brick retention

3.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick, extending forward from the rear panel to the back of the faceplate.

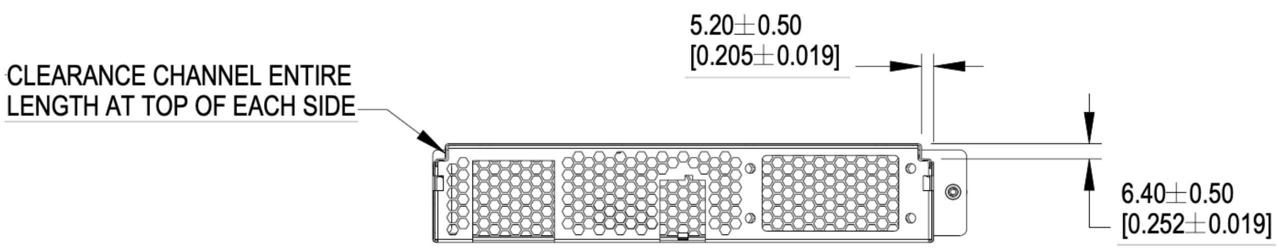


Figure 3. Mechanical guide at top edge of brick

3.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

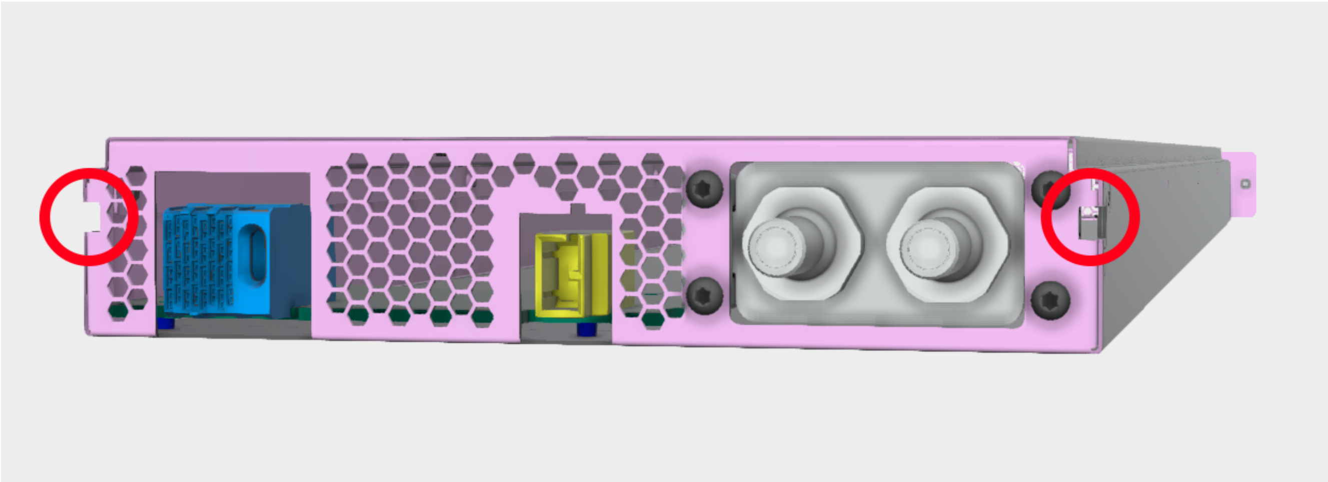


Figure 4. Brick keying, liquid cooling connections shall be defined in a future release

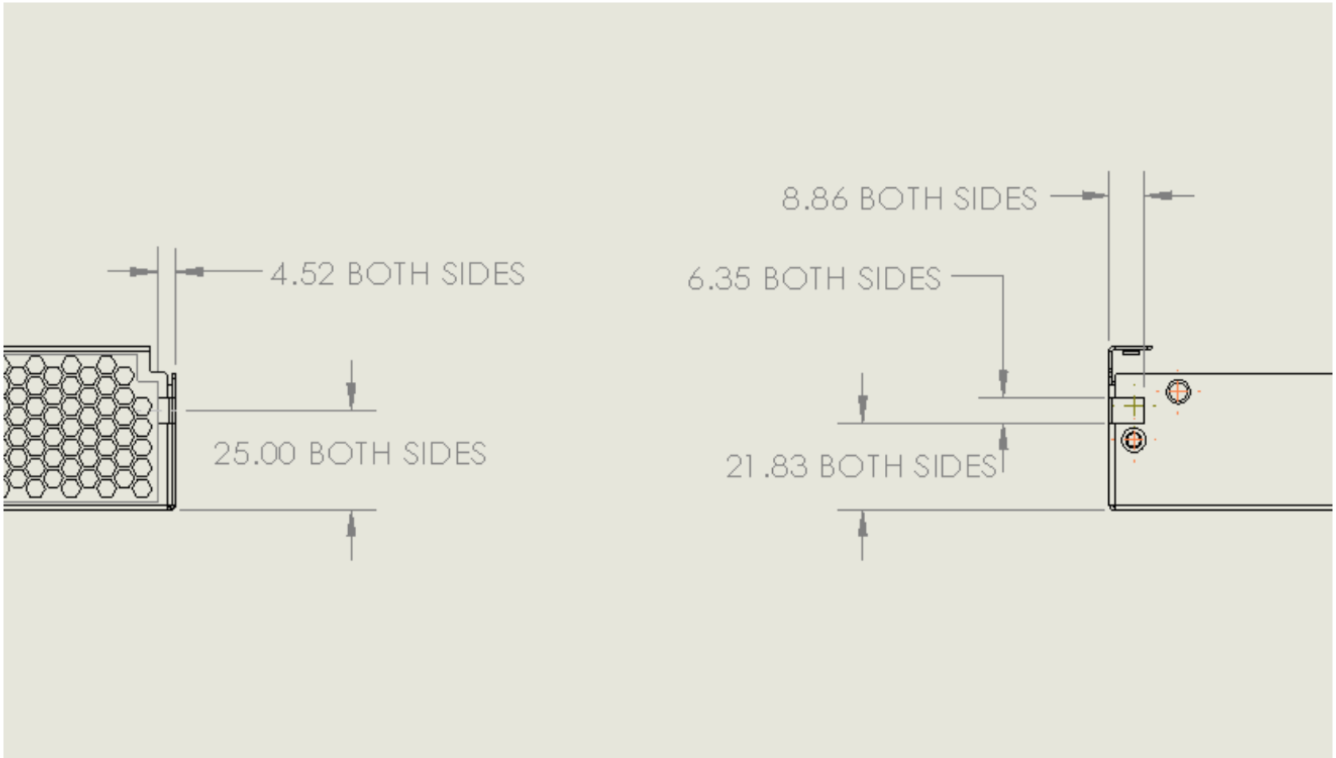


Figure 5. Brick keying dimensions

3.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

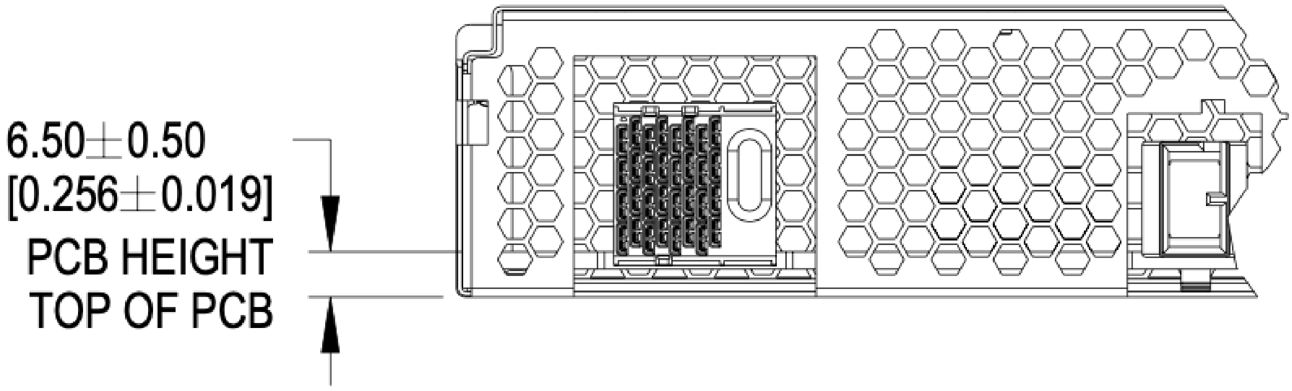


Figure 6. PCBA References

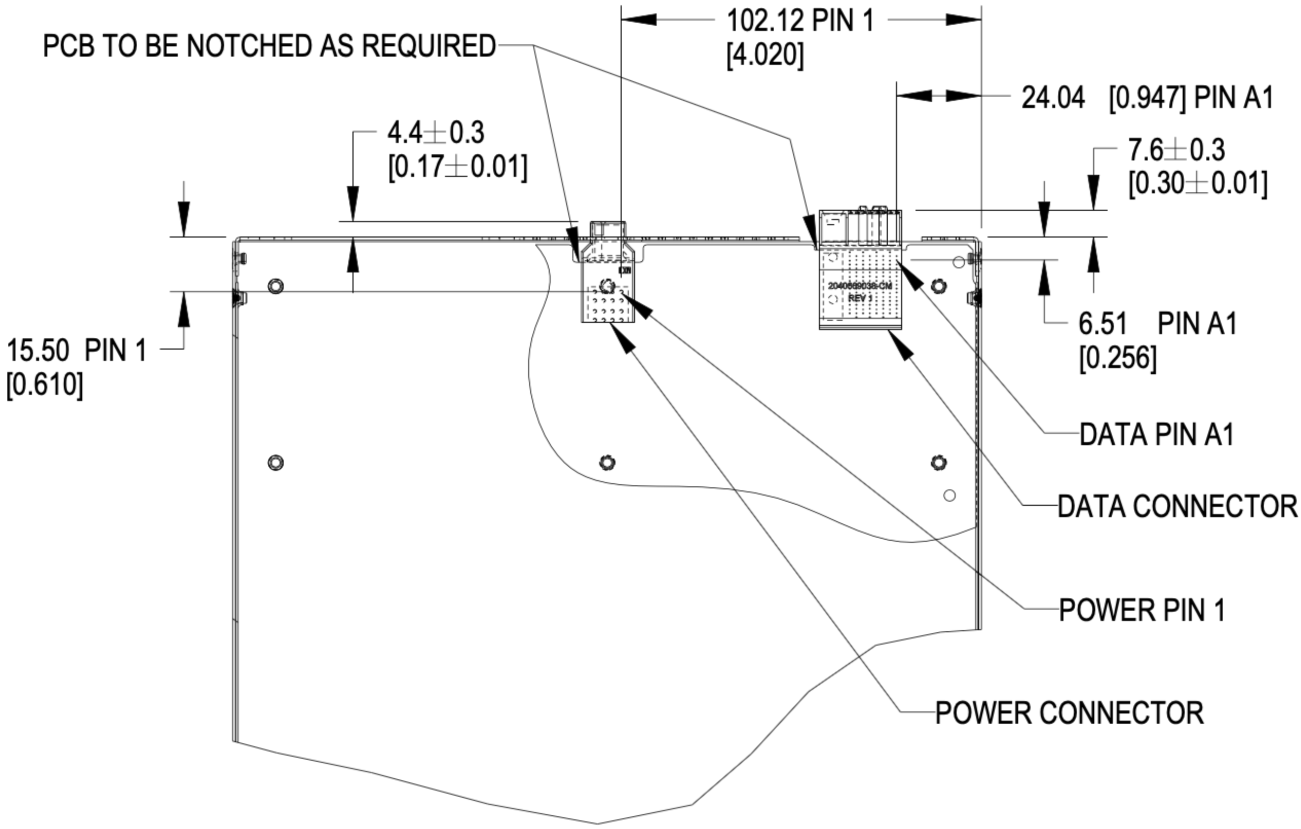


Figure 7. Brick PCB dimensions

3.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

3.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

3.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 30lbs (13.6kg). This is an increase of 5lbs from V1.

3.4. Connector Pinout

3.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

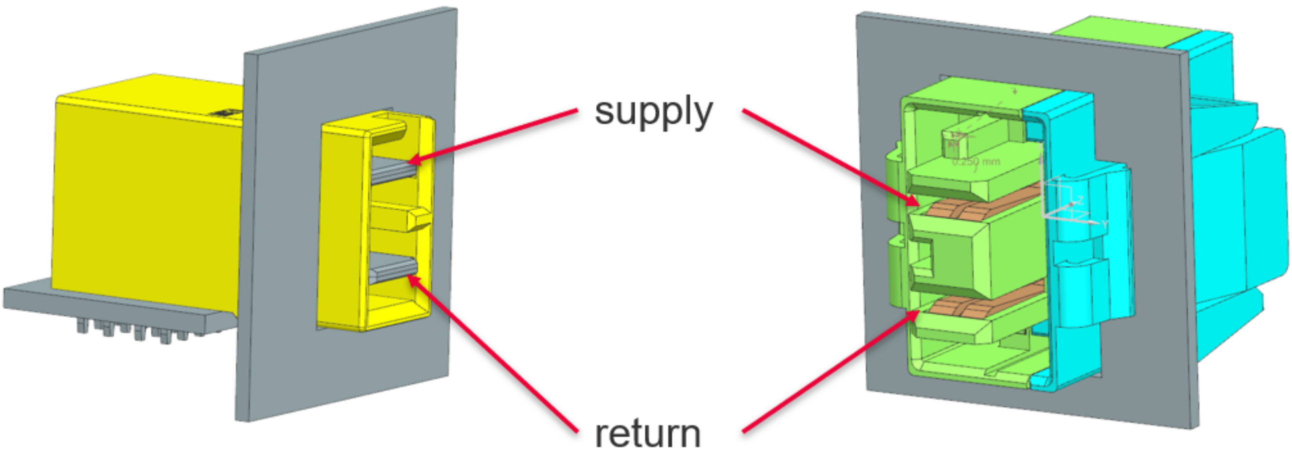


Figure 8. Power connector supply/return definition

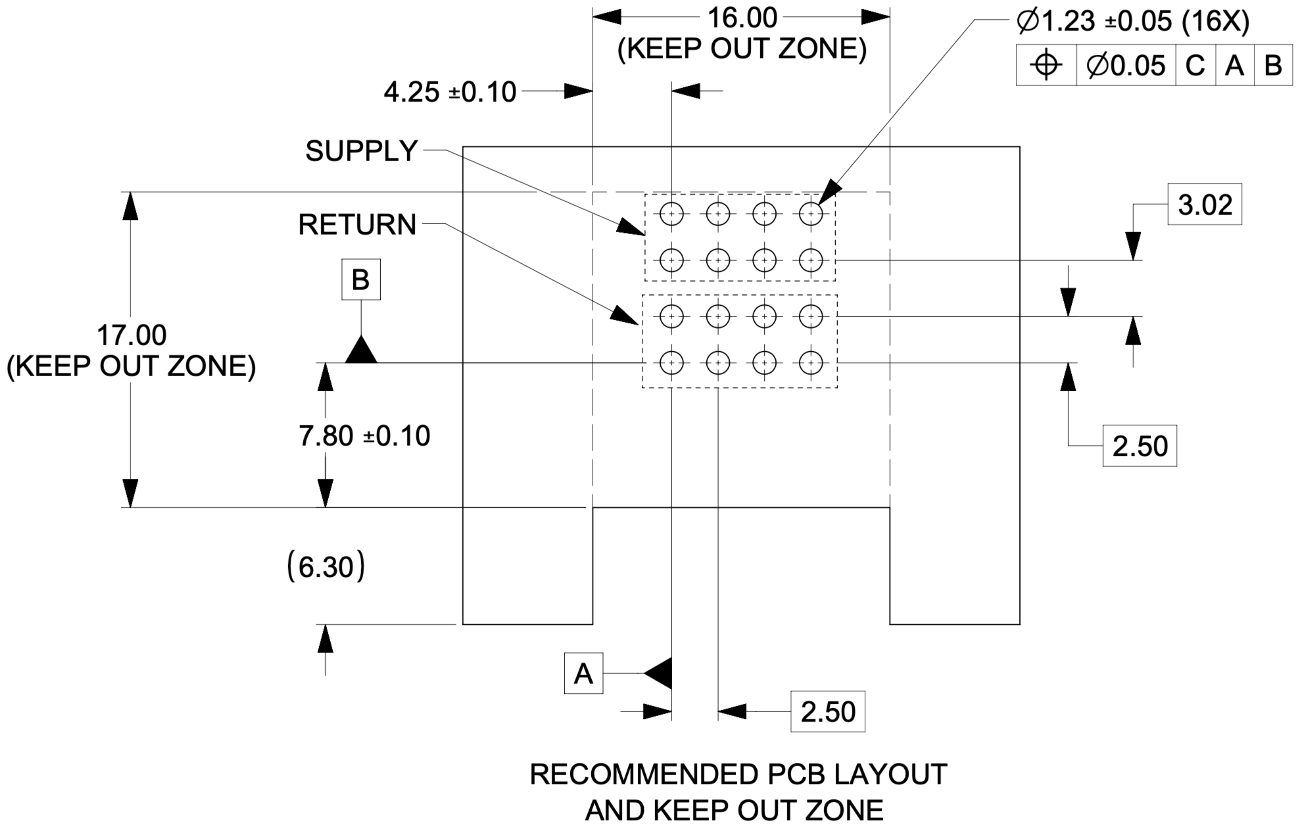


Figure 9. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

3.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

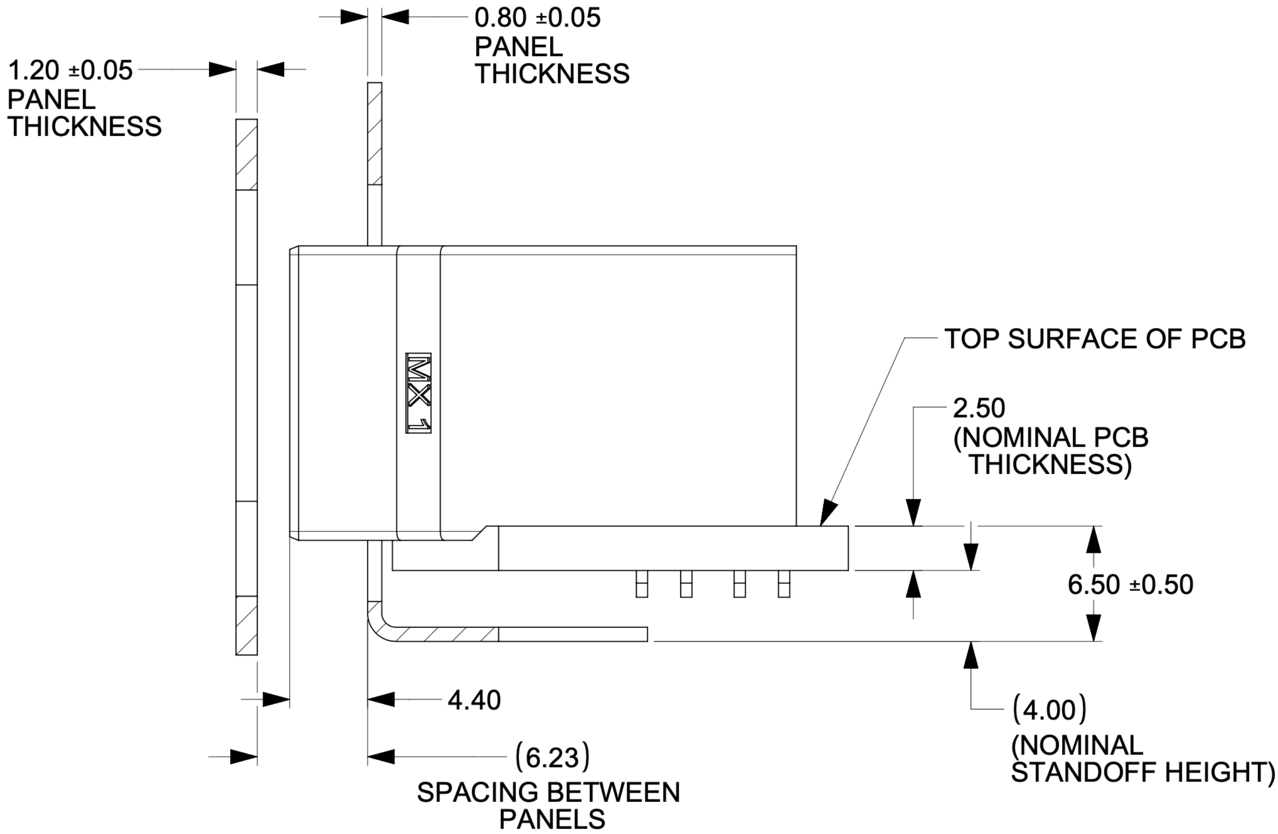


Figure 10. Power connector critical dimensions

3.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

3.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

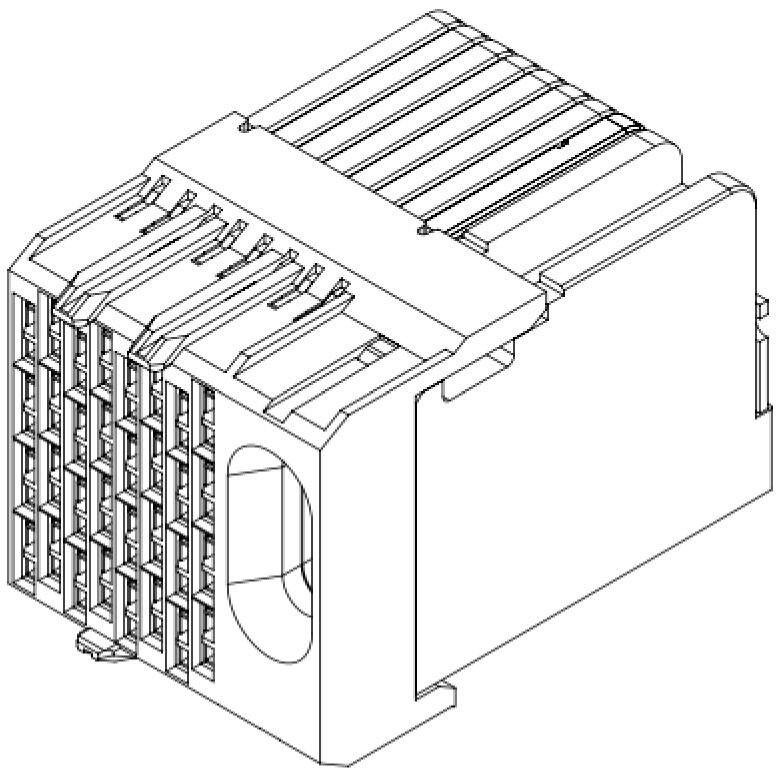


Figure 11. Impel Connector with guide feature

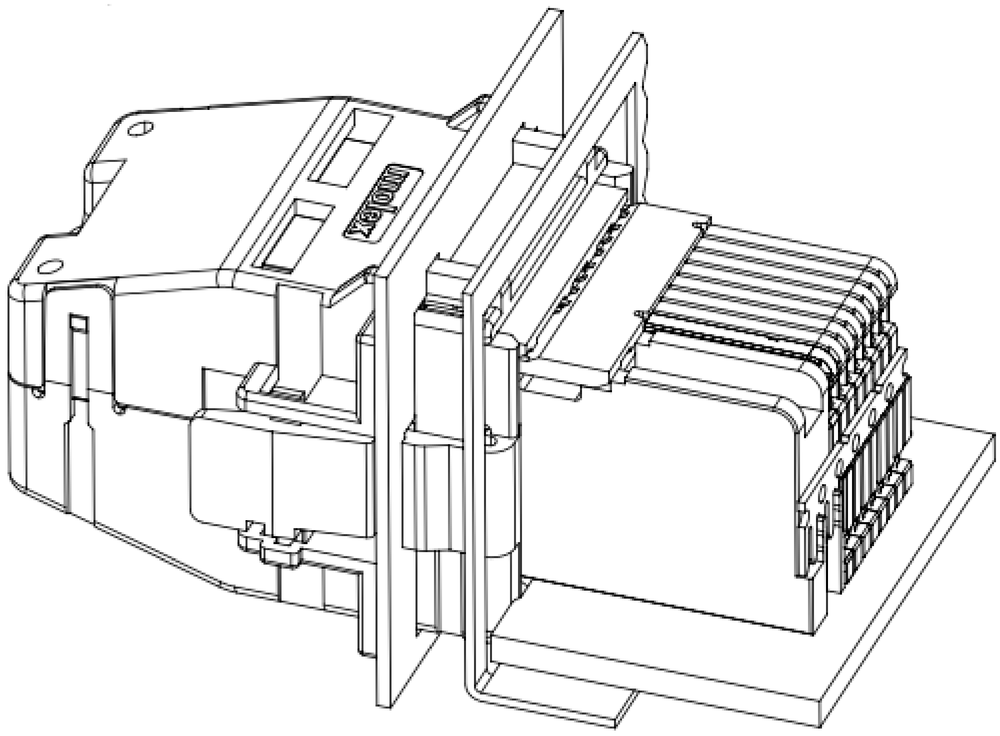


Figure 12. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

3.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

Port Count	Data Rate	Implementation Interface and order
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1
6		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	100G (4x25G)	SFP1, SPF2, SFP3, SPF4, QSFP1
6	200G (4x50G PAM4)	SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	400G (8x50G PAM4)	QSFP-DD (QSFP1+QSFP2) 200G (4x50G PAM4)
QSFP (SFP1+SFP2+SFP3+SFP4)	2	QSFP, QSFP1

3.5. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

3.5.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 2, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

3.5.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

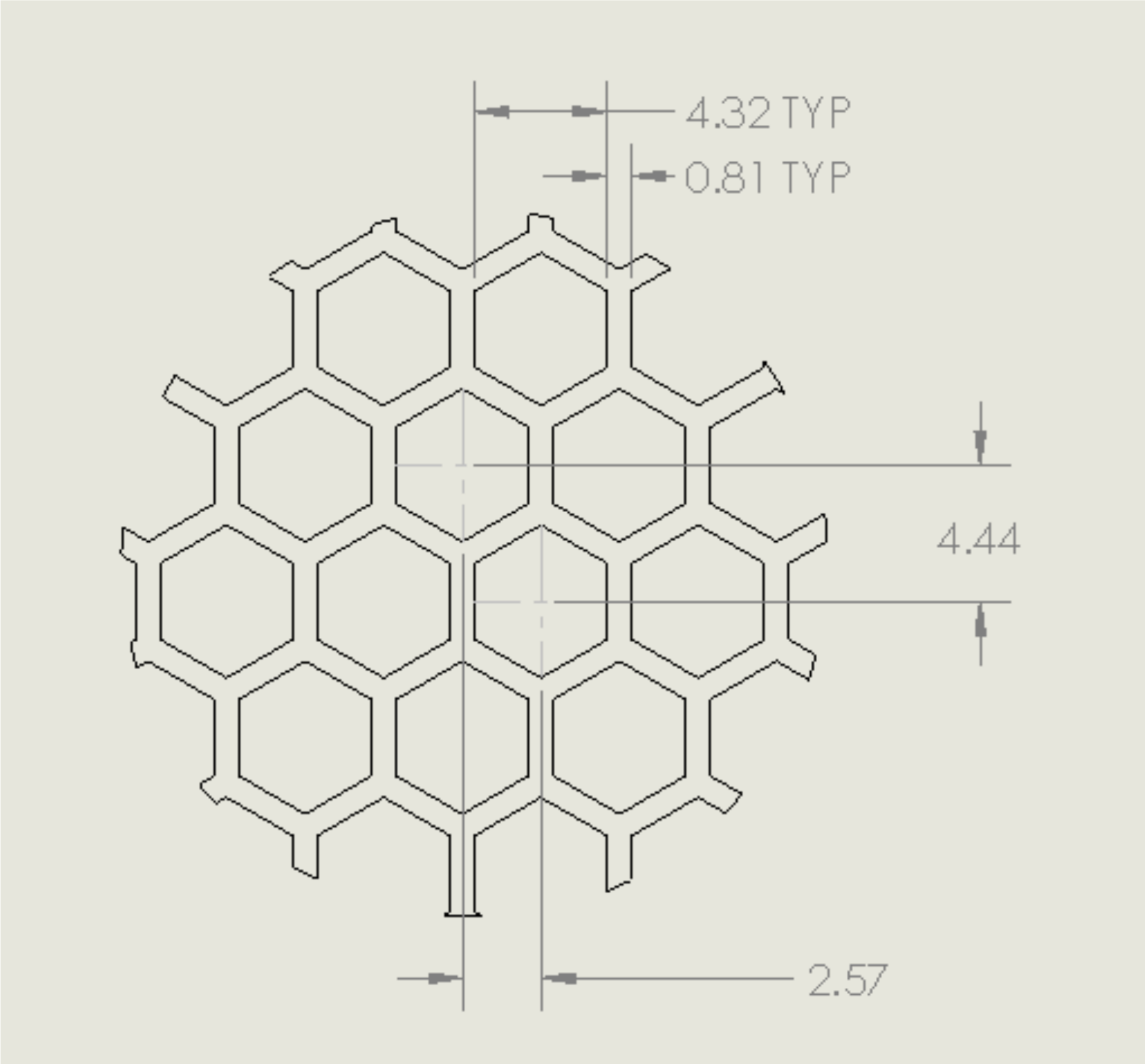


Figure 13. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

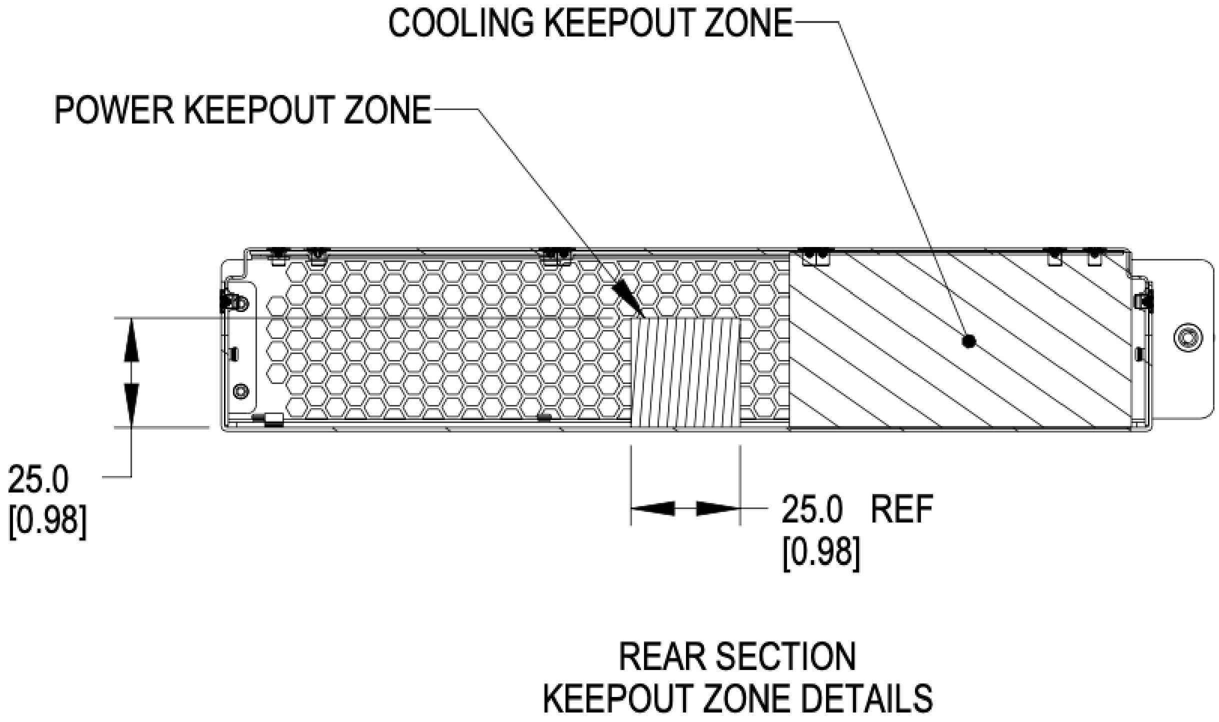


Figure 14. Brick keep-out zone

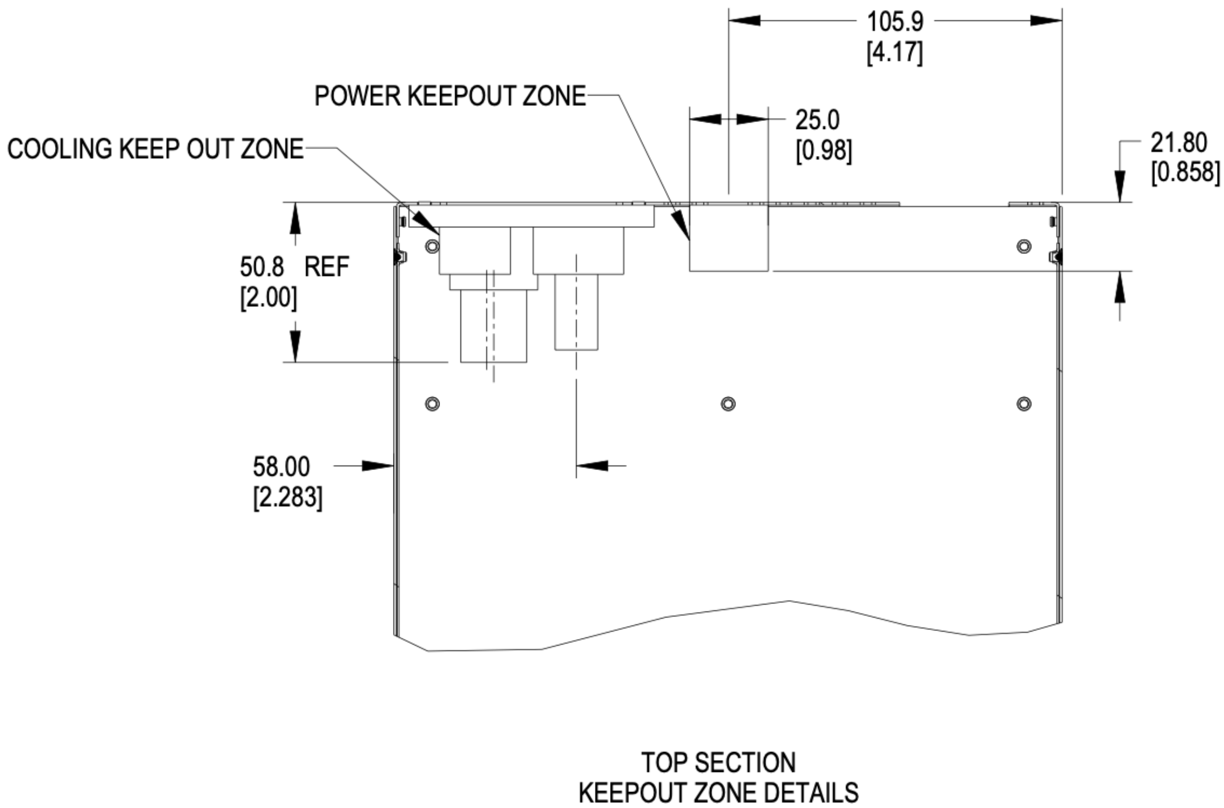


Figure 15. Brick keep-out zone top view

3.6. Brick performance

3.6.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

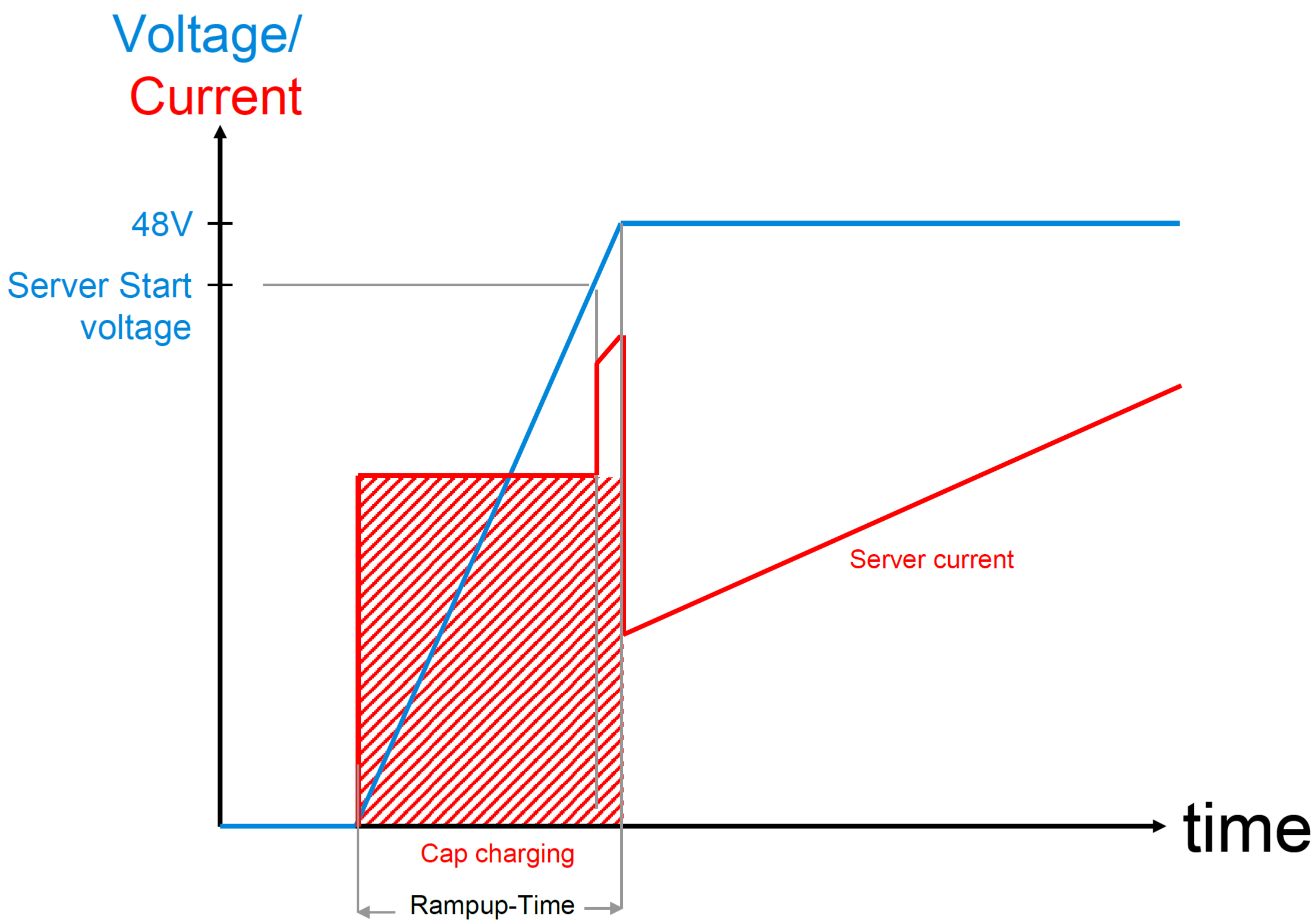


Figure 16. Server startup electrical characteristics

These values are per connector feed from a power shelf.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 4. Brick Mechanical Specification [DHHW]

4.1. DHHW Brick

The DHHW brick is commonly used as a 2U 2N server form-factor.

4.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

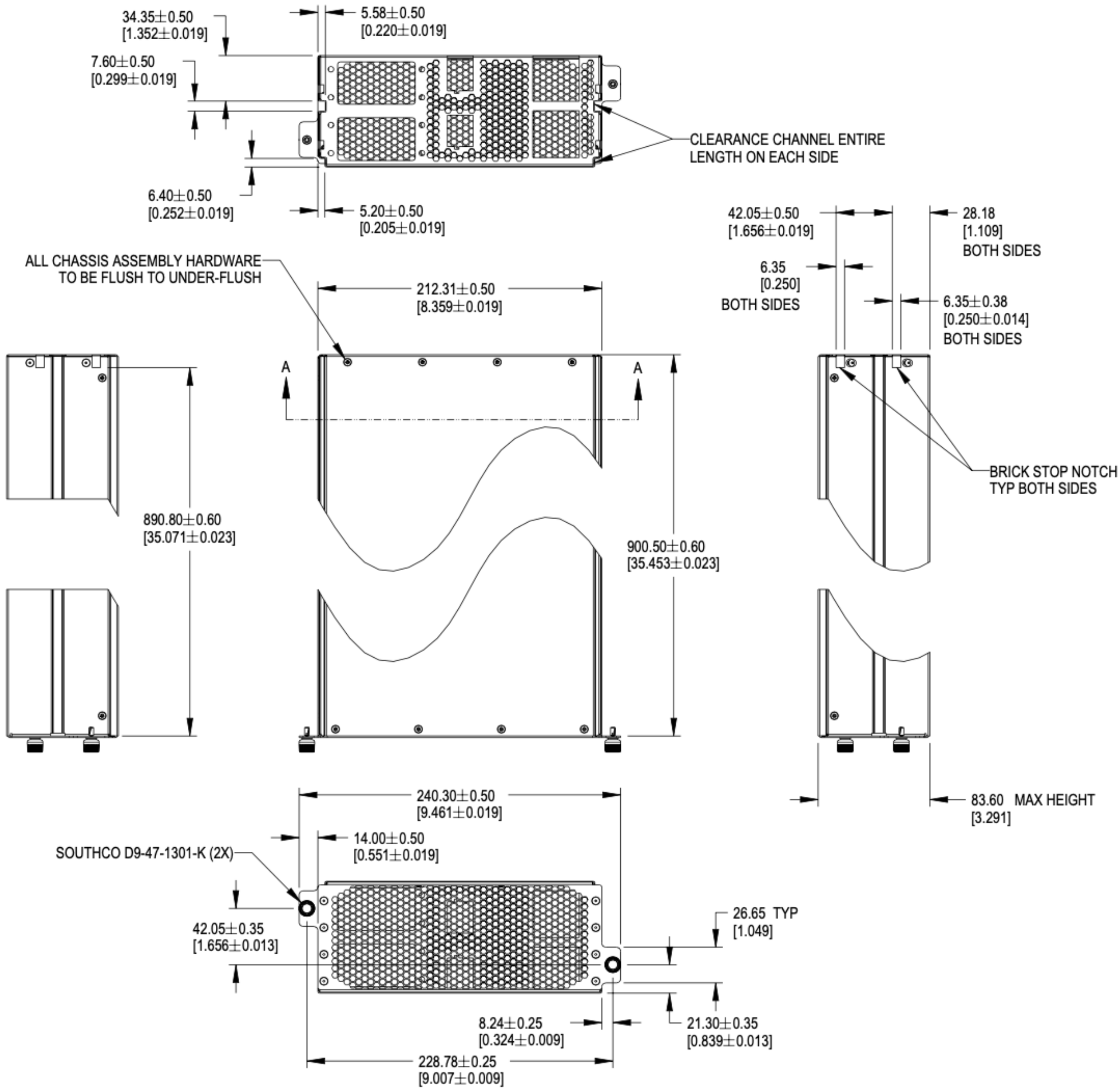


Figure 17. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

For half-width designs using the OCP DC-MHS M-DNO board specification, note the tolerance guidance in order to fit boards in a half-width chassis. This requires setting the nominal width at 209.55mm or smaller.

4.3. Mechanical features

4.3.1. Brick retention

Brick are secured into the cage with Southco D9-47-1302-K quarter-turn fasteners.

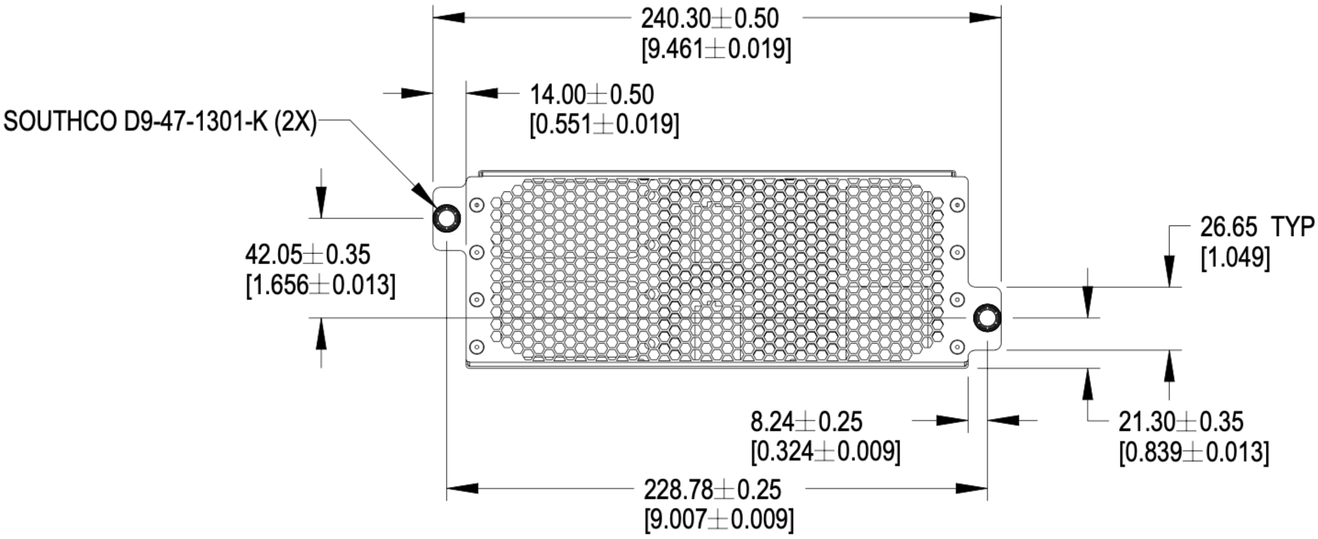


Figure 18. Brick retention

4.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick and the sidewalls, extending forward from the rear panel to the back of the faceplate.

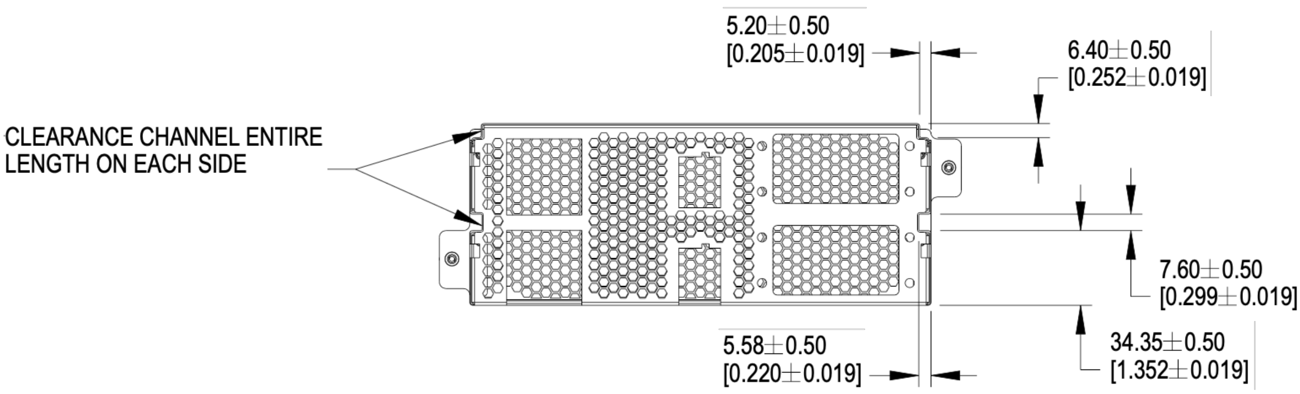


Figure 19. Mechanical guide at top edge and side walls of brick

4.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

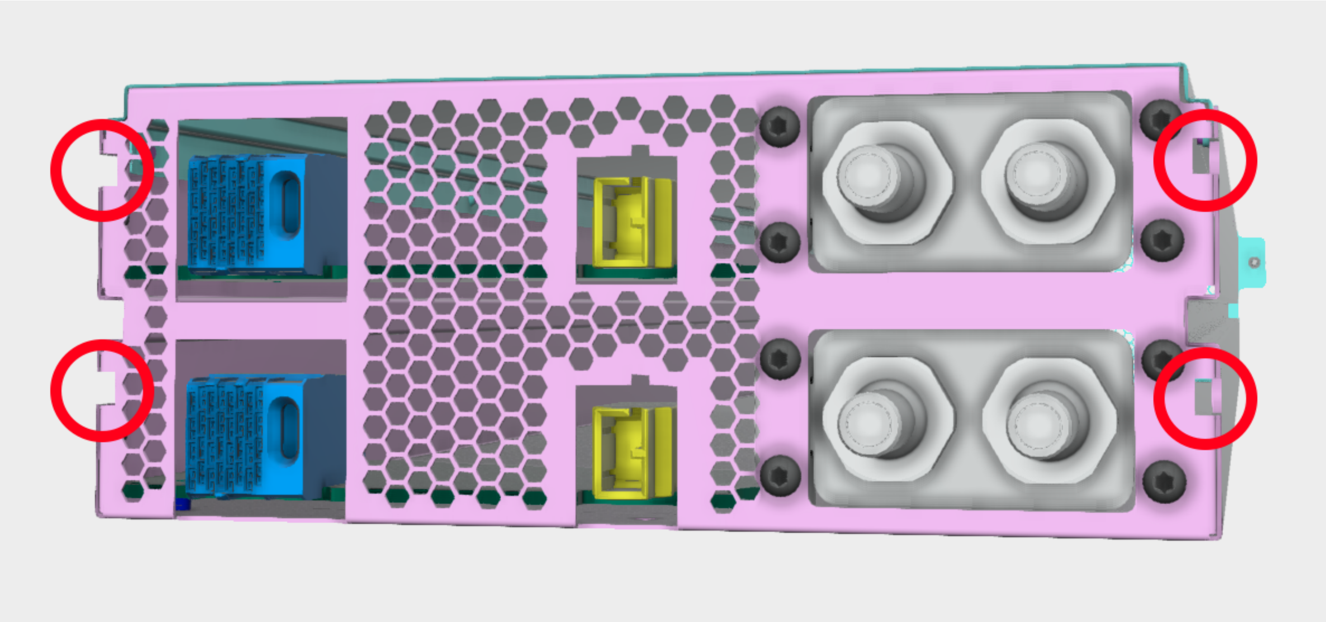


Figure 20. Brick keying, liquid cooling connections shall be defined in a future release

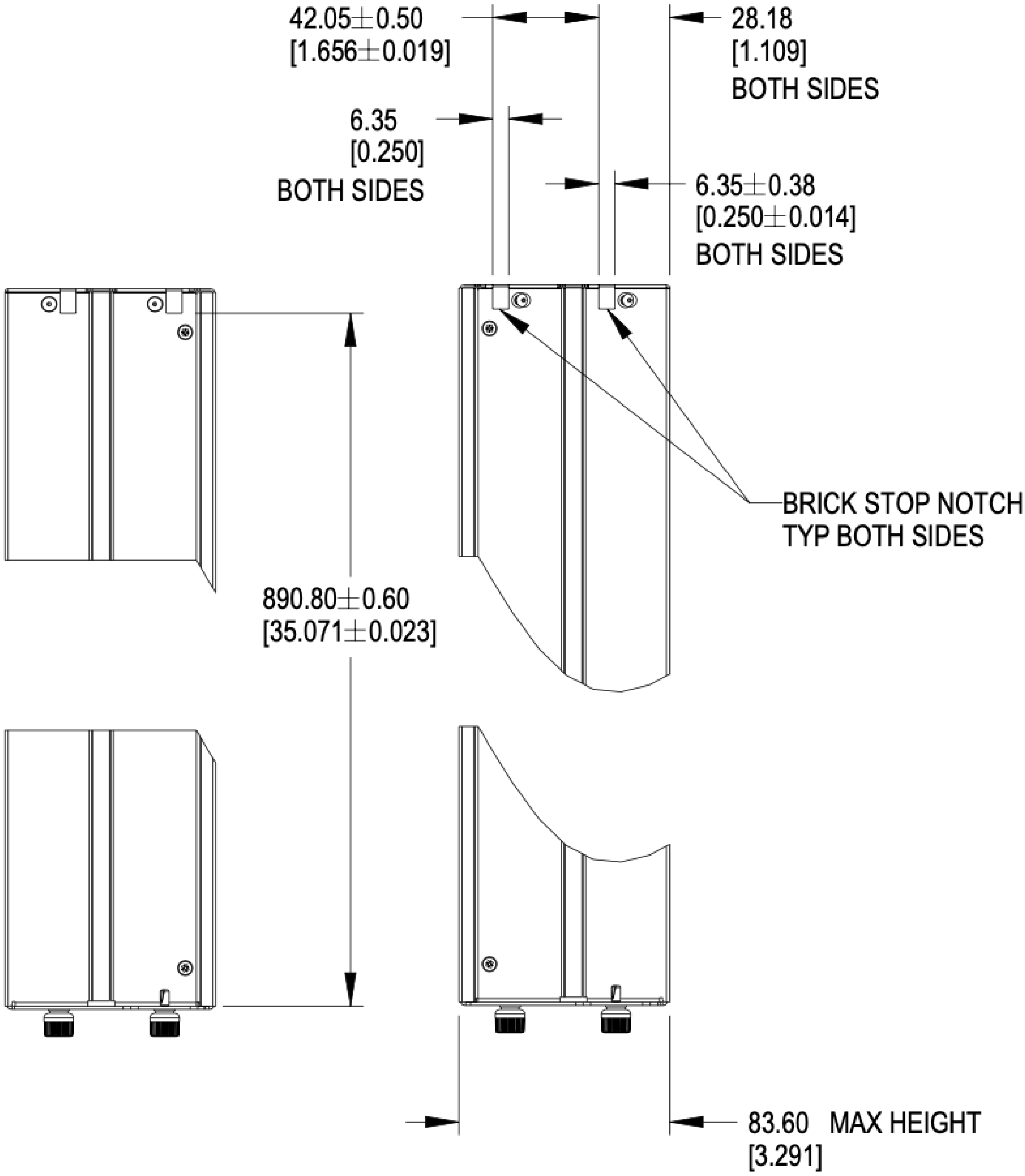


Figure 21. Brick keying dimensions

4.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

A second PCBA may be installed at 48.55mm +/- 0.5mm above the bottom surface of the chassis.

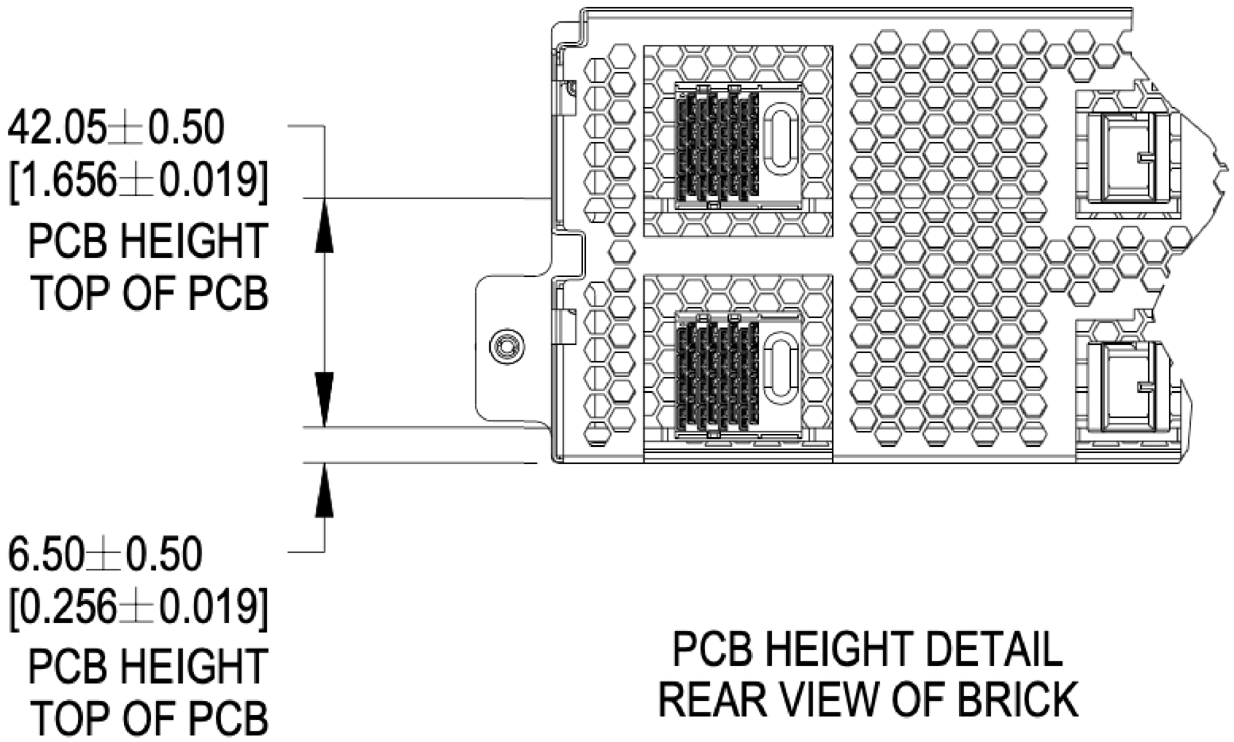


Figure 22. PCBA References

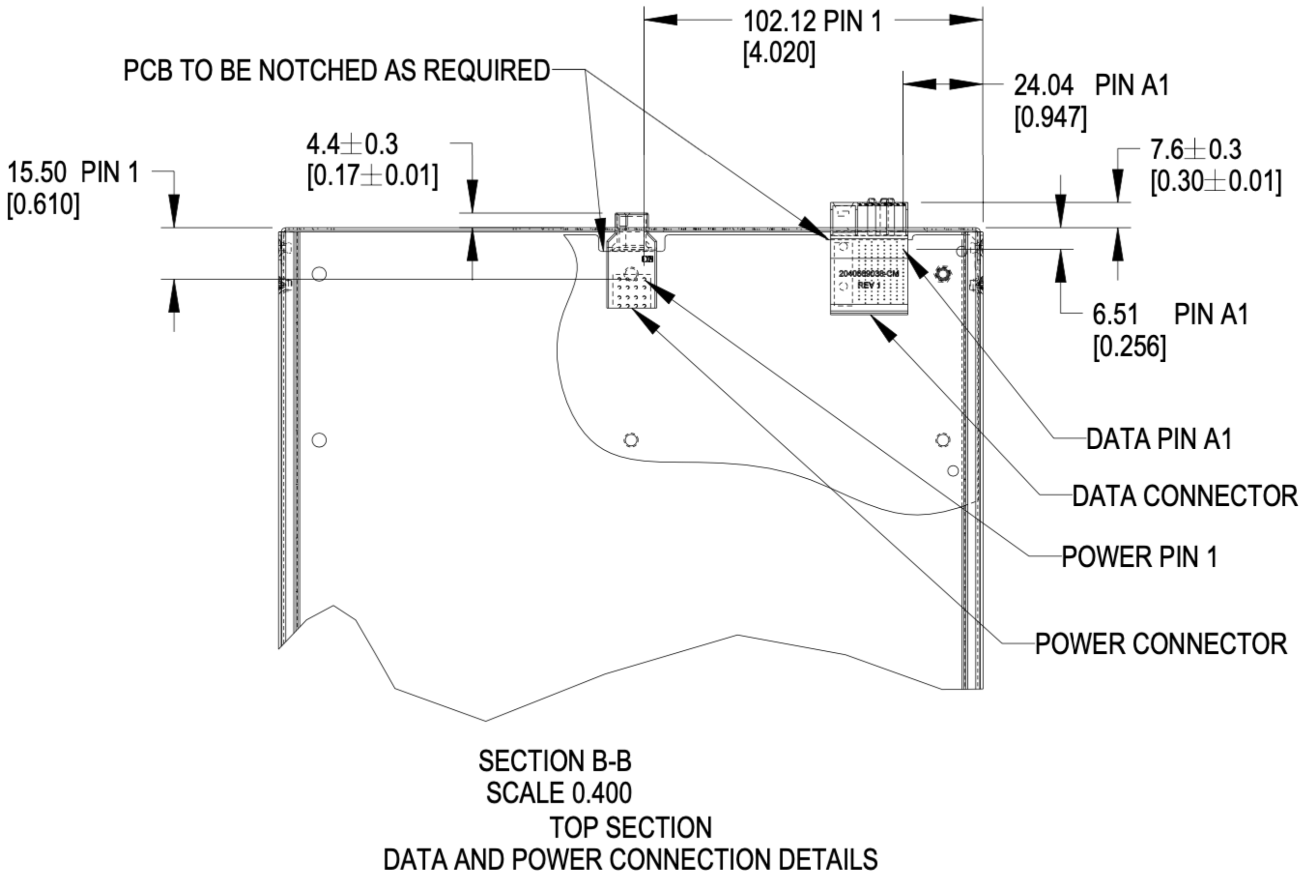


Figure 23. Brick PCB dimensions

One set of connectors are shown in the image. Two sets are available in a brick.

4.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

4.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate a second set of connectors, as required.

4.4. Connector Pinout

4.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

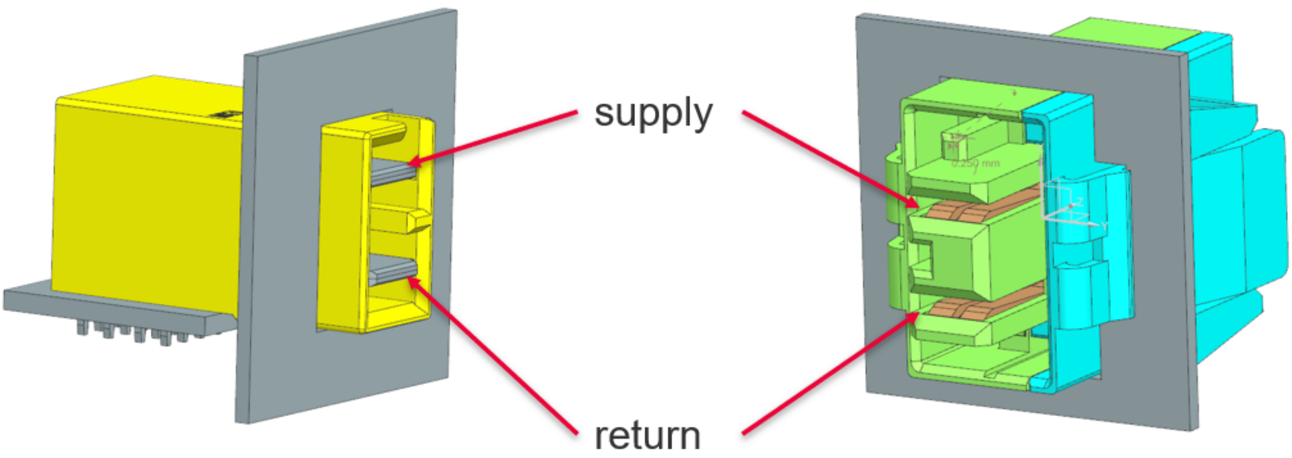


Figure 24. Power connector supply/return definition

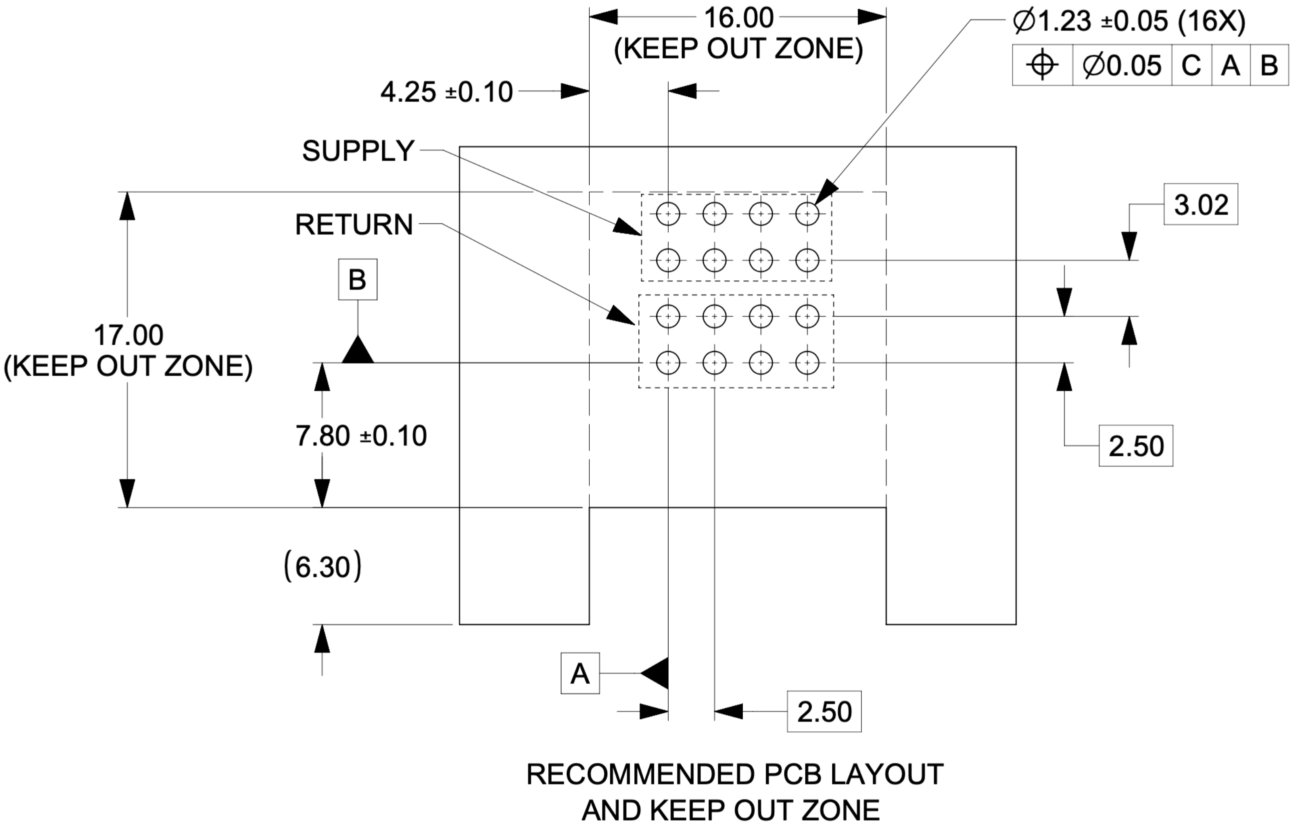


Figure 25. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

4.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

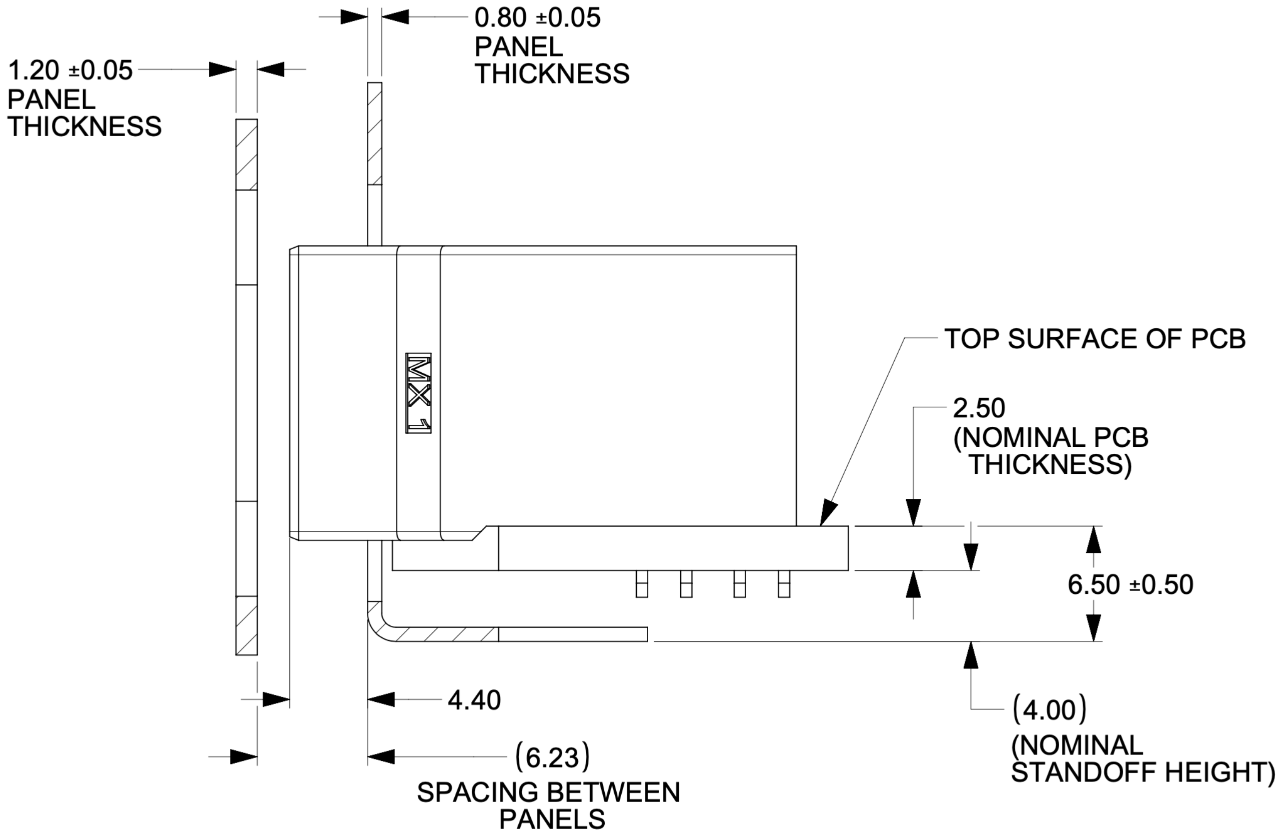


Figure 26. Power connector critical dimensions

4.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

4.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

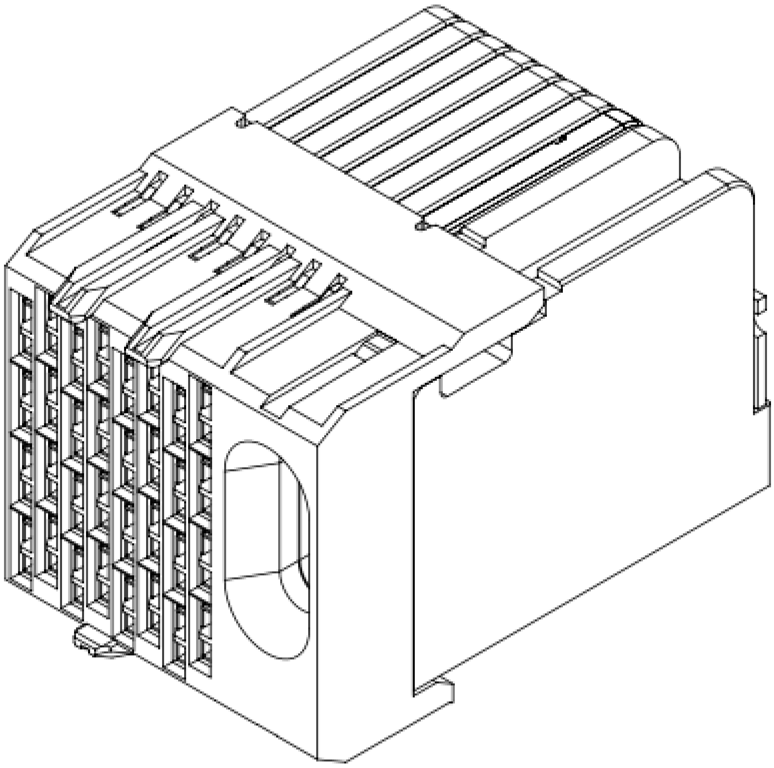


Figure 27. Impel Connector with guide feature

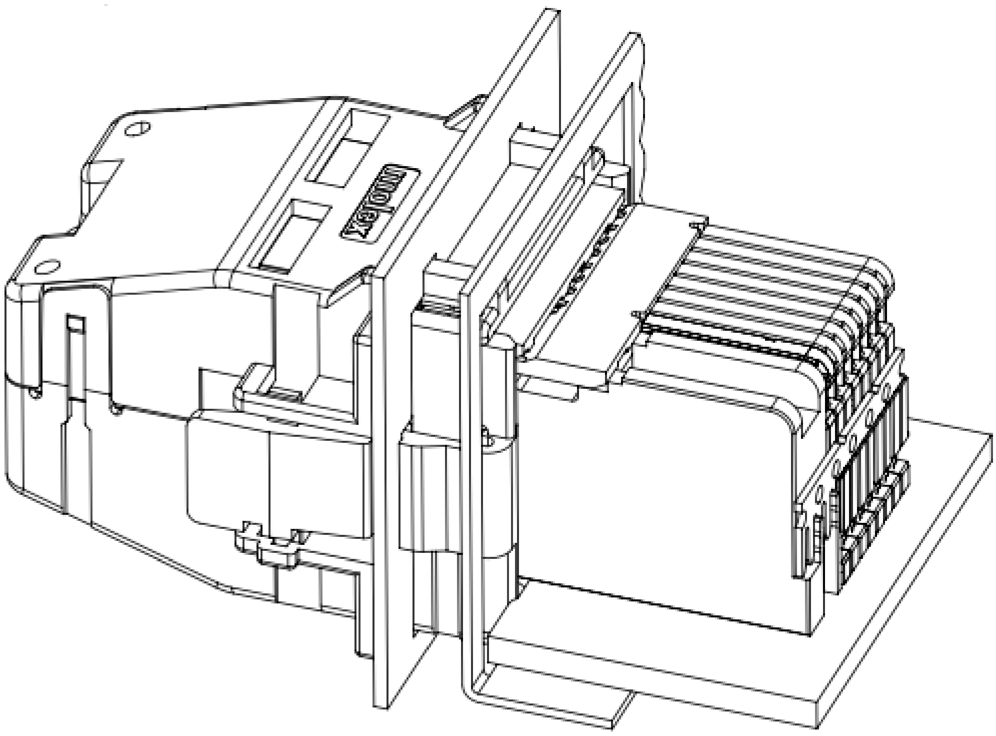


Figure 28. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

4.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

Port Count	Data Rate	Implementation Interface and order
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1
6		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	100G (4x25G) 200G (4x50G PAM4)	SFP1, SPF2, SFP3, SPF4, QSFP1
6		SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	400G (8x50G PAM4)	QSFP-DD (QSFP1+QSFP2) 200G (4x50G PAM4)
QSFP (SFP1+SFP2+SFP3+SFP4)	2	QSFP, QSFP1

4.4.6. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 60lbs (13.6kg). This is an increase of 10lbs from V1.

4.4.7. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

4.4.8. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 18, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

4.4.9. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

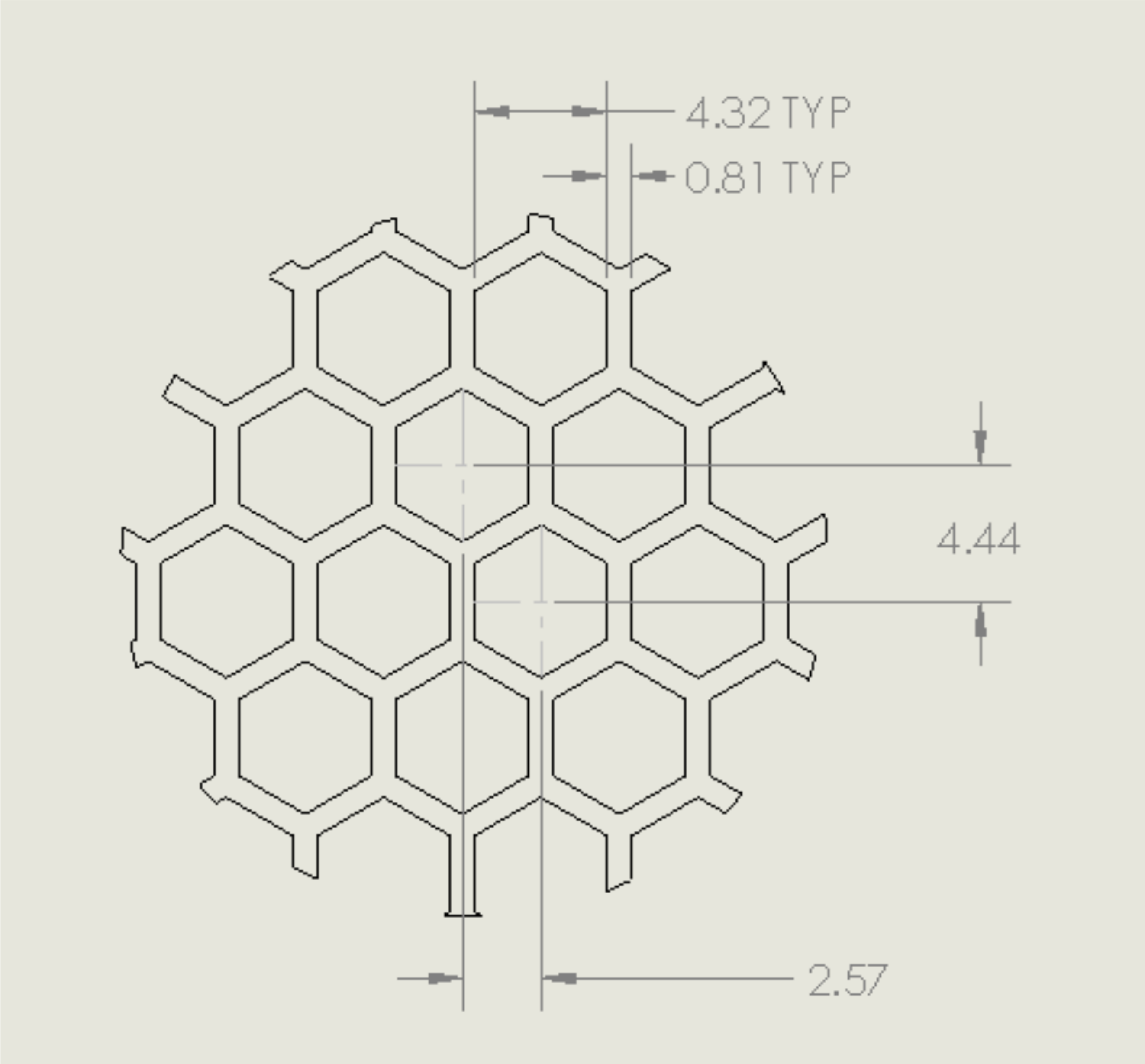


Figure 29. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

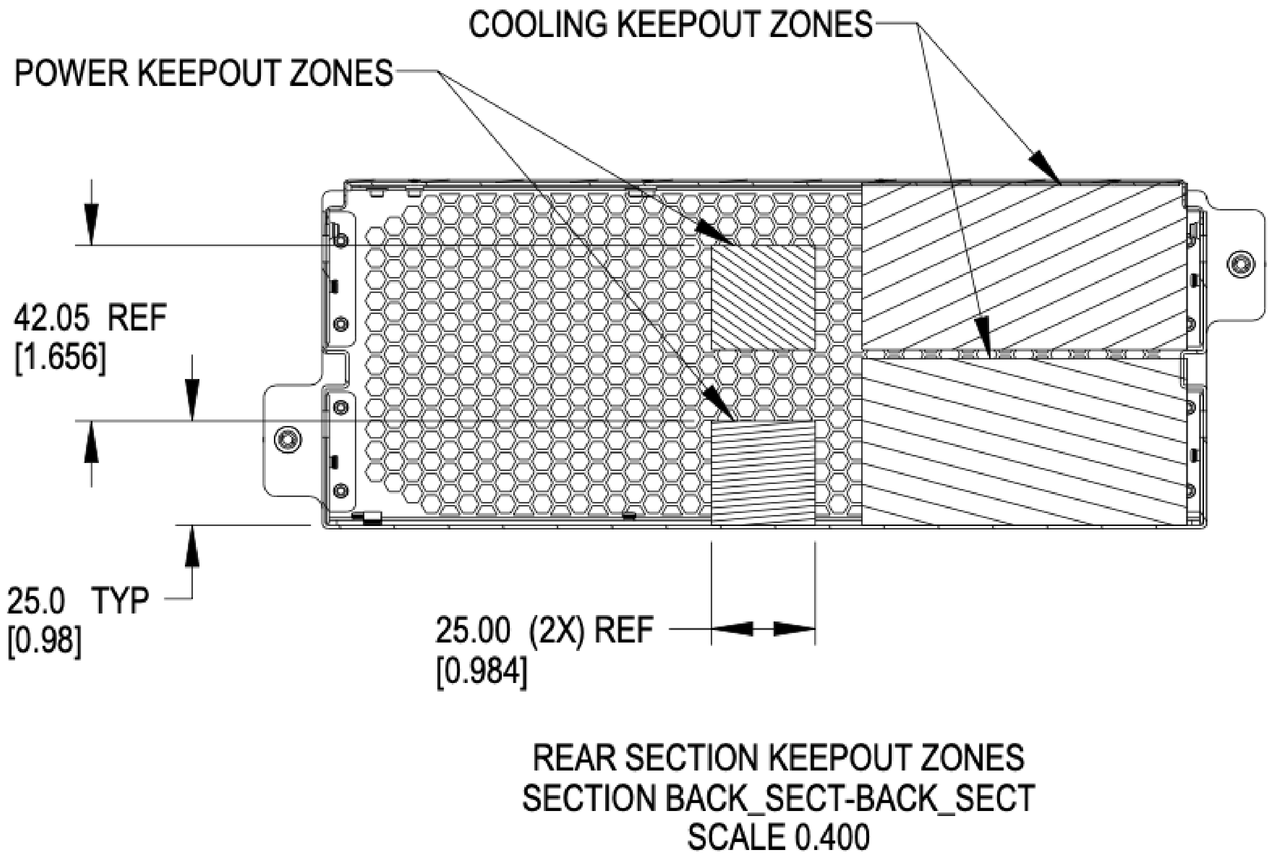


Figure 30. Brick keep-out zone

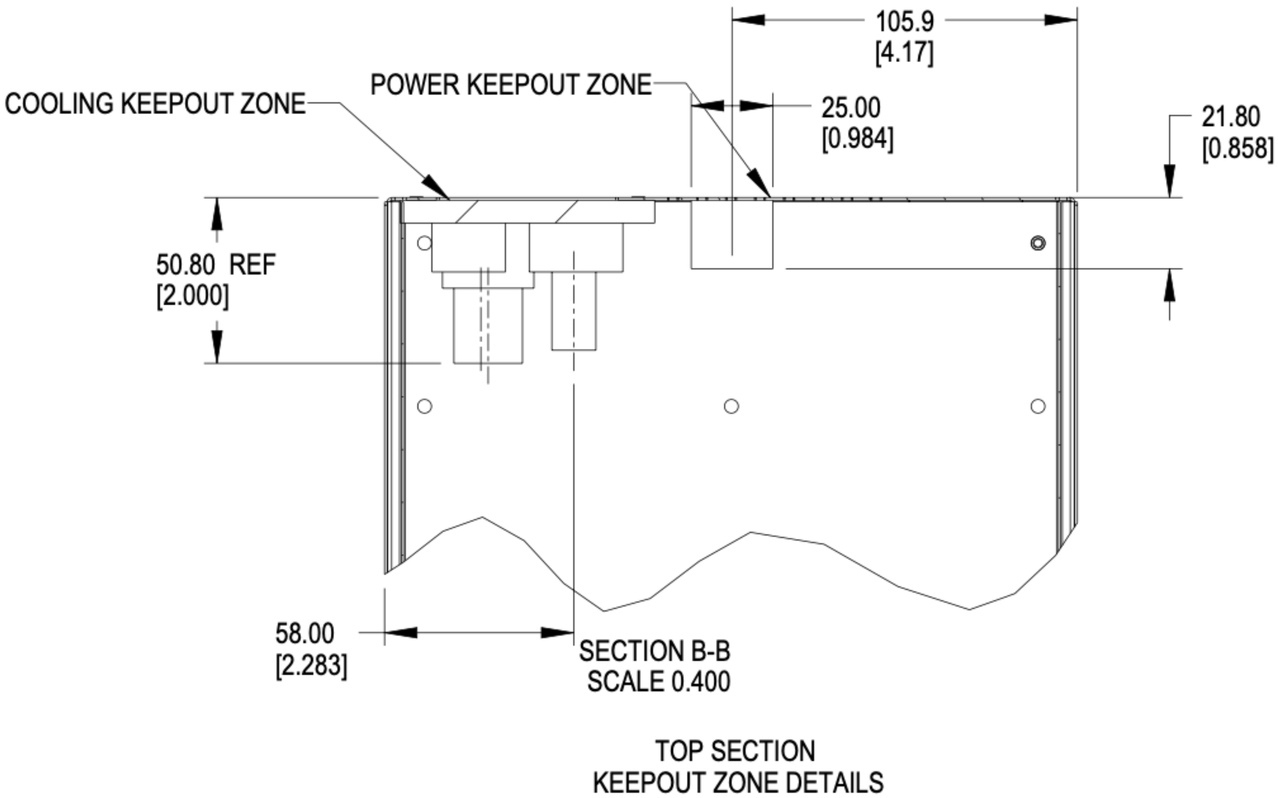


Figure 31. Brick keep-out zone top view

4.5. Brick performance

4.5.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

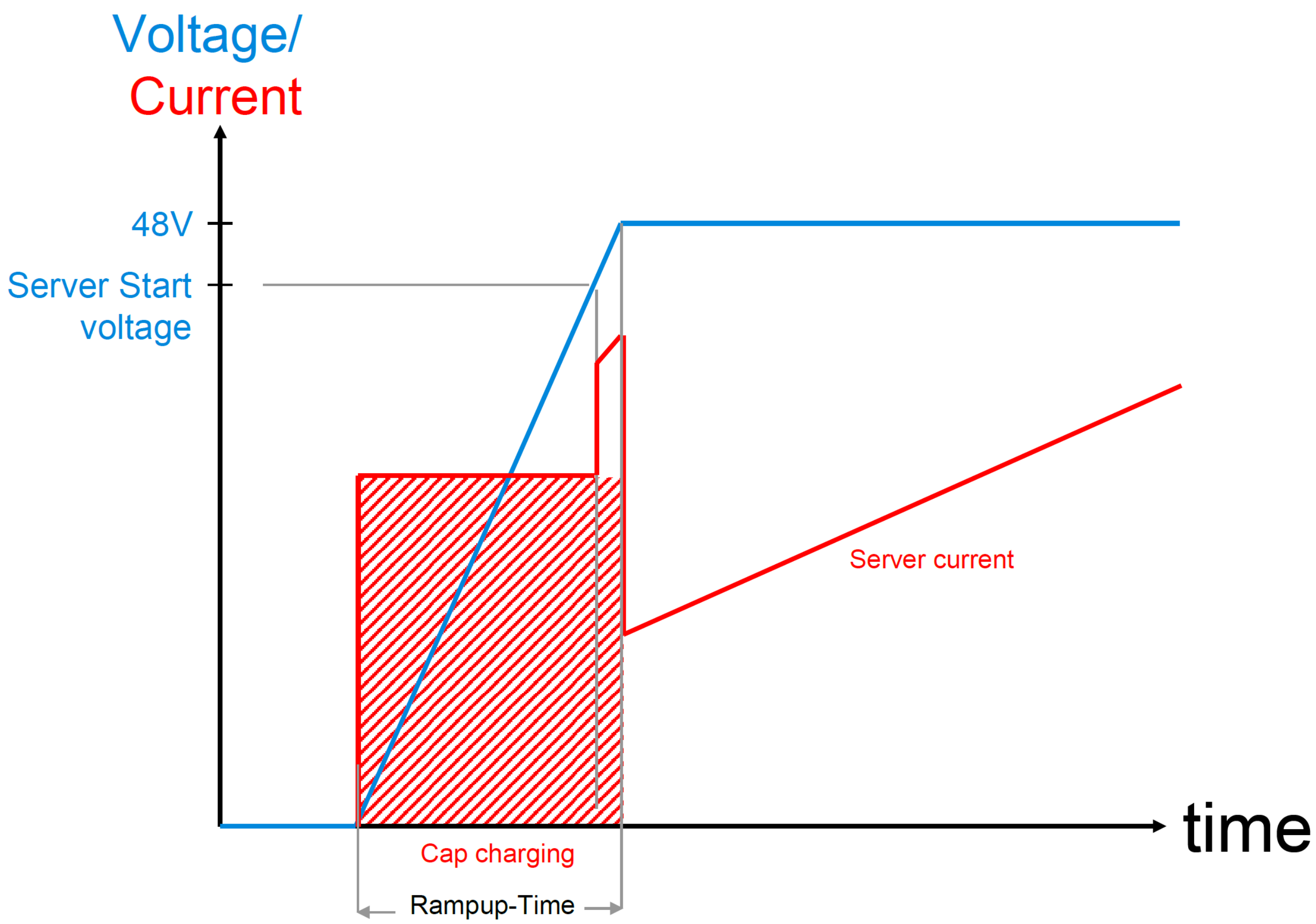


Figure 32. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Two power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 5. Brick Mechanical Specification [SHDW]

5.1. SHDW Brick

The SHDW brick is commonly used as a 1RU server form-factor.

5.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

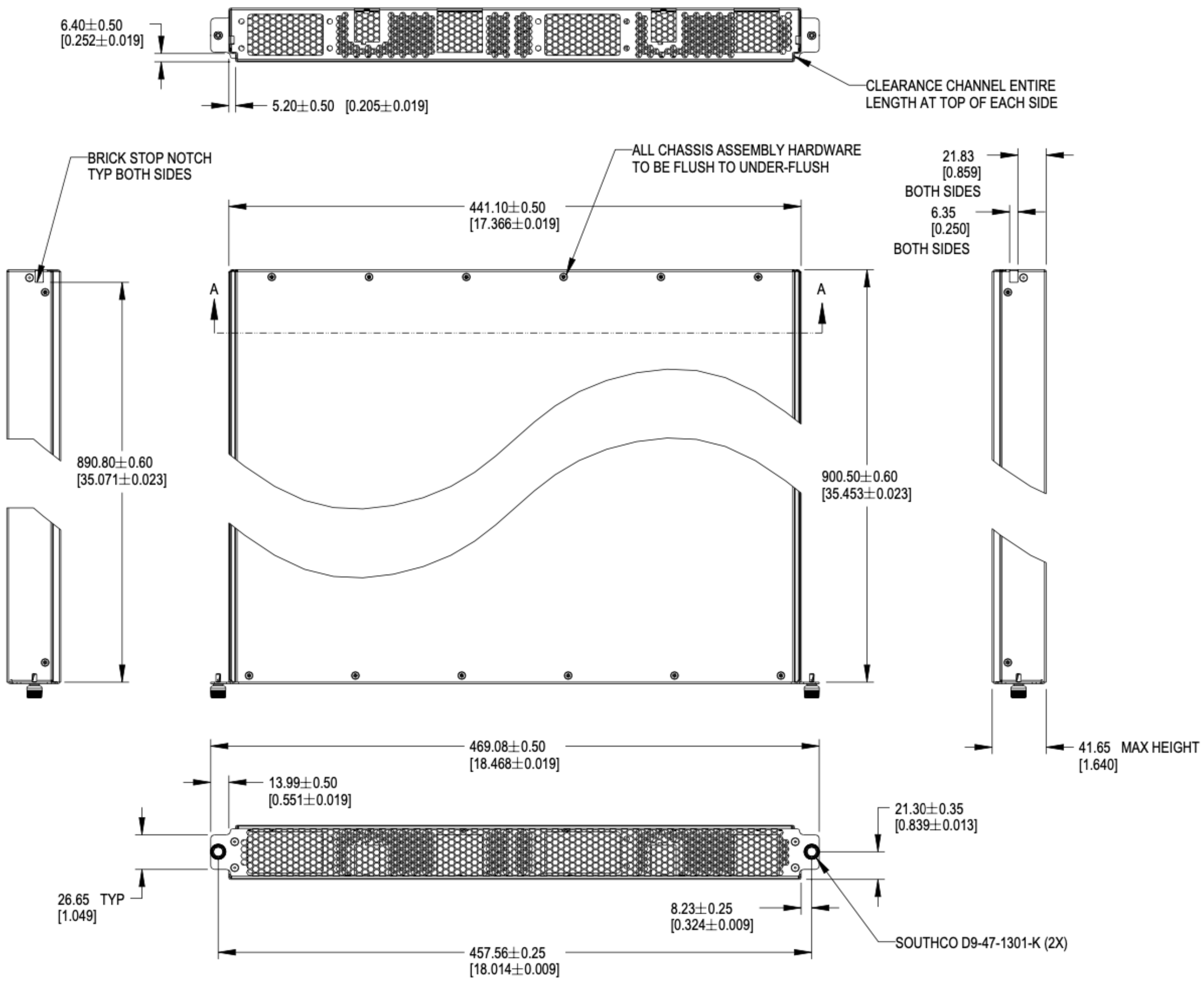


Figure 33. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

5.3. Mechanical features

5.3.1. Brick retention

The brick is secured into the cage with two Southco D9-47-1302-K quarter-turn fastener, located on the left and right side of the faceplate.

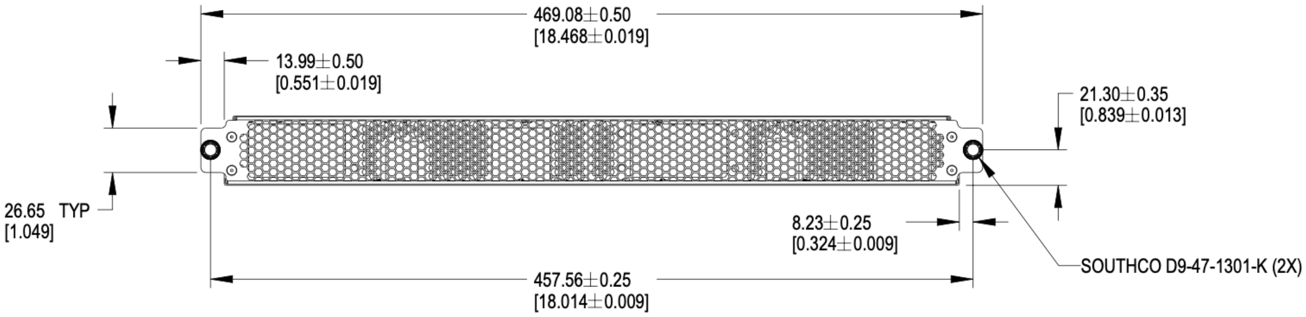


Figure 34. Brick retention

5.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick, extending forward from the rear panel to the back of the faceplate.

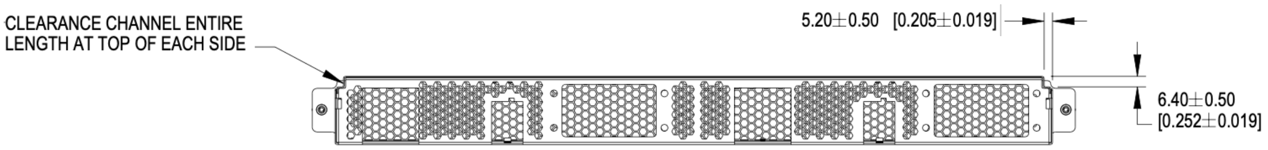


Figure 35. Mechanical guide at top edge of brick

5.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

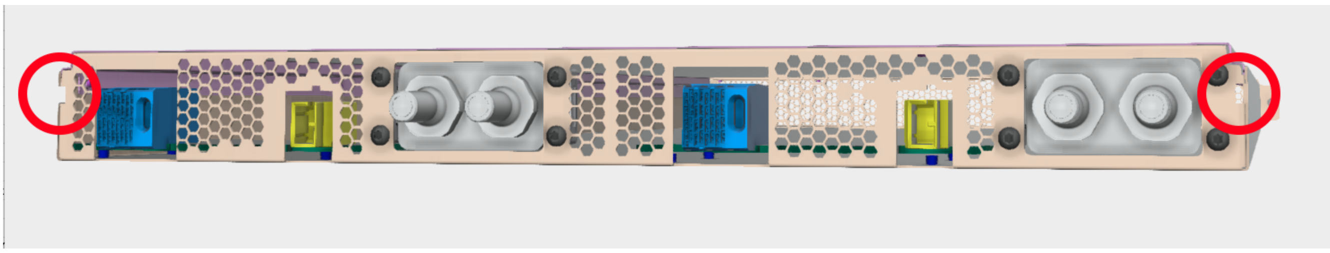


Figure 36. Brick keying, liquid cooling connections shall be defined in a future release

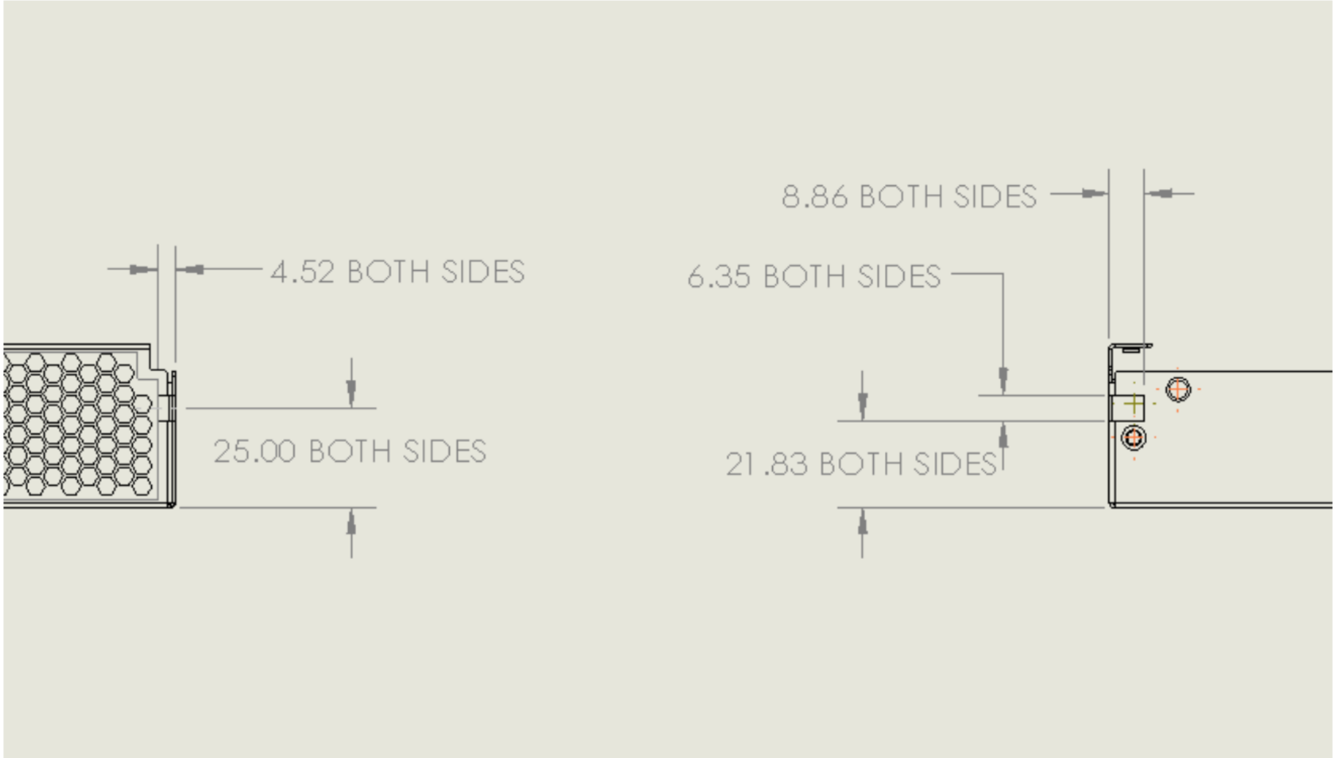


Figure 37. Brick keying dimensions

5.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

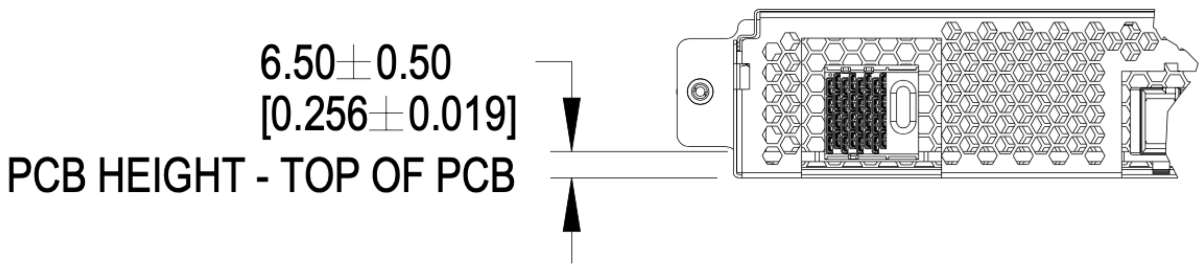


Figure 38. PCBA References

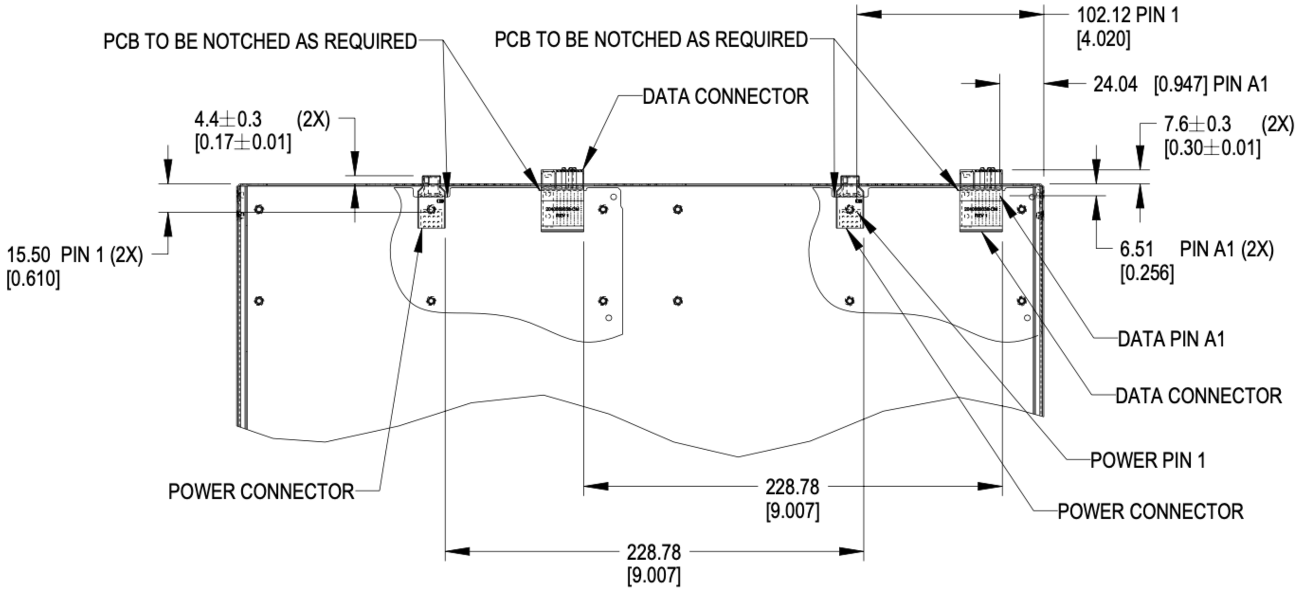


Figure 39. Brick PCB dimensions

5.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

5.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate a second set of connectors, as required.

5.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 60lbs (13.6kg). This is an increase of 10lbs from V1.

5.4. Connector Pinout

5.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

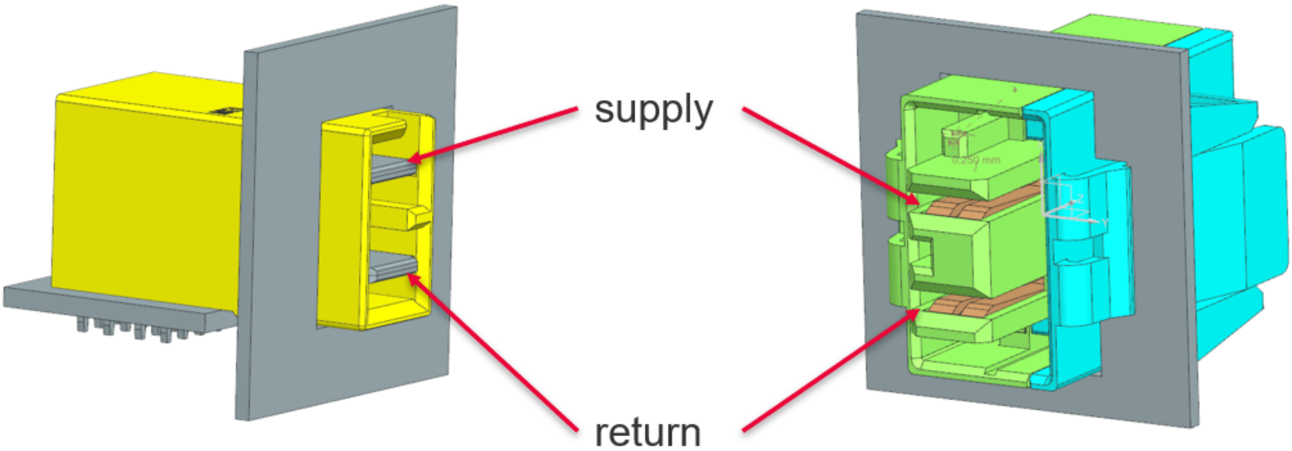


Figure 40. Power connector supply/return definition

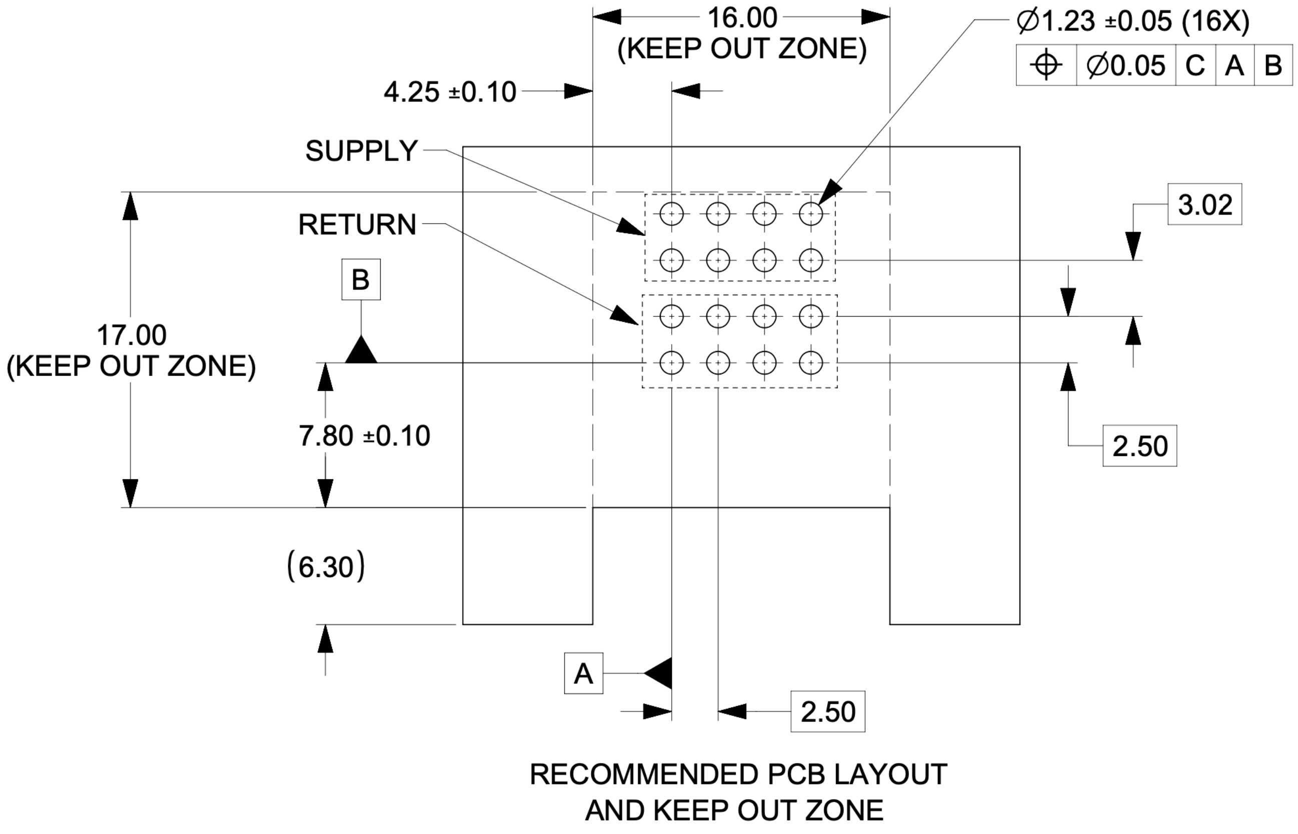


Figure 41. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

5.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

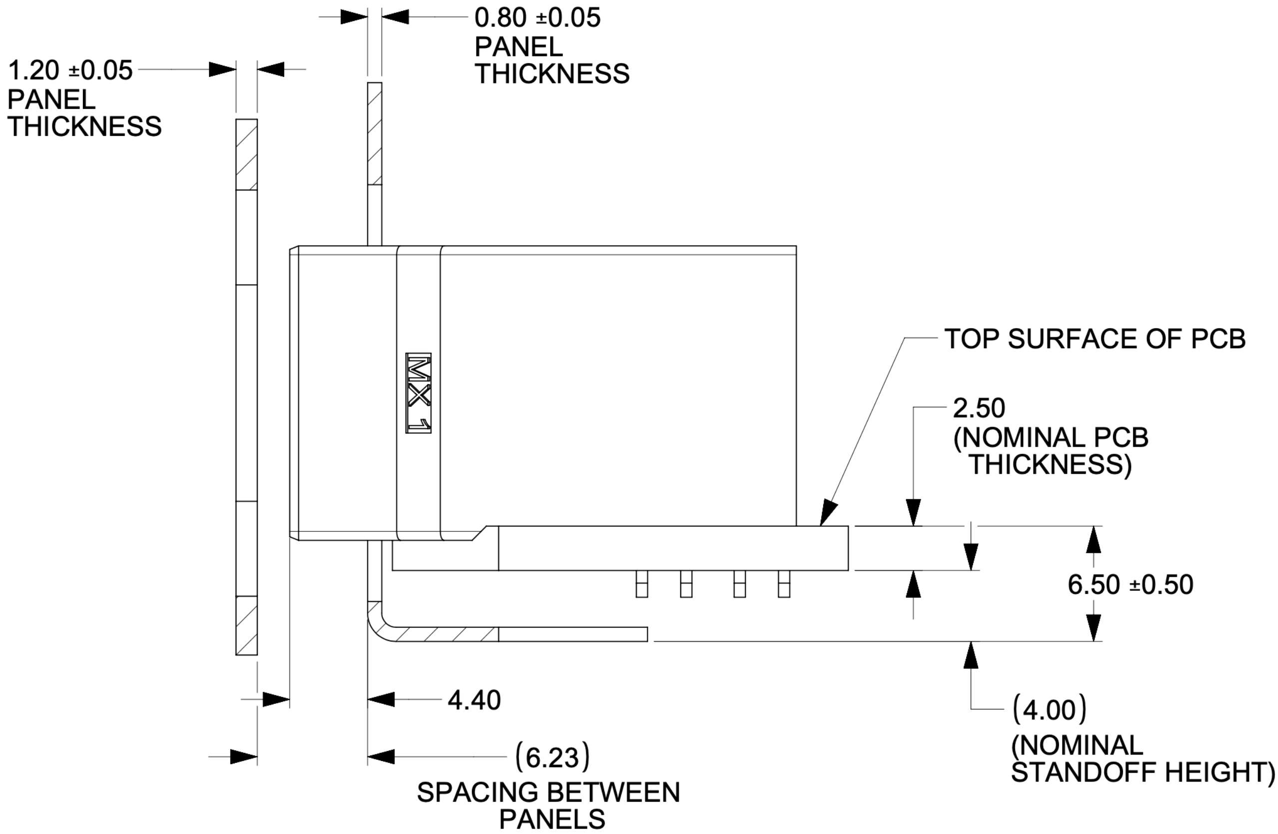


Figure 42. Power connector critical dimensions

5.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

5.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

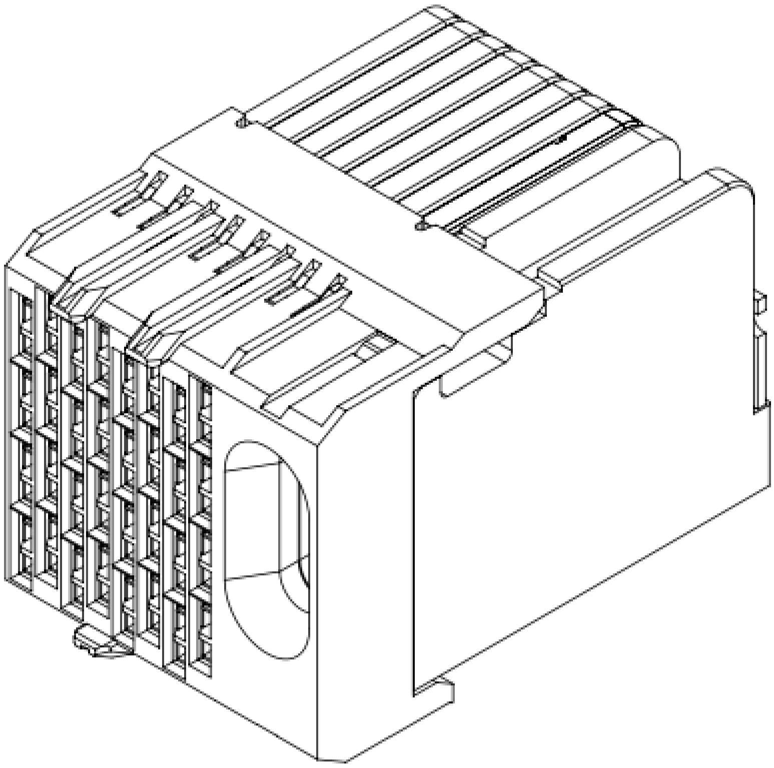


Figure 43. Impel Connector with guide feature

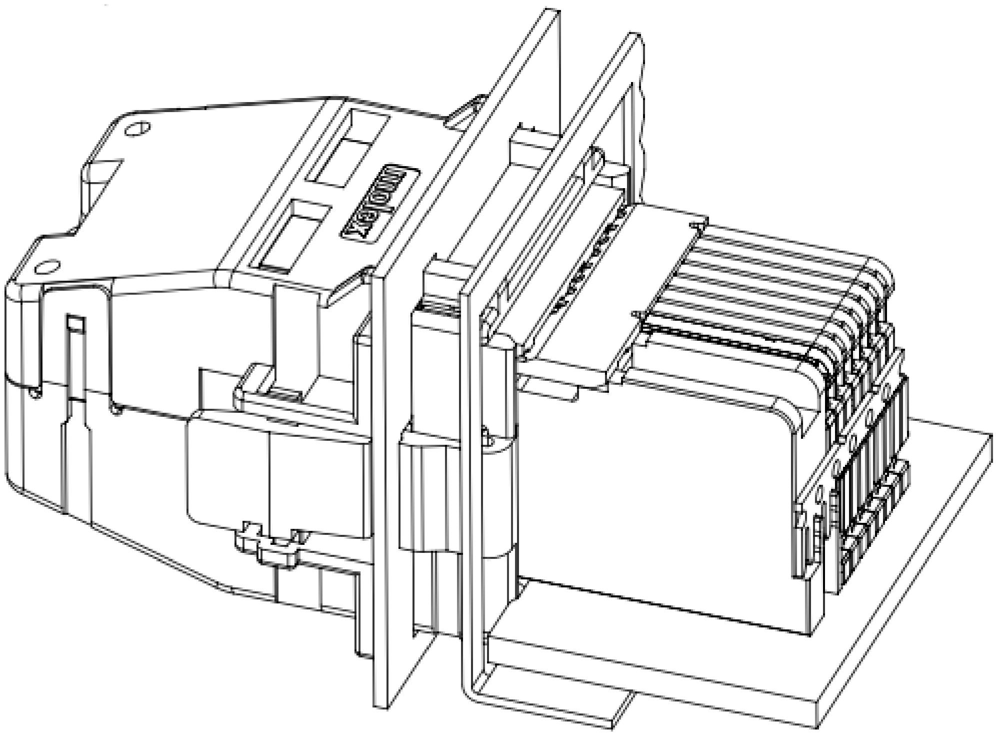


Figure 44. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

5.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

<table>
<tr><th>Port Count</th><th>Data Rate</th><th>Implementation Interface and order</th></tr>
<tr><td>1</td><td rowspan="6">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td rowspan="2">100G (4x25G)

200G (4x50G PAM4)</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>400G (8x50G PAM4)</td><td>QSFP-DD (QSFP1+QSFP2)

200G (4x50G PAM4)</td></tr>
<tr><td>QSFP (SFP1+SFP2+SFP3+ SFP4)</td><td>2</td><td>QSFP, QSFP1</td></tr>
</table>

5.5. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

5.5.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 34, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

5.5.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

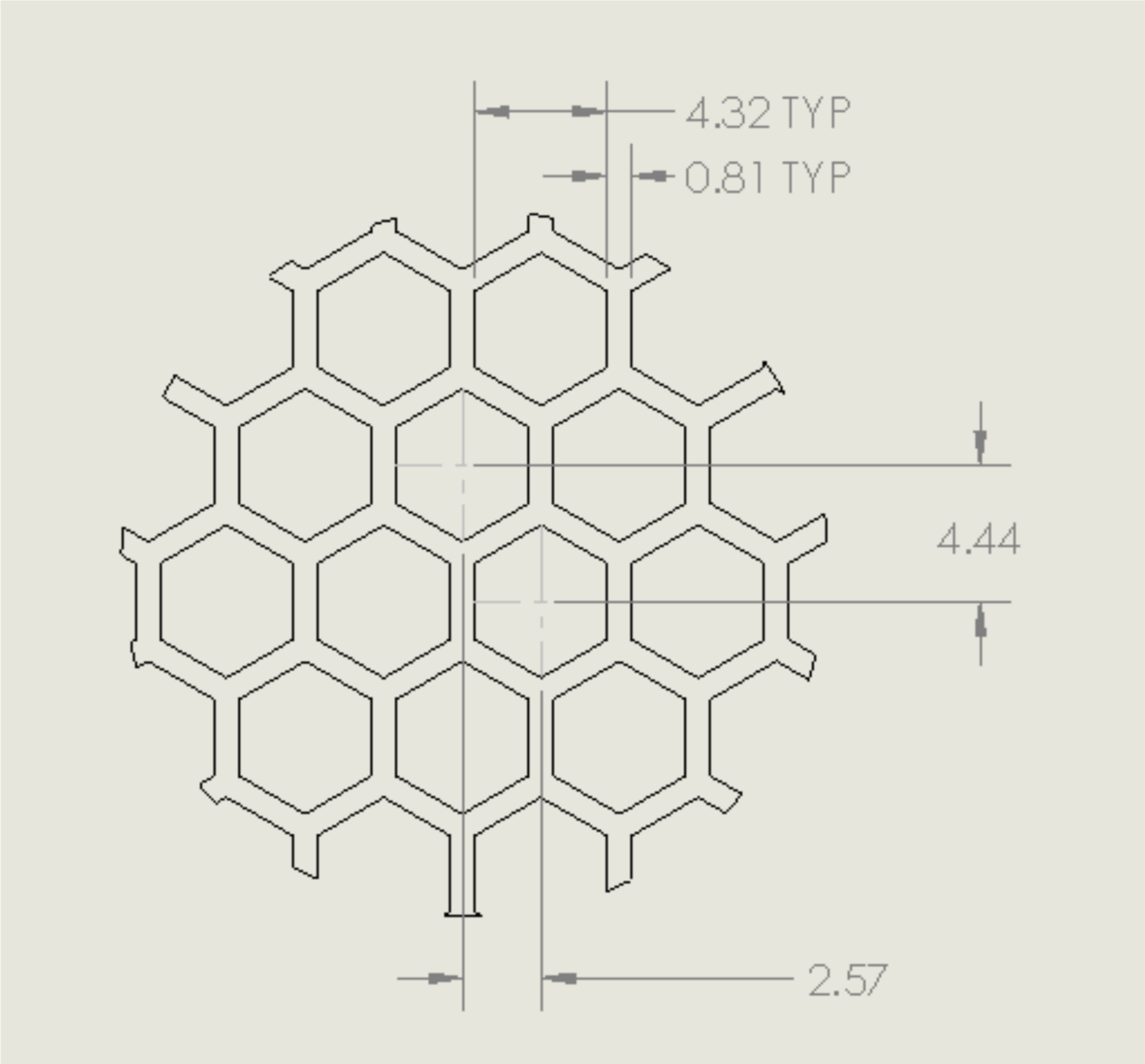


Figure 45. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

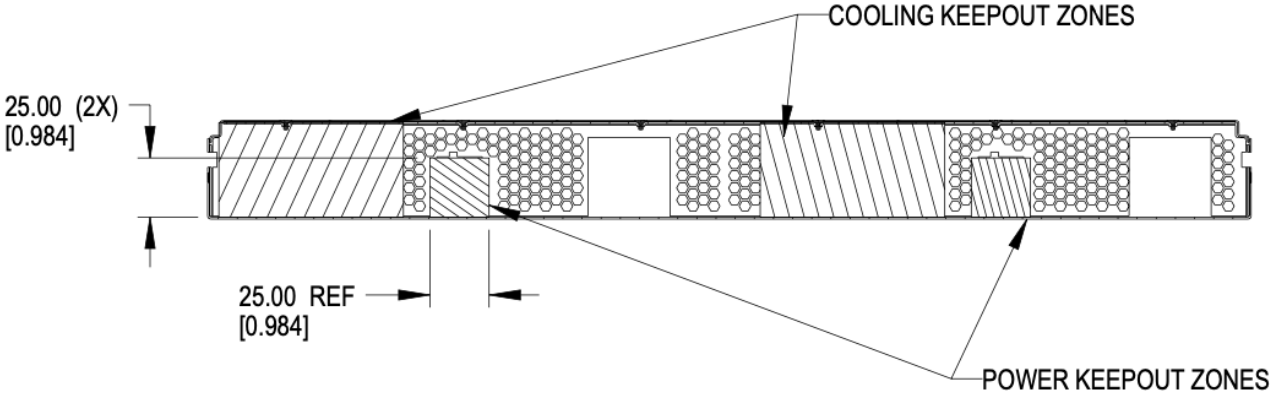


Figure 46. Brick keep-out zone

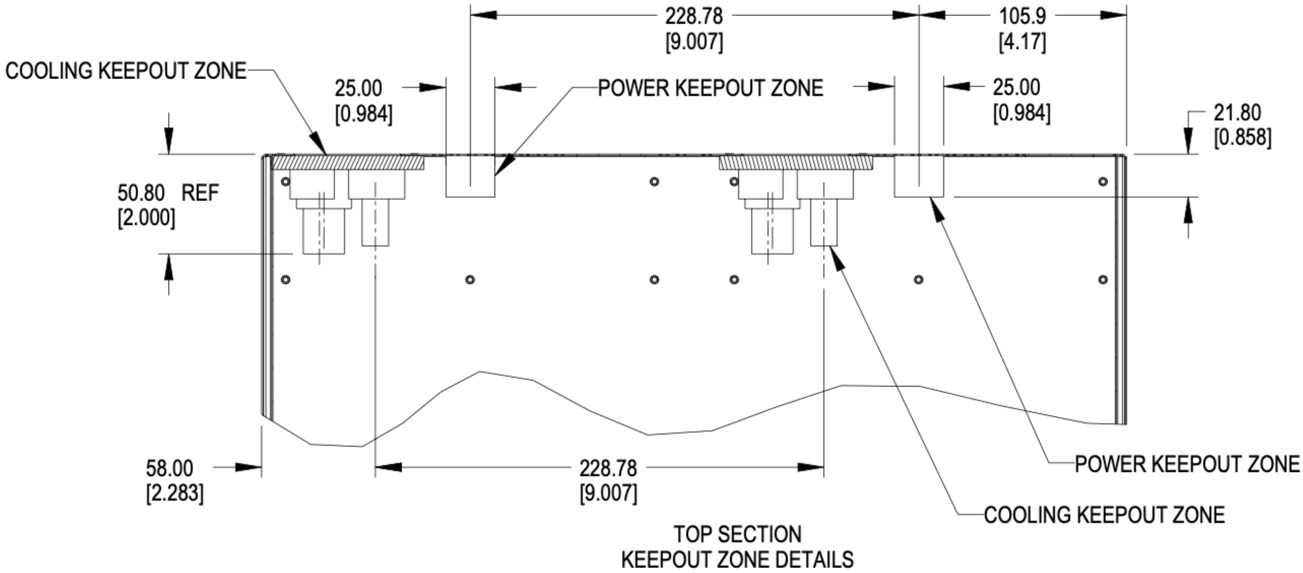


Figure 47. Brick keep-out zone top view

5.6. Brick performance

5.6.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

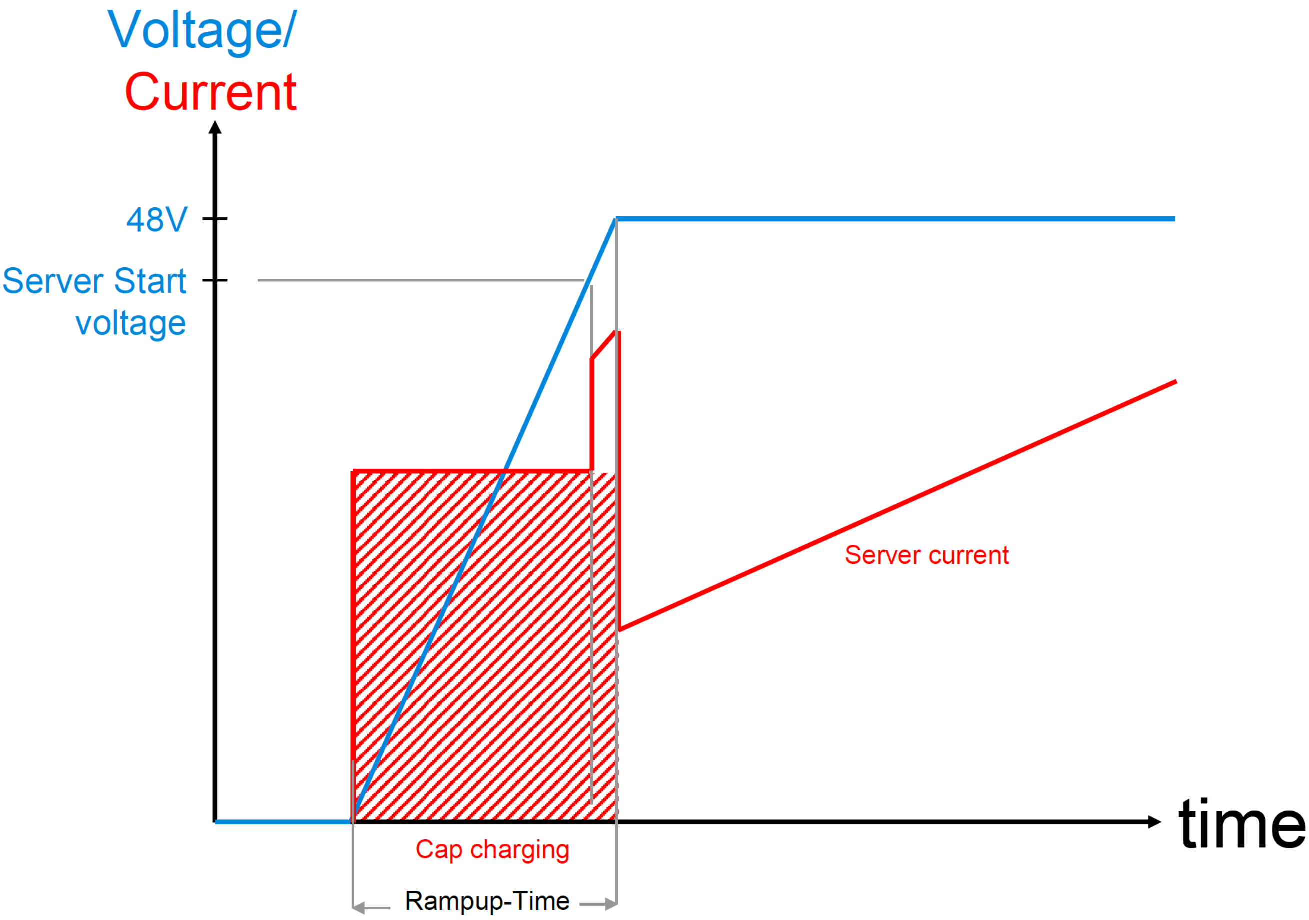


Figure 48. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Two power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 6. Brick Mechanical Specification [DHDW]

6.1. DHDW Brick

The DHDW brick is commonly used as a 2U server form-factor.

6.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

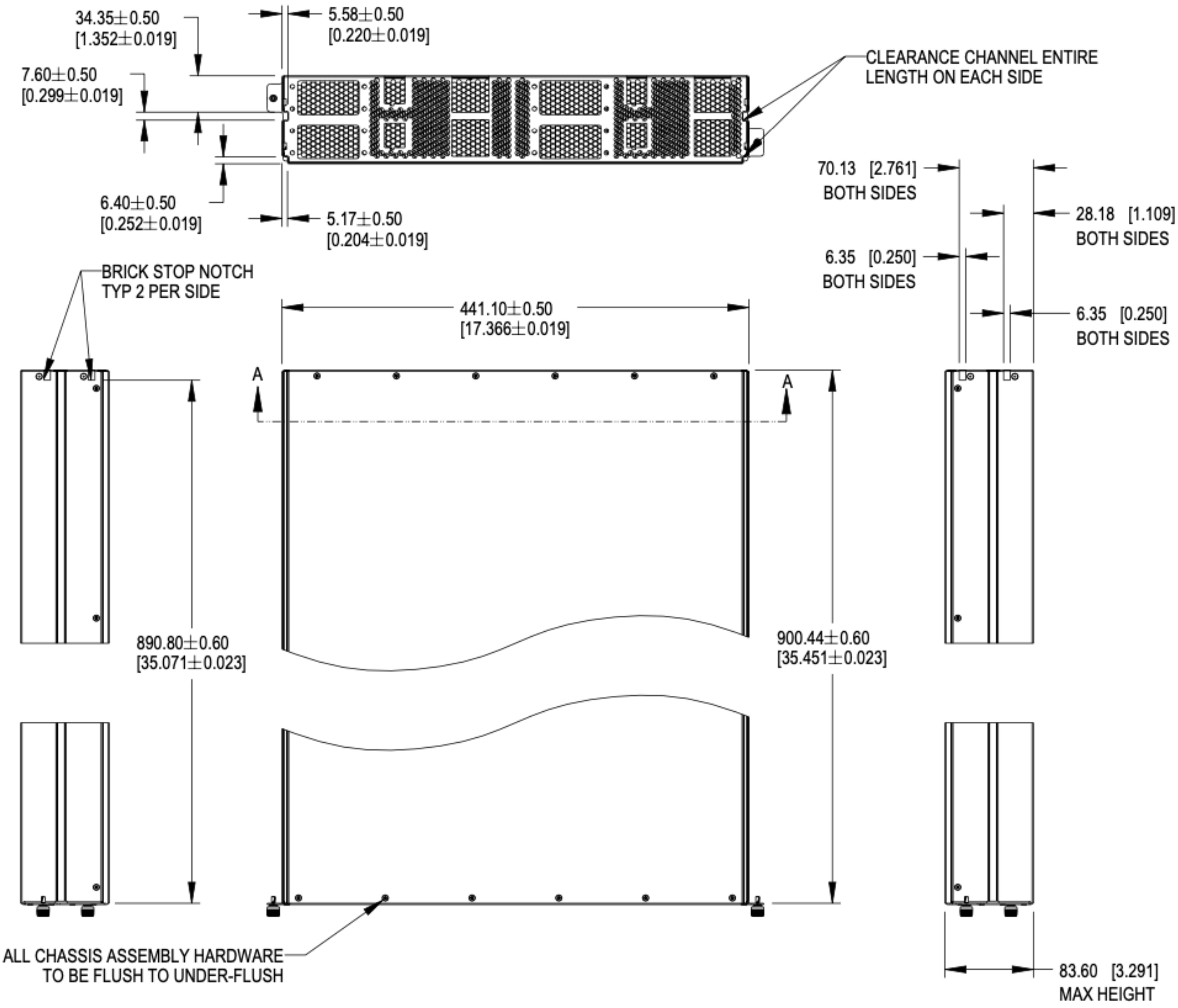


Figure 49. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

6.3. Mechanical features

6.3.1. Brick retention

The brick is secured into the cage with up to four Southco D9-47-1302-K quarter-turn fasteners, located on the left and right side of the faceplate.

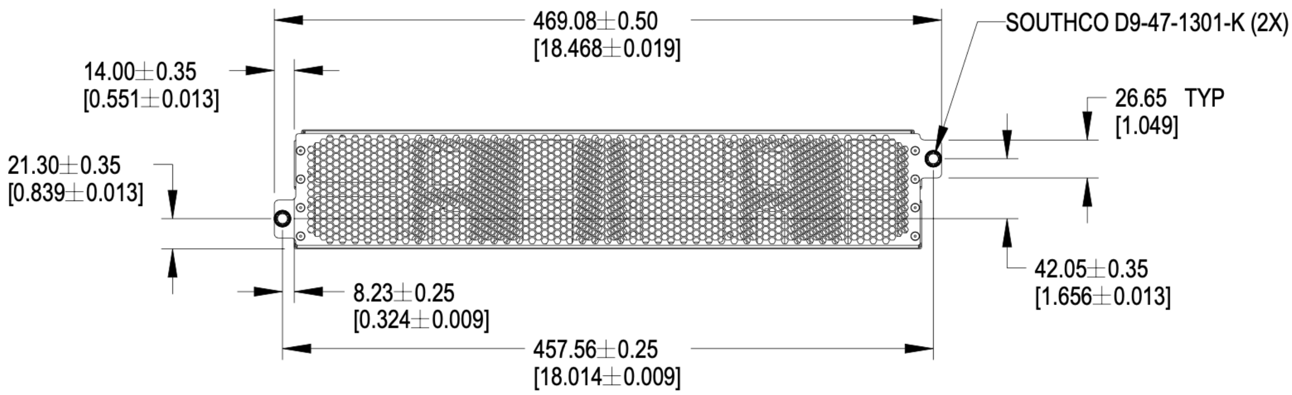


Figure 50. Brick retention

6.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick and the sidewalls, extending forward from the rear panel to the back of the faceplate.

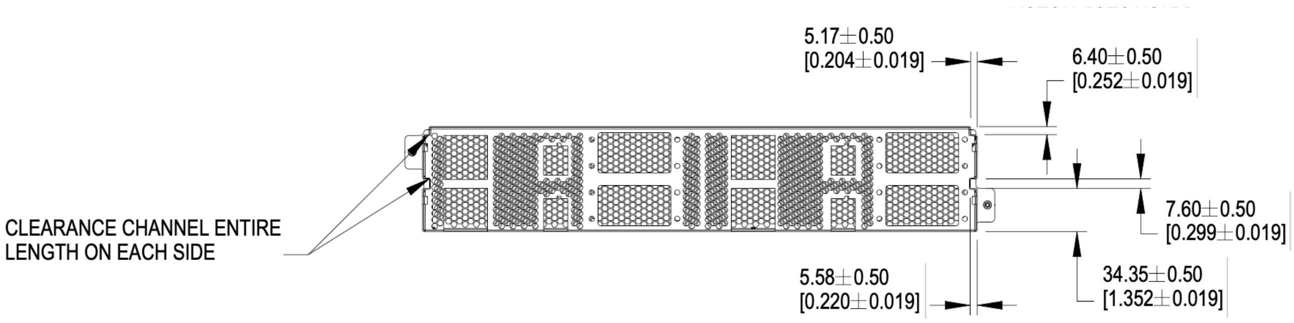


Figure 51. Mechanical guide at top edge and side walls of brick

6.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

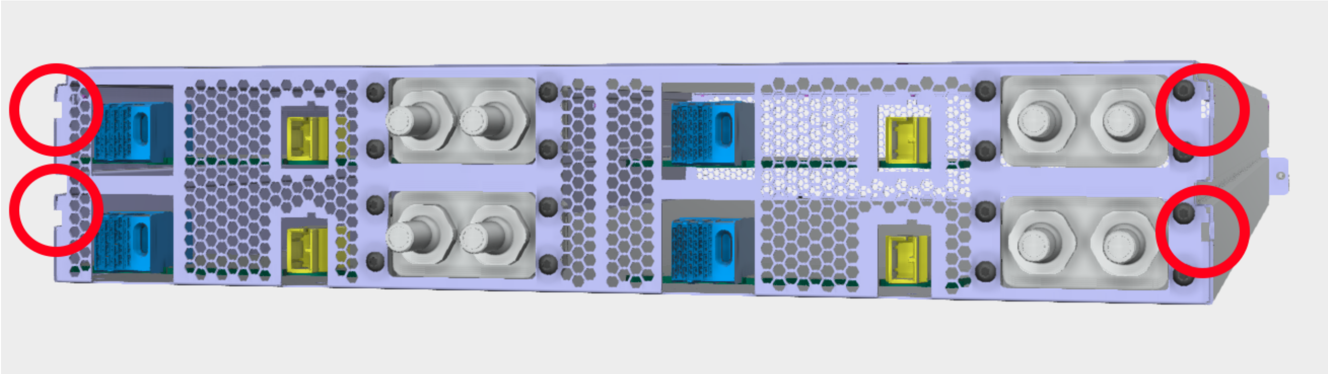


Figure 52. Brick keying, liquid cooling connections shall be defined in a future release

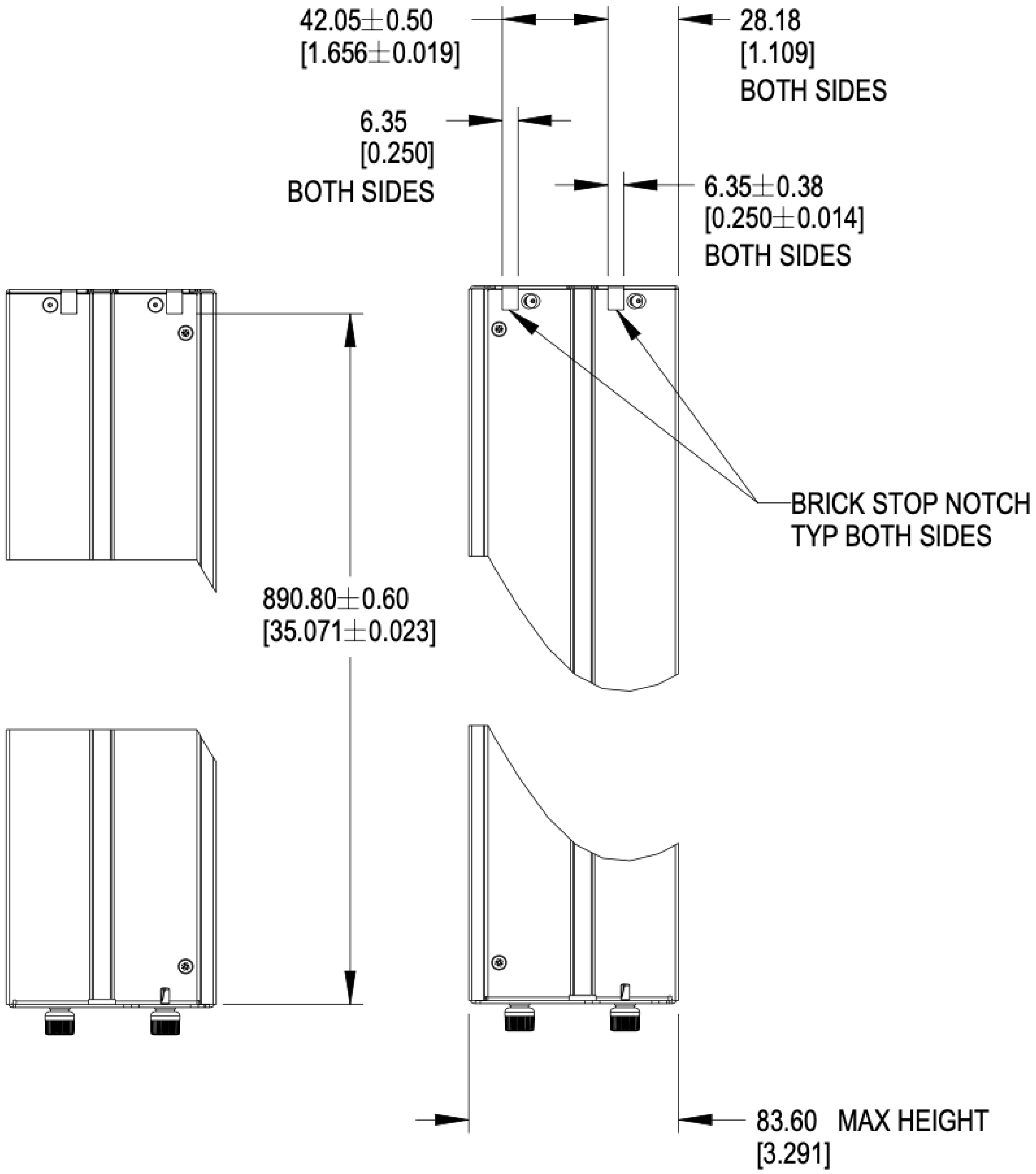


Figure 53. Brick keying dimensions

6.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

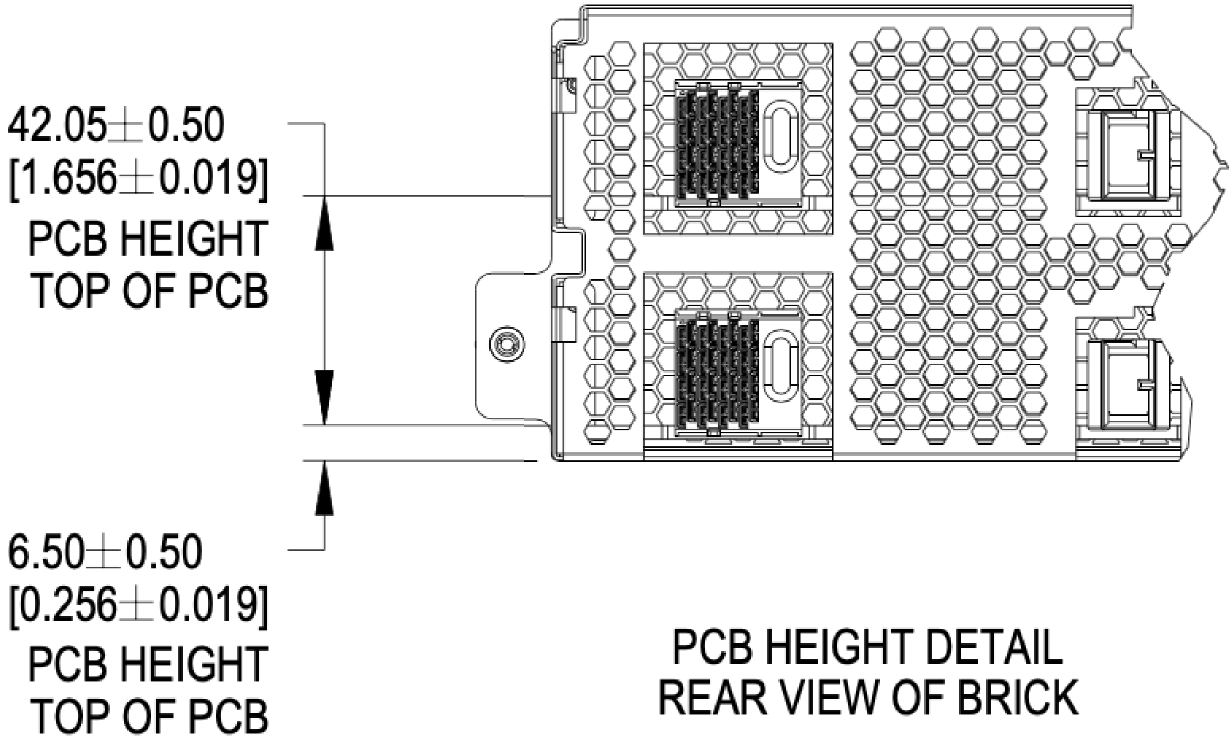


Figure 54. PCBA References

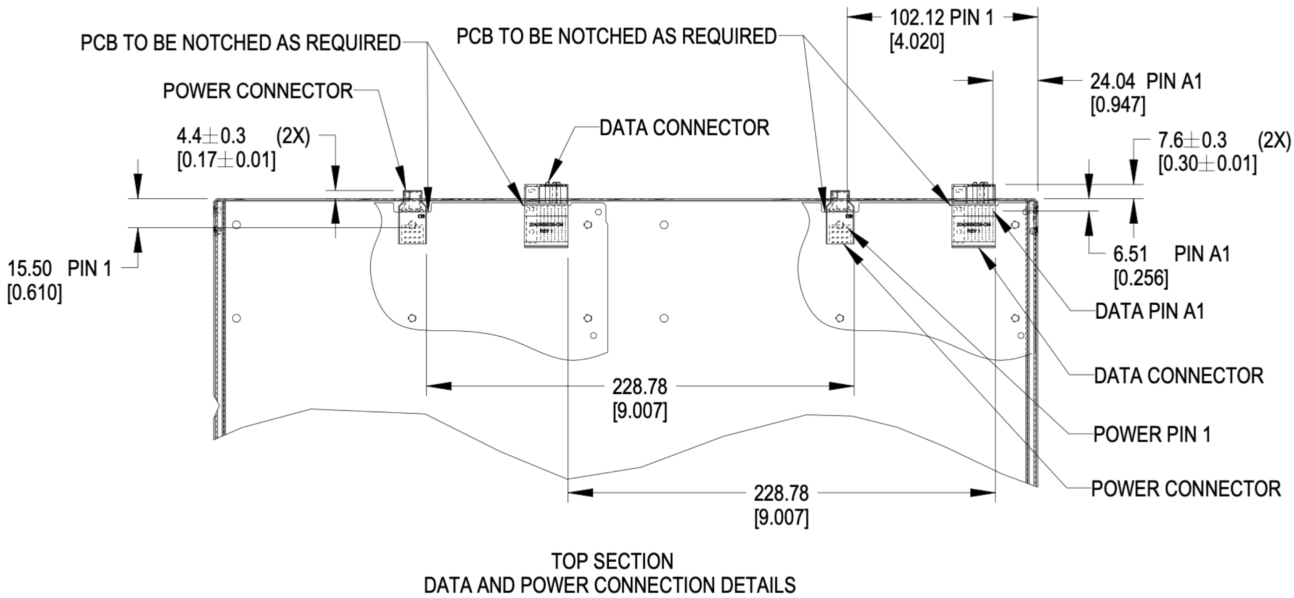


Figure 55. Brick PCB dimensions

6.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

6.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate up to four sets of connectors, as required.

6.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 120lbs (54.43kg). This is an increase of 20lbs from V1.

6.4. Connector Pinout

6.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

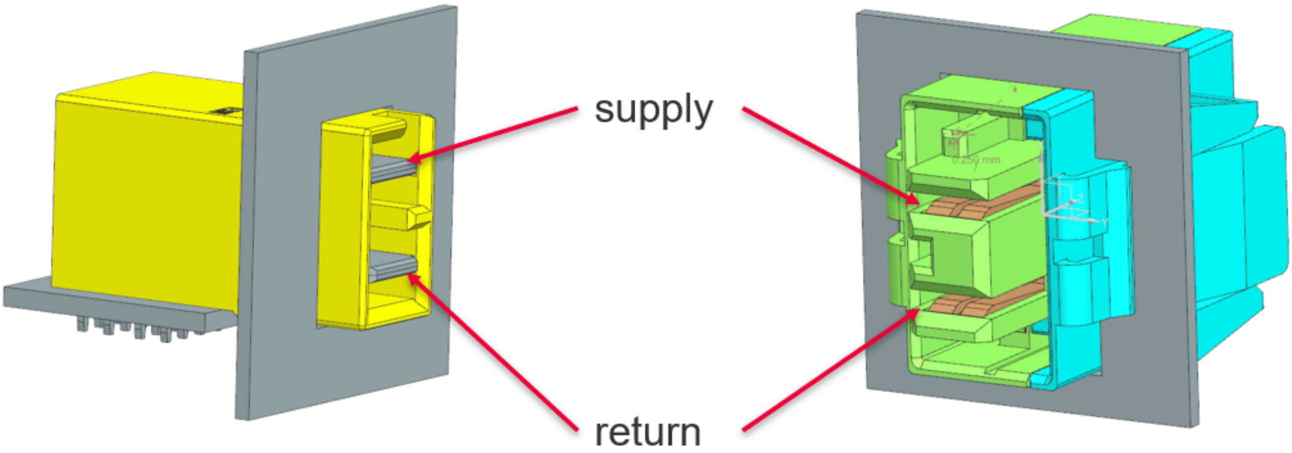


Figure 56. Power connector supply/return definition

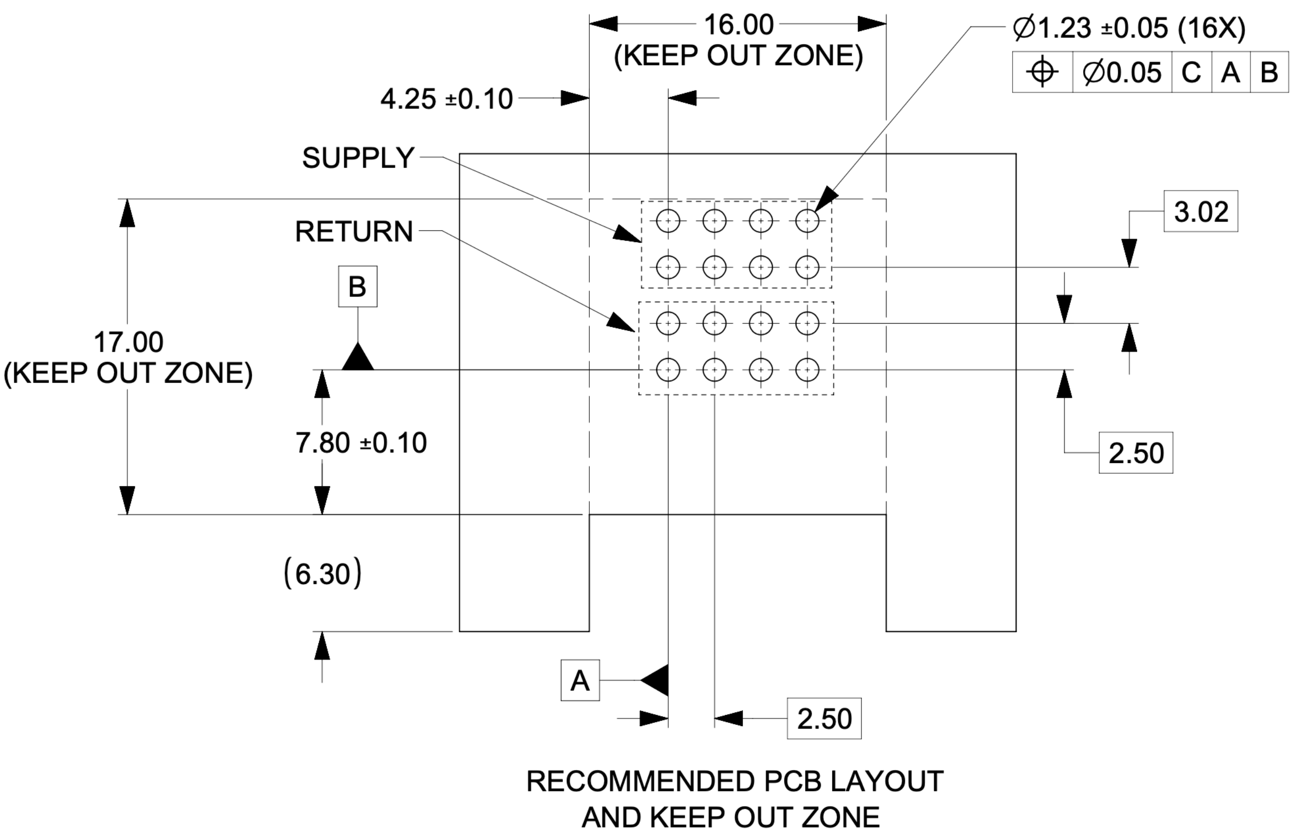


Figure 57. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

6.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

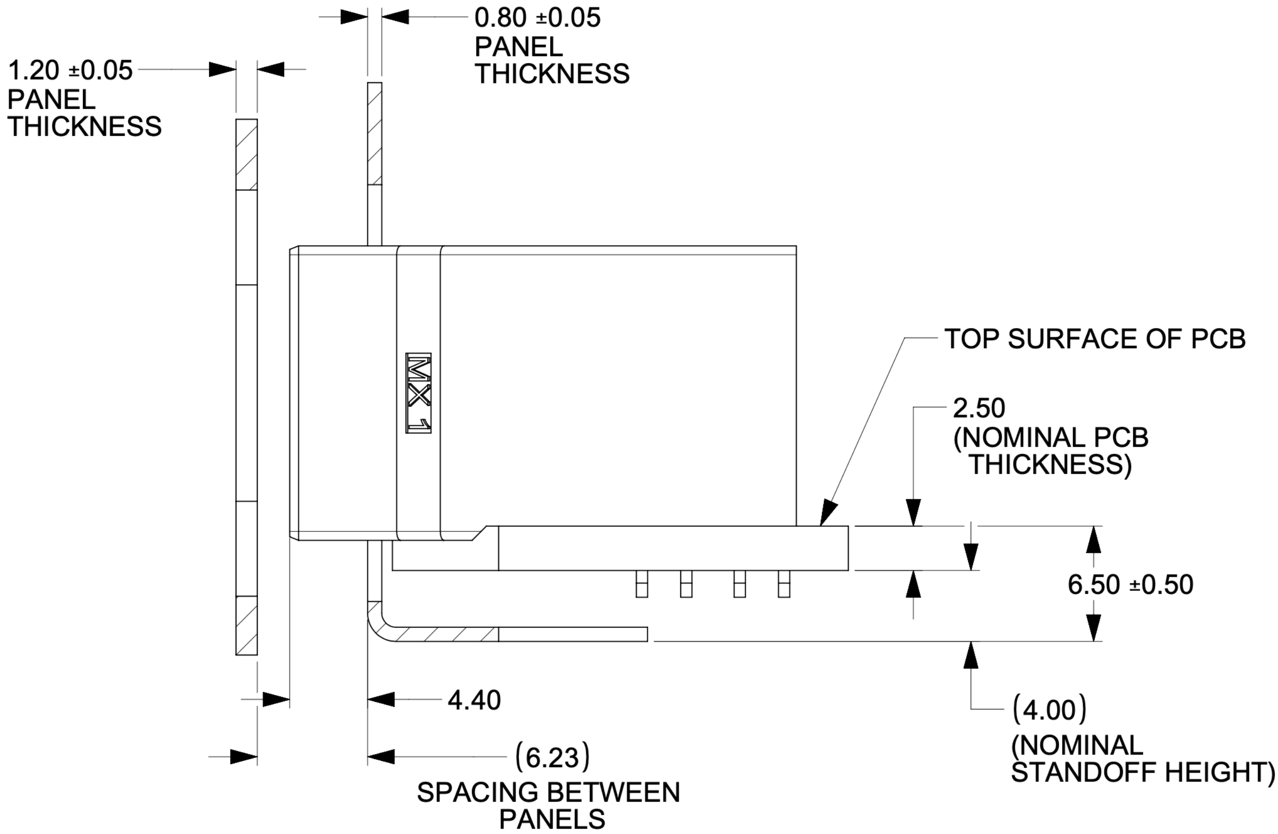


Figure 58. Power connector critical dimensions

6.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

6.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

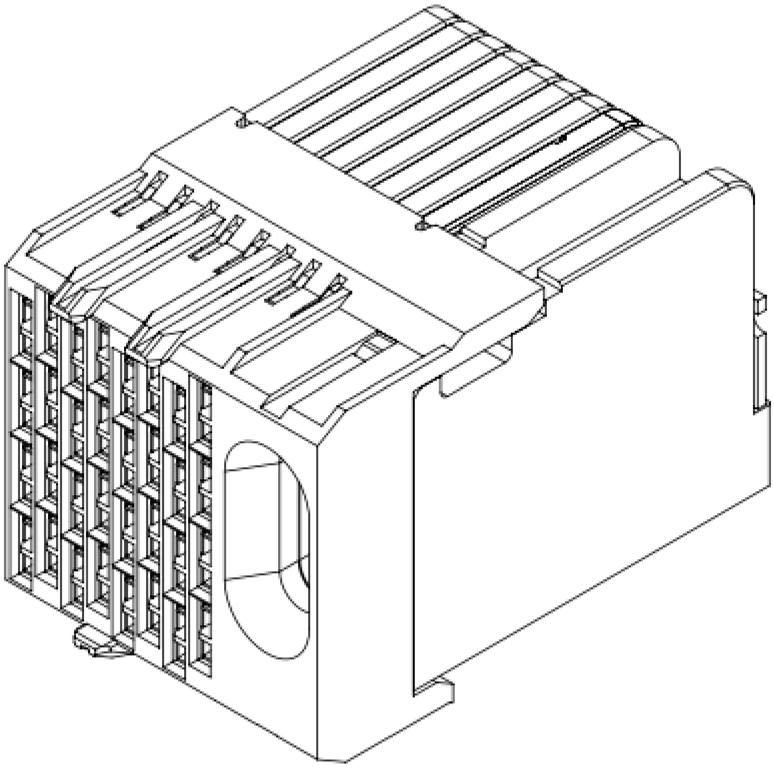


Figure 59. Impel Connector with guide feature

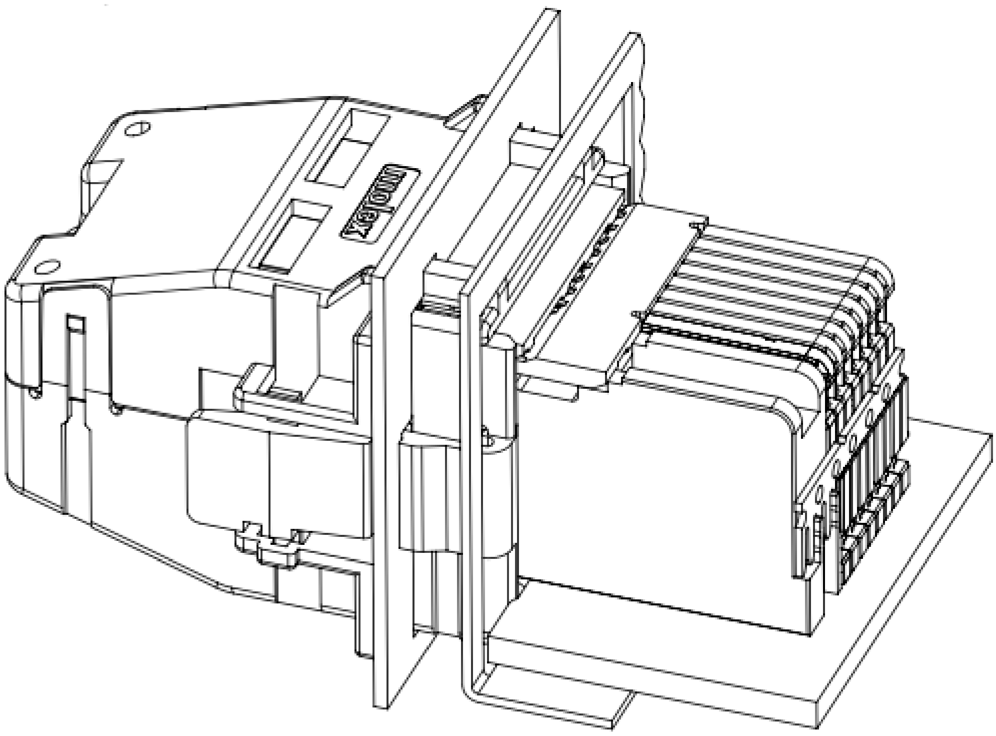


Figure 60. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

6.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

Port Count	Data Rate	Implementation Interface and order
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1
6		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	100G (4x25G) 200G (4x50G PAM4)	SFP1, SPF2, SFP3, SPF4, QSFP1
6		SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	400G (8x50G PAM4)	QSFP-DD (QSFP1+QSFP2) 200G (4x50G PAM4)
QSFP (SFP1+SFP2+SFP3+ SFP4)	2	QSFP, QSFP1

6.5. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

6.5.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 50, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

6.5.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

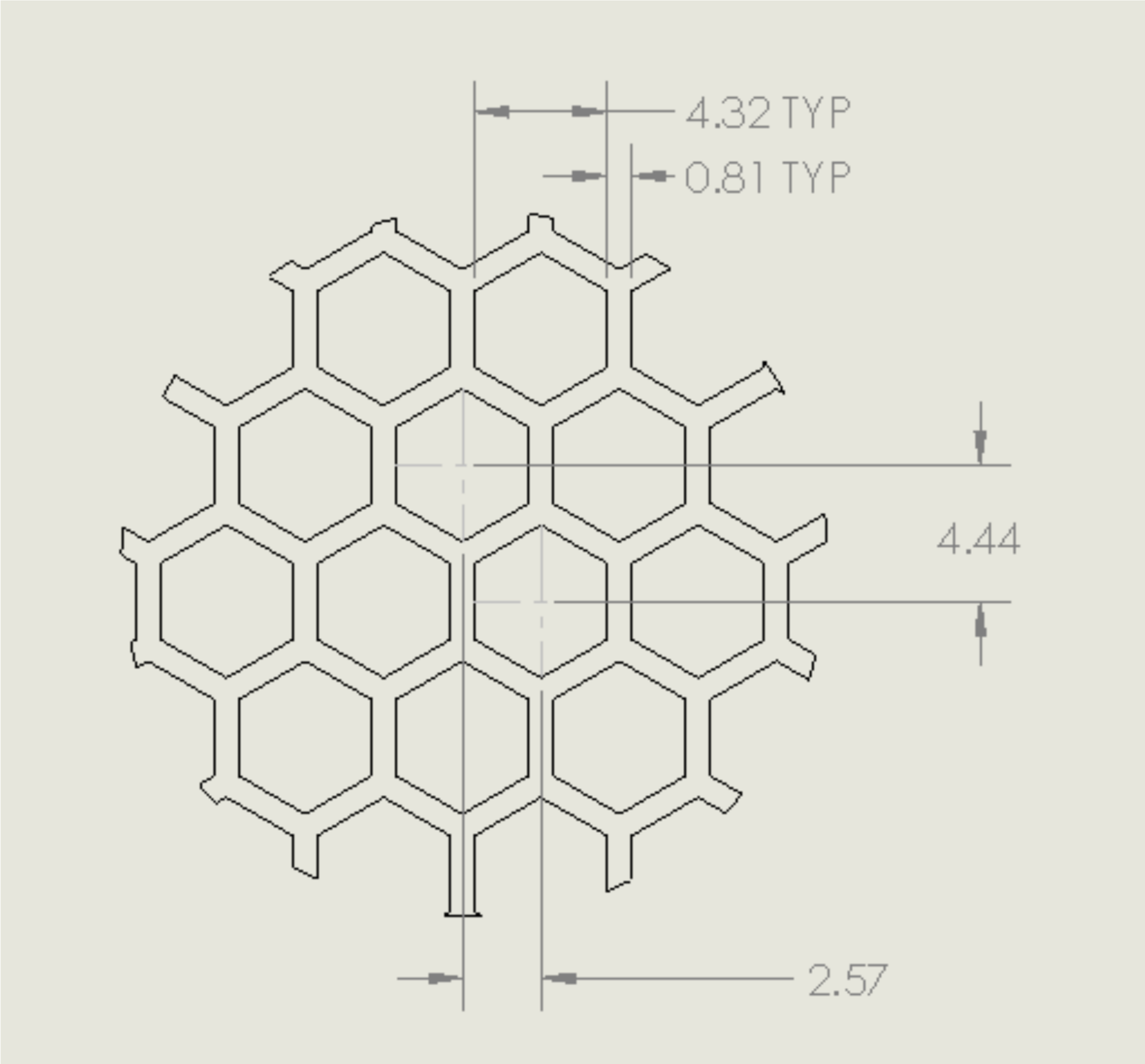


Figure 61. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

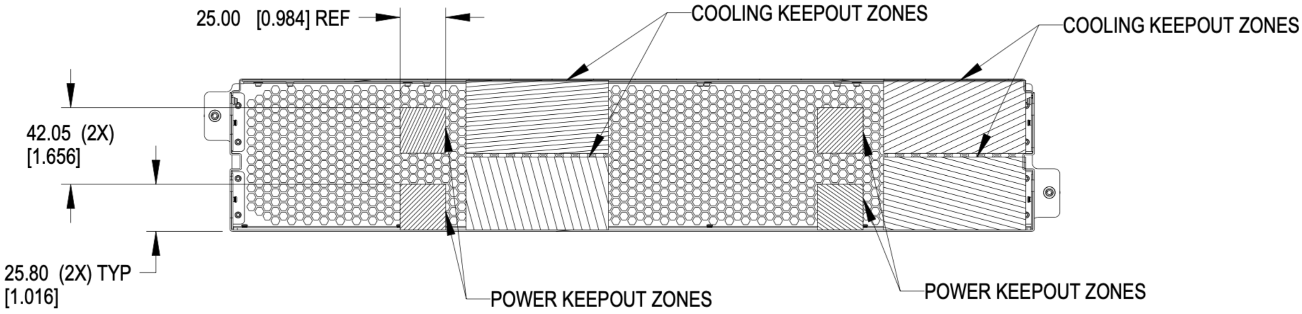


Figure 62. Brick keep-out zone

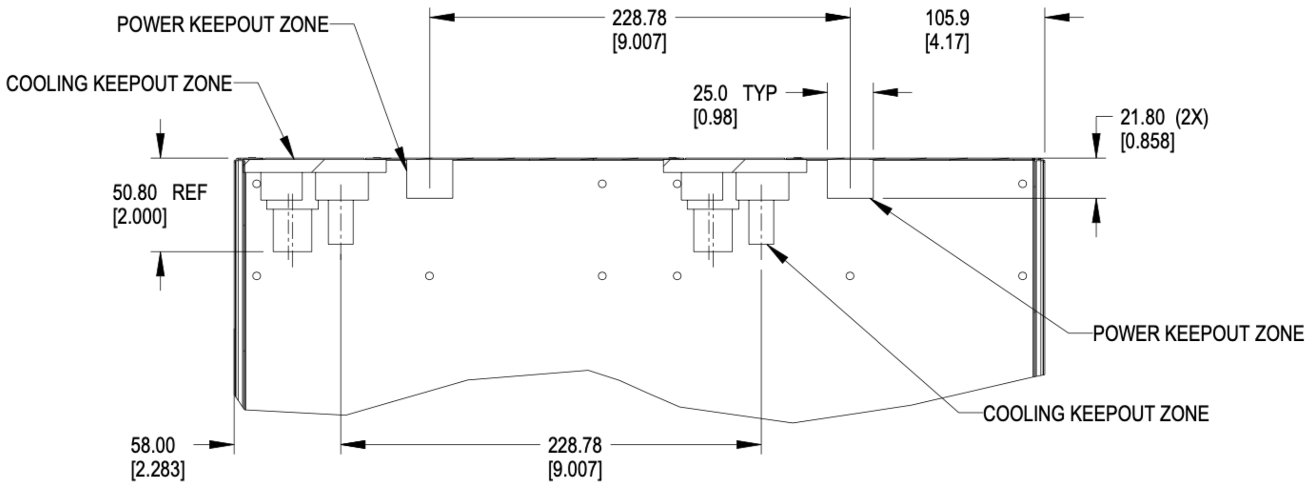


Figure 63. Brick keep-out zone top view

6.6. Brick performance

6.6.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

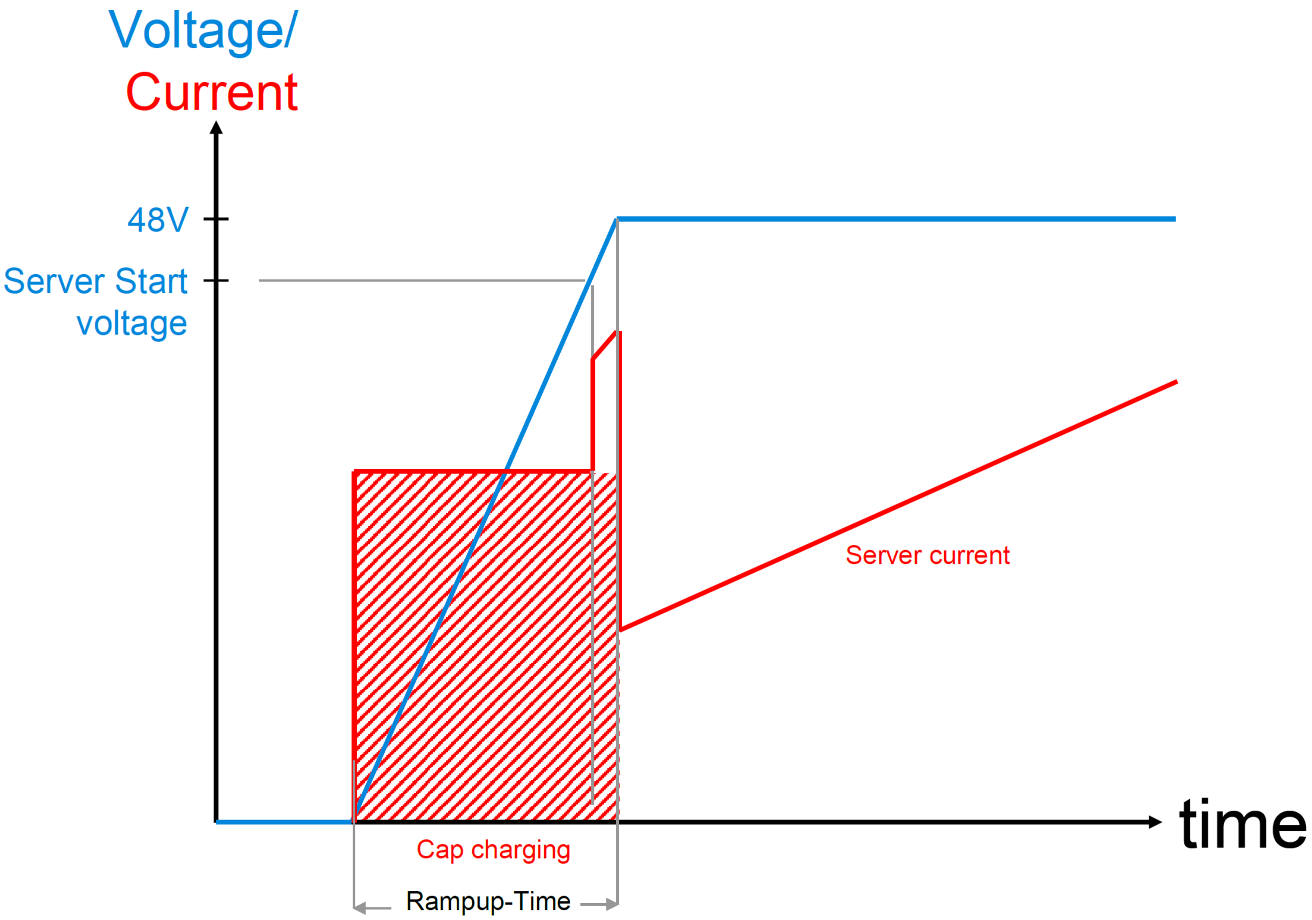


Figure 64. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Four power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Data connector pin assignments

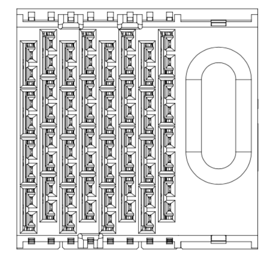
WHEN VIEWED FROM PCB DAUGHTER CARD CONNECTOR REAR BRICK PANEL
DIRECTLY FACING CONNECTOR. PIN AC1 is BOTTOM LEFT CORNER

2022 December 4

	C1	C2	C3	C4	C5	C6	C7	C8	
H	DP4-	DP8-	DP12-	DP16-	DP20-	DP24-	DP28-	DP32-	Key
G	DP4+	DP8+	DP12+	DP16+	DP20+	DP24+	DP28+	DP32+	
F	DP3-	DP7-	DP11-	DP15-	DP19-	DP23-	DP27-	DP31-	
E	DP3+	DP7+	DP11+	DP15+	DP19+	DP23+	DP27+	DP31+	
D	DP2-	DP6-	DP10-	DP14-	DP18-	DP22-	DP26-	DP30-	
C	DP2+	DP6+	DP10+	DP14+	DP18+	DP22+	DP26+	DP30+	
B	DP1-	DP5-	DP9-	DP13-	DP17-	DP21-	DP25-	DP29-	
A	DP1+	DP5+	DP9+	DP13+	DP17+	DP21+	DP25+	DP29+	

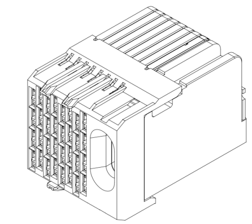
GROUND SHIELD TO GROUND

TOP



KEY

TOP



KEY

	C1	C2	C3	C4	C5	C6	C7	C8	
H, G, F, E	RJ45	RJ45	SFP	SFP	QSFP		QSFP		Key
D, C, B, A			SFP	SFP					

	C1	C2	C3	C4	C5	C6	C7	C8	
H, G	RJ45_1	RJ45_2	SFP2 RX	SFP4 RX	QSFP1		QSFP2		Key
F, E			SFP1 RX	SFP3 RX					
D, C			SFP2 TX	SFP4 TX					
B, A			SFP1 TX	SFP3 TX					

	C1	C2	C3	C4	C5	C6	C7	C8	
H	RJ45 PIN8	RJ45 PIN8	SFP2	SFP4	QSFP1	QSFP1	QSFP2	QSFP2	Key
G	RJ45 PIN7	RJ45 PIN7	LN1 RX	LN1 RX	LN2 RX	LN1 RX	LN2 RX	LN1 RX	
F	RJ45 PIN5	RJ45 PIN5	SFP1	SFP3	QSFP1	QSFP1	QSFP2	QSFP2	
E	RJ45 PIN4	RJ45 PIN4	LN1 RX	LN1 RX	LN4 RX	LN3 RX	LN4 RX	LN3 RX	
D	RJ45 PIN6	RJ45 PIN6	SFP2	SFP4	QSFP1	QSFP1	QSFP2	QSFP2	
C	RJ45 PIN3	RJ45 PIN3	LN1 TX	LN1 TX	LN3 TX	LN4 TX	LN3 TX	LN4 TX	
B	RJ45 PIN2	RJ45 PIN2	SFP1	SFP3	QSFP1	QSFP1	QSFP2	QSFP2	
A	RJ45 PIN1	RJ45 PIN1	LN1 TX	LN1 TX	LN1 TX	LN2 TX	LN1 TX	LN2 TX	

GROUND SHIELD TO GROUND

	C1	C2	C3	C4	C5	C6	C7	C8	
H	RJ45_1_BI_DD-	RJ45_2_BI_DD-	SFP2_LN1_RX_N	SFP4_LN1_RX_N	QSFP1_LN2_RX_N	QSFP1_LN1_RX_N	QSFP2_LN2_RX_N	QSFP2_LN1_RX_N	Key
G	RJ45_1_BI_DD+	RJ45_2_BI_DD+	SFP2_LN1_RX_P	SFP4_LN1_RX_P	QSFP1_LN2_RX_P	QSFP1_LN1_RX_P	QSFP2_LN2_RX_P	QSFP2_LN1_RX_P	
F	RJ45_1_BI_DC-	RJ45_2_BI_DC-	SFP1_LN1_RX_N	SFP3_LN1_RX_N	QSFP1_LN4_RX_N	QSFP1_LN3_RX_N	QSFP2_LN4_RX_N	QSFP2_LN3_RX_N	
E	RJ45_1_BI_DC+	RJ45_2_BI_DC+	SFP1_LN1_RX_P	SFP3_LN1_RX_P	QSFP1_LN4_RX_P	QSFP1_LN3_RX_P	QSFP2_LN4_RX_P	QSFP2_LN3_RX_P	
D	RJ45_1_BI_DB-	RJ45_2_BI_DB-	SFP2_LN1_TX_N	SFP4_LN1_TX_N	QSFP1_LN3_TX_N	QSFP1_LN4_TX_N	QSFP2_LN3_TX_N	QSFP2_LN4_TX_N	
C	RJ45_1_BI_DB+	RJ45_2_BI_DB+	SFP2_LN1_TX_P	SFP4_LN1_TX_P	QSFP1_LN3_TX_P	QSFP1_LN4_TX_P	QSFP2_LN3_TX_P	QSFP2_LN4_TX_P	
B	RJ45_1_BI_DA-	RJ45_2_BI_DA-	SFP1_LN1_TX_N	SFP3_LN1_TX_N	QSFP1_LN1_TX_N	QSFP1_LN2_TX_N	QSFP2_LN1_TX_N	QSFP2_LN2_TX_N	
A	RJ45_1_BI_DA+	RJ45_2_BI_DA+	SFP1_LN1_TX_P	SFP3_LN1_TX_P	QSFP1_LN1_TX_P	QSFP1_LN2_TX_P	QSFP2_LN1_TX_P	QSFP2_LN2_TX_P	

GROUND SHIELD TO GROUND

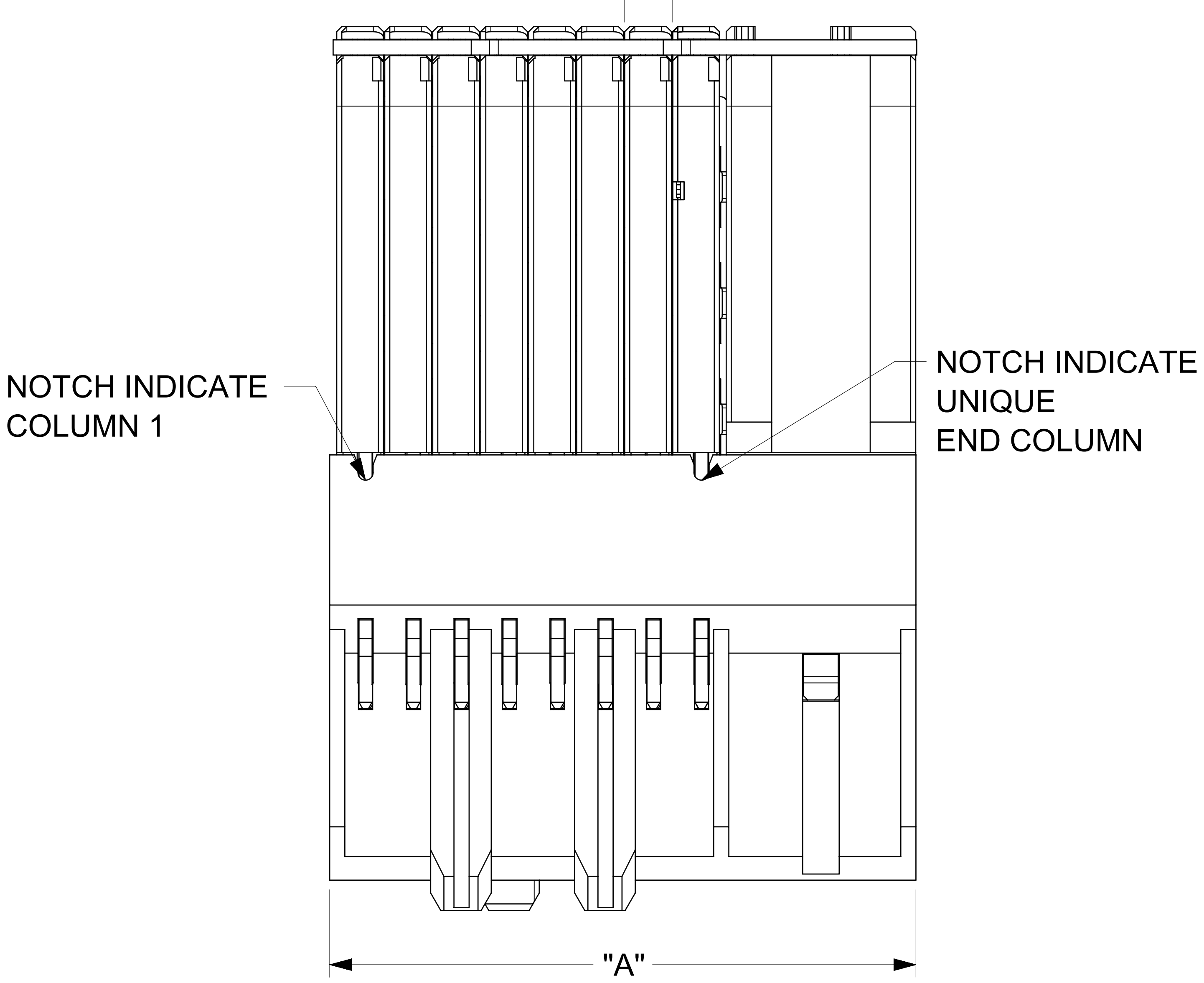
AB = CD = Tx FOR SFP & QSFP
EF = GH = Rx FOR SFP & QSFP

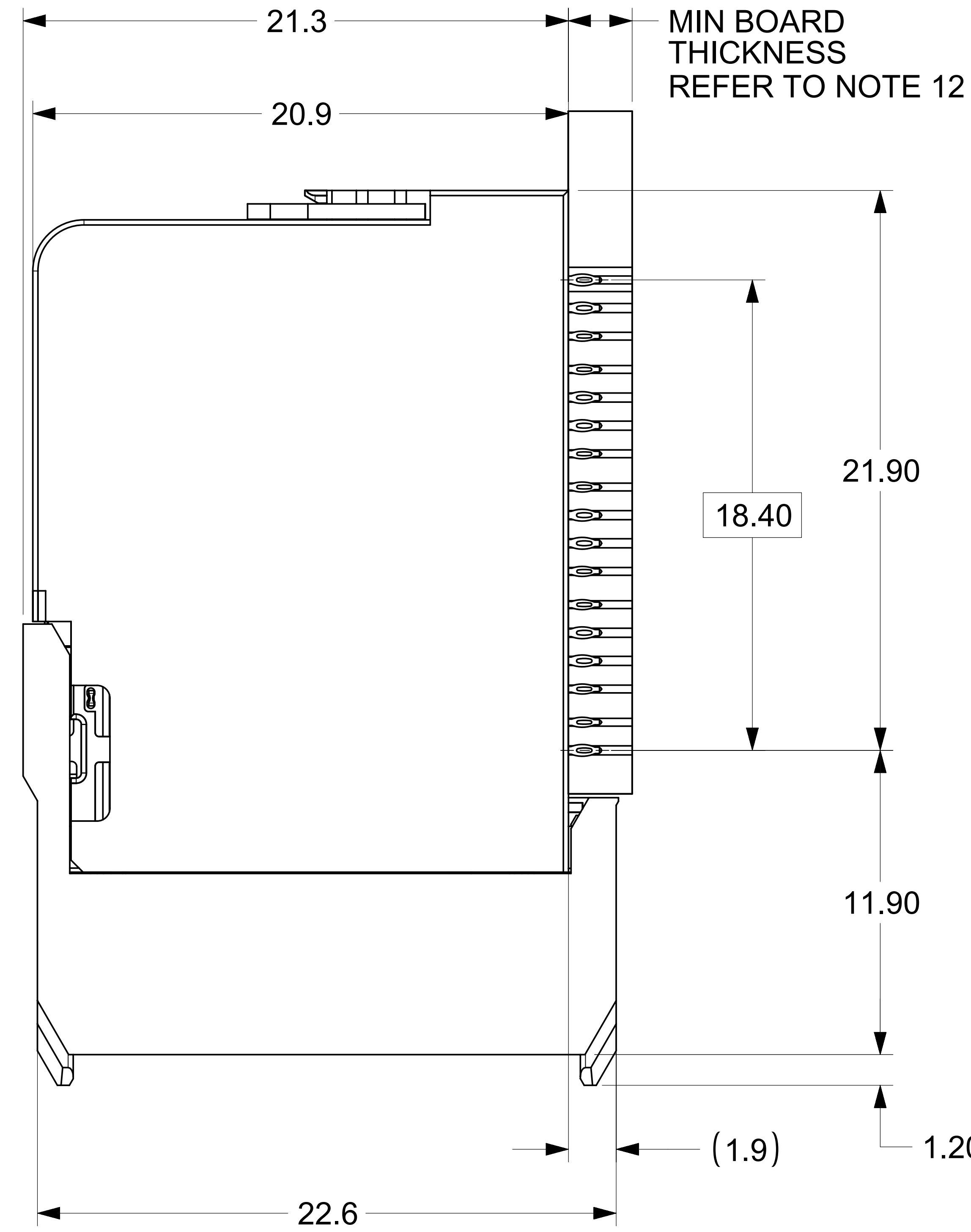
AB = CD = Tx FOR SFP & QSFP
EF = GH = Rx FOR SFP & QSFP

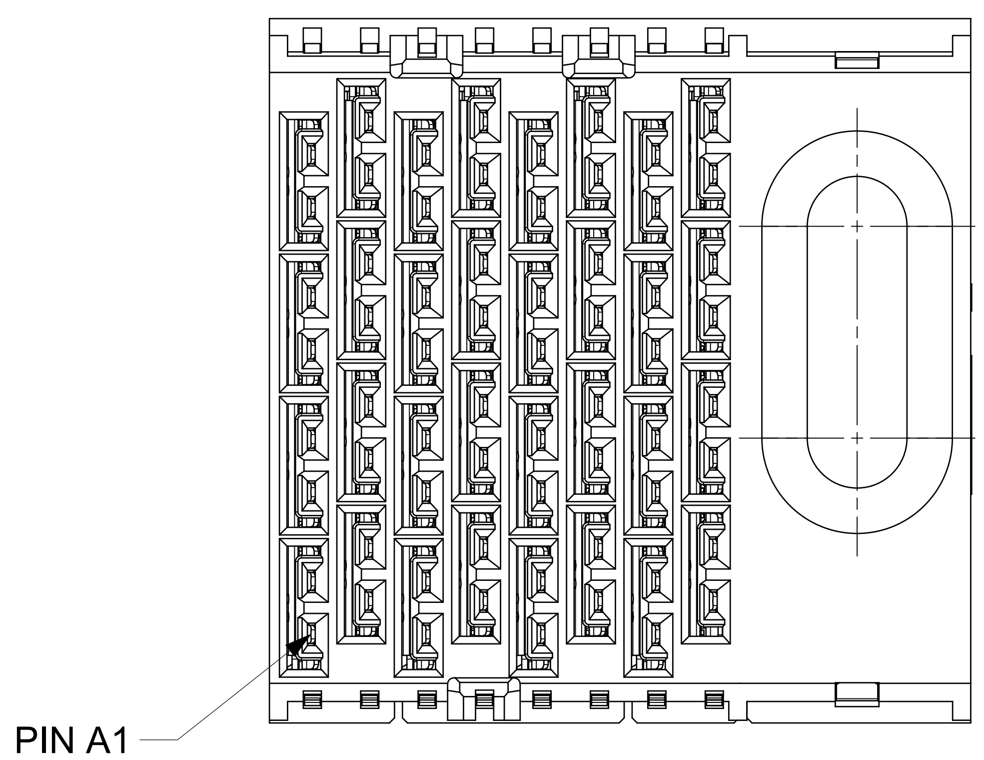
Chapter 7. Private Appendix

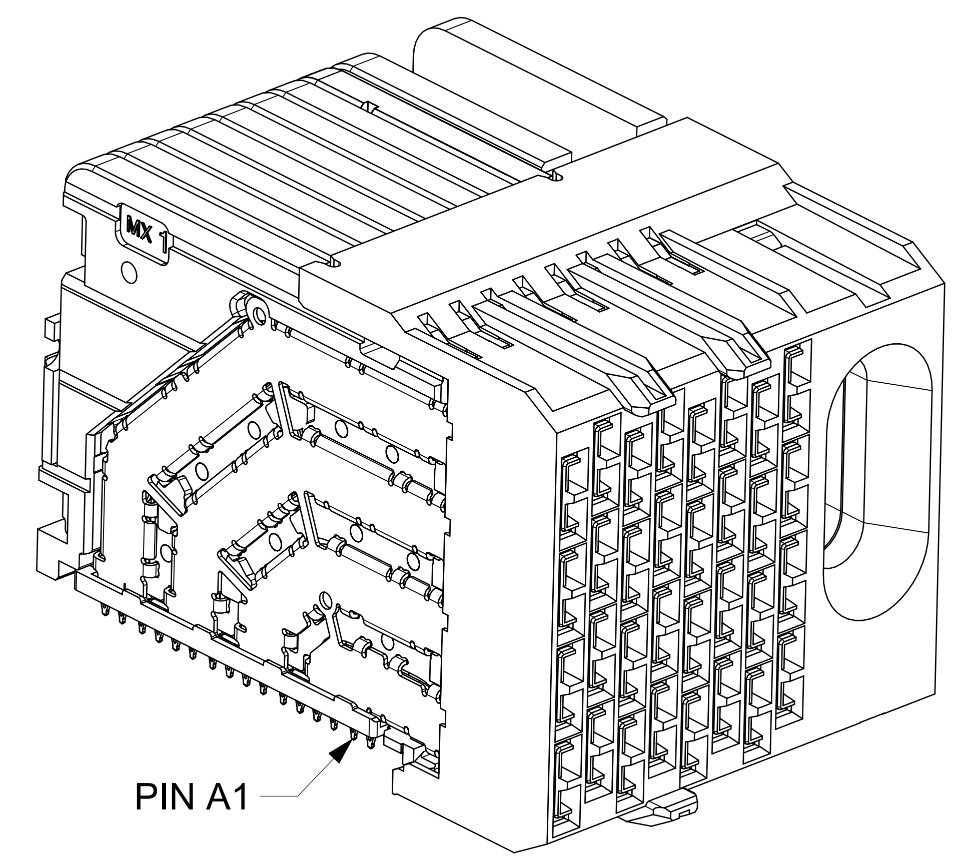
The following material is included for the convenience of Open19 members only. No redistribution to the public is permitted.

7.1. Data connector sales drawing









NOTES:

1. MATERIALS: HOUSING - LIQUID CRYSTAL POLYMER (LCP)
   GLASS-FILLED, UL94V-0
   TERMINALS - HIGH PERFORMANCE COPPER ALLOY
2. FINISH: 30uIN GOLD IN CONTACT AREA.
   SELECTIVE TIN ON PCB TAILS.
   NICKEL OVERALL.
3. REFER TO MOLEX PRODUCT SPECIFICATION 172740-9990 FOR PERFORMANCE SPECIFICATIONS.
4. THIS PART CONFORMS TO CLASS B REQUIREMENTS OF MOLEX COSMETIC SPEC PS-45499-002.
5. PRODUCT WILL BE PACKAGED PER PK-78888-085.
6. SEE SHEET 2 FOR HOLE PATTERN DIMENSIONS.
7. MATES WITH IMPEL CABLE SERIES 730650.
8. REFER TO IMPEL PLUS PCB ROUTING GUIDE AS-172730-990 FOR ANTIPAD, ROUTING RECOMMENDATIONS, AND ADDITIONAL PCB INFORMATION.
9. WHEN STACKING MODULES END TO END, THE MINIMUM COLUMN TO COLUMN HOLE SPACING IS 10.52 MM.
10. COMPONENT PLACEMENT AND TRACE KEEP OUT ZONE; DO NOT PLACE ANY OTHER COMPONENT OR ROUTE ANY TOP LAYER TRACES IN THIS AREA.
11. CONNECTOR IS SUPPLIED WITH TWO 2-32 THREAD FORMING SCREWS. REFER TO TABLE ON SHEET 2 FOR MOLEX PART NUMBER.
12. WHEN USING THE SUPPLIED 2-32 THREAD FORMING SCREWS, REFER TO TABLE ON SHEET 2 FOR THE APPLICABLE BOARD THICKNESS RANGE.

THIS DESIGN IS BASED ON DESIGN OBJECTIVES AND IS STRICTLY TENTATIVE. IT MAY CHANGE BASED ON RESULTS OF ADDITIONAL DESIGN REVIEWS & VERIFICATIONS.

DIMENSION UNITS	SCALE
mm	NTS

GENERAL TOLERANCES (UNLESS SPECIFIED)

ANGULAR TOL	±	°
4 PLACES	±	
3 PLACES	±	
2 PLACES	±	0.13
1 PLACE	±	0.25
0 PLACES	±	

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6

EC NO: 714548
DRWN: JLTAN06 2022/07/06
CHK'D: YLLIM 2022/07/18
APPR: DWLEE01 2022/07/18

INITIAL REVISION:
DRWN: JLTAN06 2022/04/12
APPR: DWLEE01 2022/06/27

THIRD ANGLE PROJECTION	DRAWING	SERIES
	C-SIZE	204066

molex

IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)

PRODUCT CUSTOMER DRAWING

DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2040669001-SD	PSD	000	A1

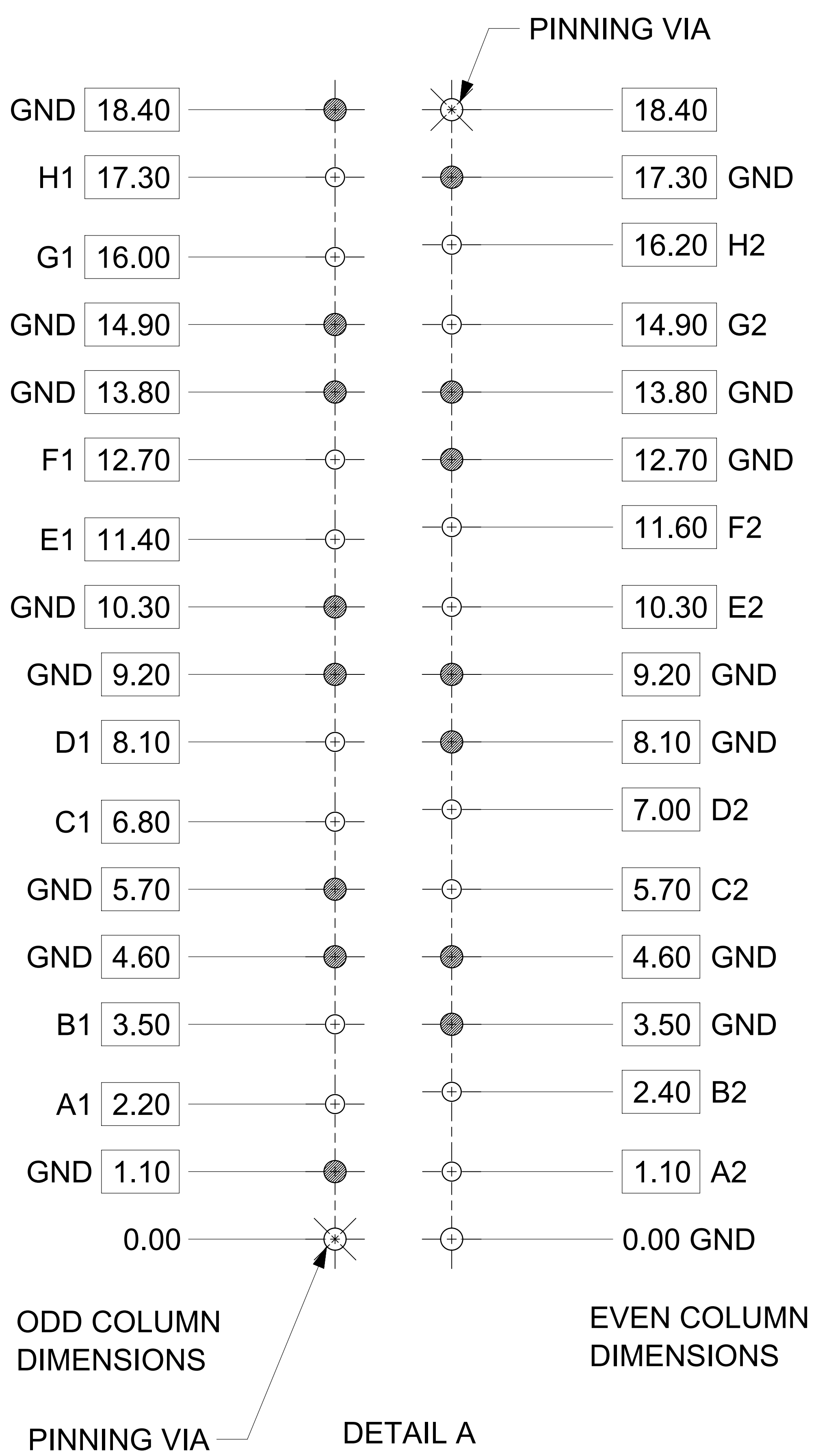
MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
SEE TABLE	GENERAL MARKET	1 OF 3

DOCUMENT STATUS	P1	RELEASE DATE	2022/07/18	10:32:07

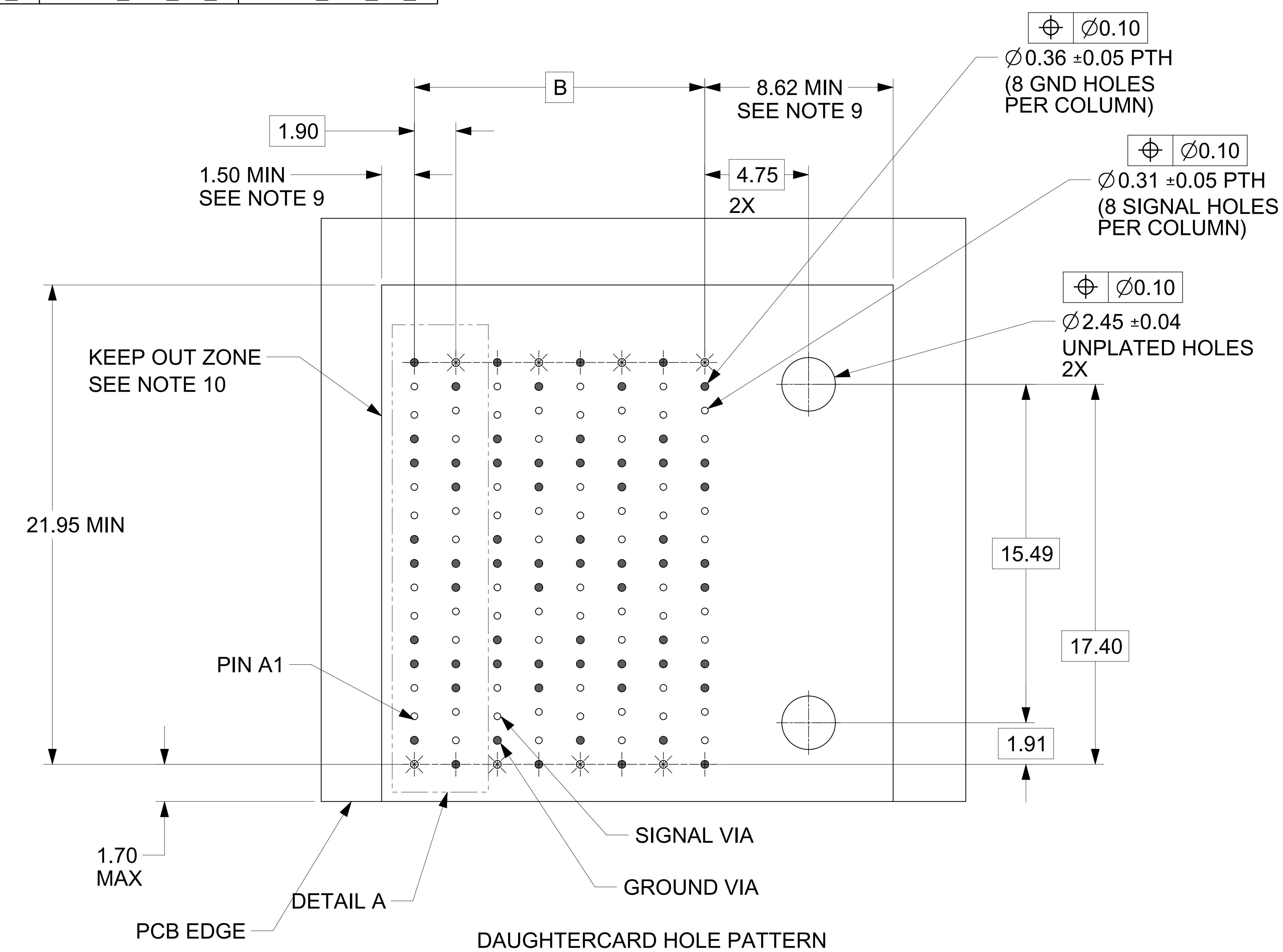
MATERIAL NUMBER	NUMBER OF COLUMNS	NUMBER OF DIFF PAIR	DIM "A"	DIM "B"	SUPPLIED SCREW PN	APPLICABLE BOARD THICKNESS
2040669038	8	32	23.40 MAX	13.30	737263100	2.20 TO 4.10 MM
2040669138	8	32	23.40 MAX	13.30	737260015	1.00 TO 1.90 MM

BRICK (4X8) PIN OUT MAPPING

PIN	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5	COLUMN 6	COLUMN 7	COLUMN 8
H	BI_DD-	NC	SFP2_LN1_RX_N	SFP4_LN1_RX_N	QSFP1_LN2_RX_N	QSFP1_LN1_RX_N	QSFP2_LN2_RX_N	QSFP2_LN1_RX_N
G	BI_DD+		SFP2_LN1_RX_P	SFP4_LN1_RX_P	QSFP1_LN2_RX_P	QSFP1_LN1_RX_P	QSFP2_LN2_RX_P	QSFP2_LN1_RX_P
F	BI_DC-	NC	SFP1_LN1_RX_N	SFP3_LN1_RX_N	QSFP1_LN4_RX_N	QSFP1_LN3_RX_N	QSFP2_LN4_RX_N	QSFP2_LN3_RX_N
E	BI_DC+		SFP1_LN1_RX_P	SFP3_LN1_RX_P	QSFP1_LN4_RX_P	QSFP1_LN3_RX_P	QSFP2_LN4_RX_P	QSFP2_LN3_RX_P
D	BI_DB-	NC	SFP2_LN1_TX_N	SFP4_LN1_TX_N	QSFP1_LN3_TX_N	QSFP1_LN4_TX_N	QSFP2_LN3_TX_N	QSFP2_LN4_TX_N
C	BI_DB+		SFP2_LN1_TX_P	SFP4_LN1_TX_P	QSFP1_LN3_TX_P	QSFP1_LN4_TX_P	QSFP2_LN3_TX_P	QSFP2_LN4_TX_P
B	BI_DA-	NC	SFP1_LN1_TX_N	SFP3_LN1_TX_N	QSFP1_LN1_TX_N	QSFP1_LN2_TX_N	QSFP2_LN1_TX_N	QSFP2_LN2_TX_N
A	BI_DA+		SFP1_LN1_TX_P	SFP3_LN1_TX_P	QSFP1_LN1_TX_P	QSFP1_LN2_TX_P	QSFP2_LN1_TX_P	QSFP2_LN2_TX_P



DETAIL A
SCALE 10:1



DAUGHTERCARD HOLE PATTERN
(CONNECTOR SIDE)

DIMENSION UNITS: mm
SCALE: NTS

CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6

GENERAL TOLERANCES (UNLESS SPECIFIED)

ANGULAR TOL	± °
4 PLACES	±
3 PLACES	±
2 PLACES	± 0.13
1 PLACE	± 0.25
0 PLACES	±

EC NO: 714548
DRWN: JLTAN06 2022/07/06
CHK'D: YLLIM 2022/07/18
APPR: DWLEE01 2022/07/18
INITIAL REVISION:
DRWN: JLTAN06 2022/04/12
APPR: DWLEE01 2022/06/27

molex

IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)

PRODUCT CUSTOMER DRAWING

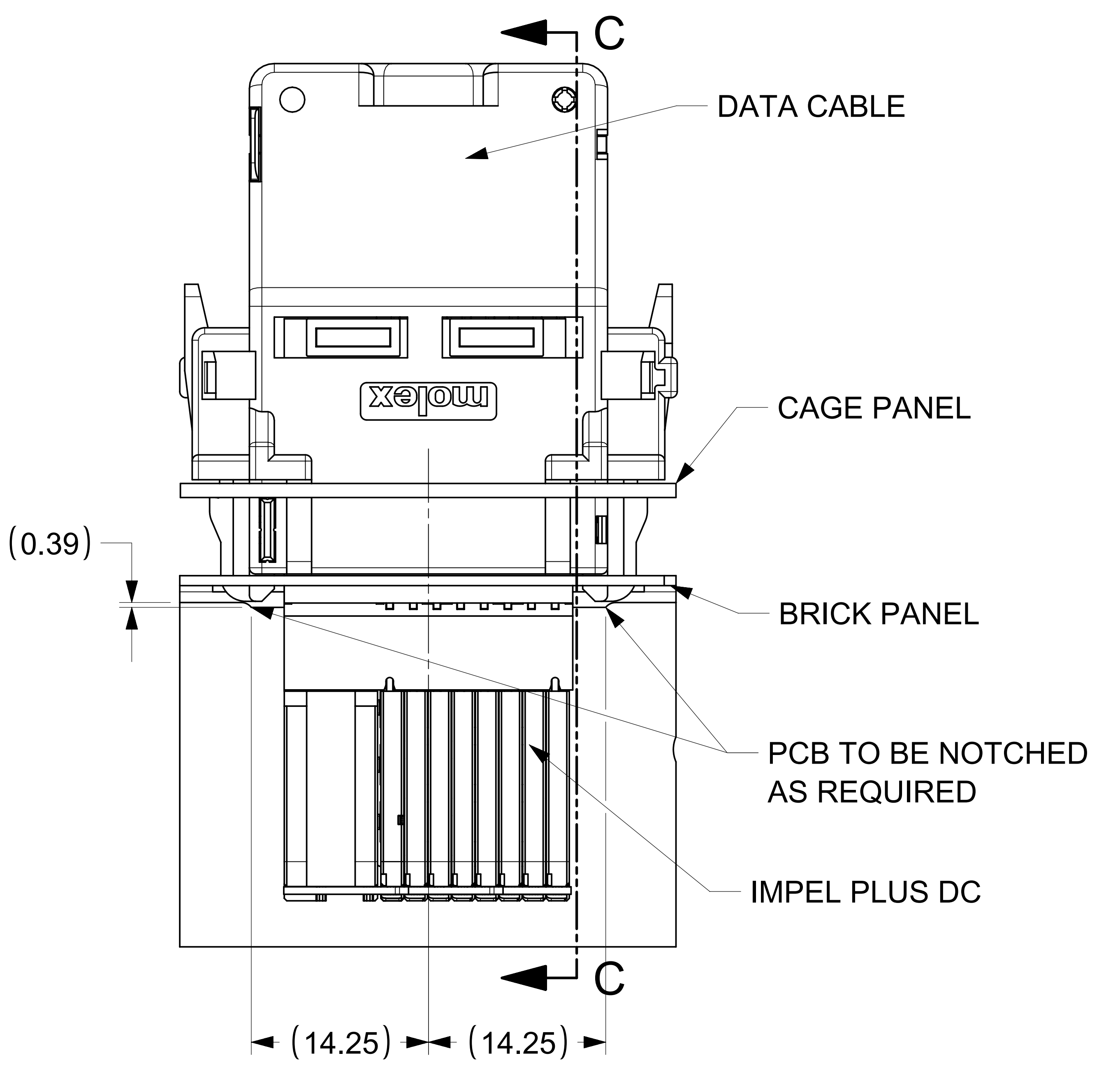
DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2040669001-SD	PSD	000	A1

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

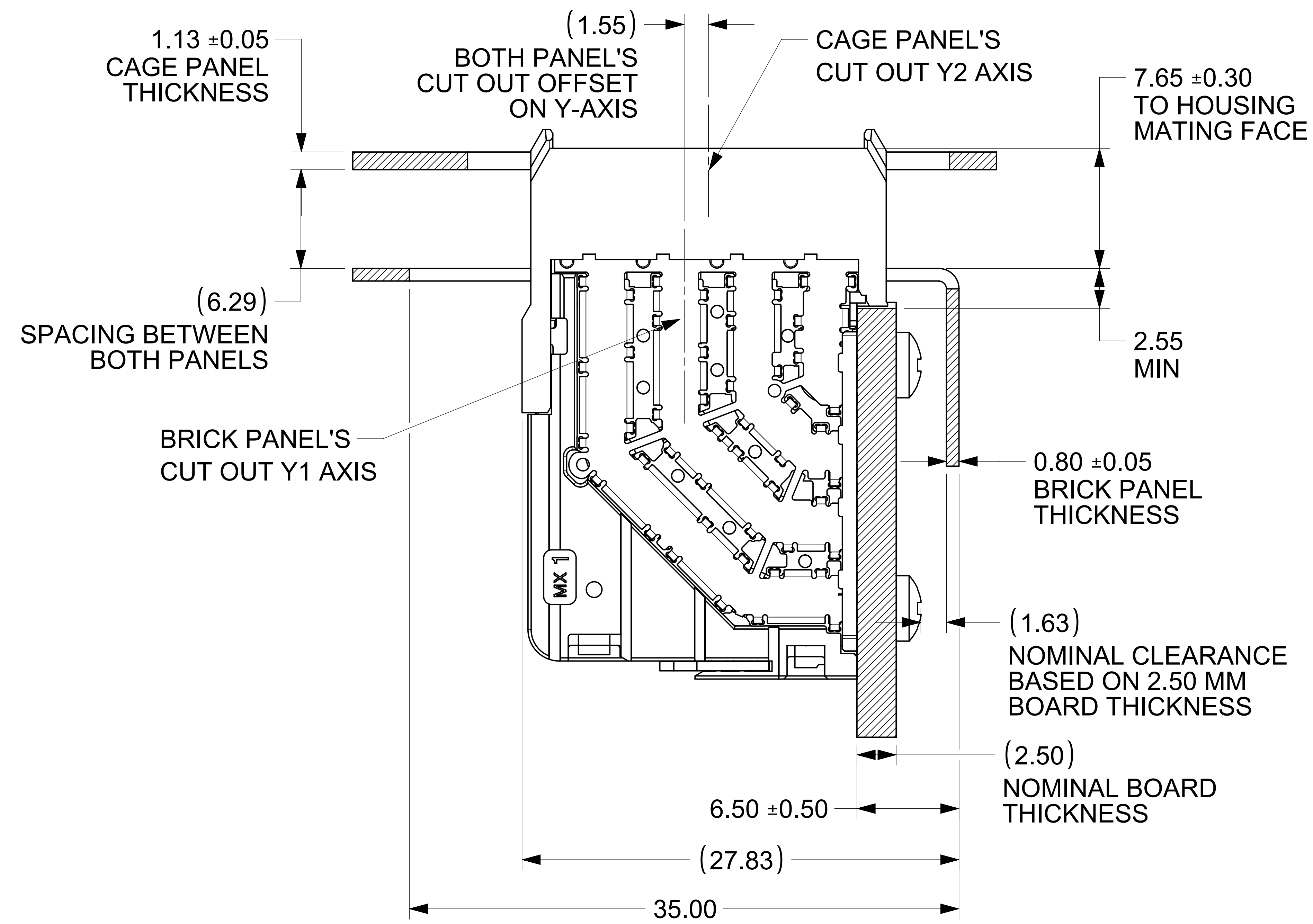
THIRD ANGLE PROJECTION	DRAWING	SERIES	MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
	C-SIZE	204066	SEE TABLE	GENERAL MARKET	2 OF 3

DOCUMENT STATUS	P1	RELEASE DATE	2022/07/18 10:32:07

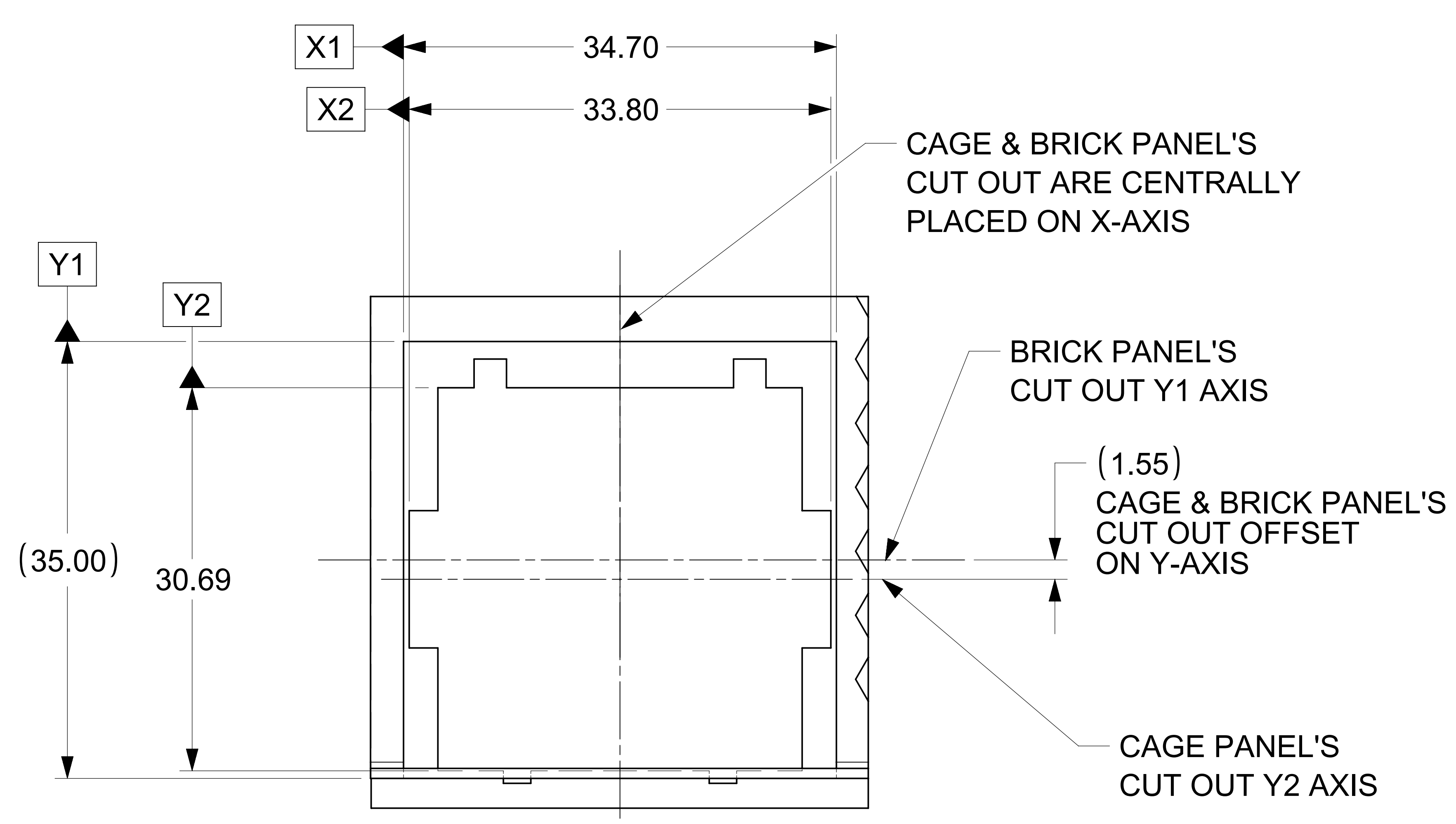
IMPEL PLUS DC MATED TO DATA CABLE



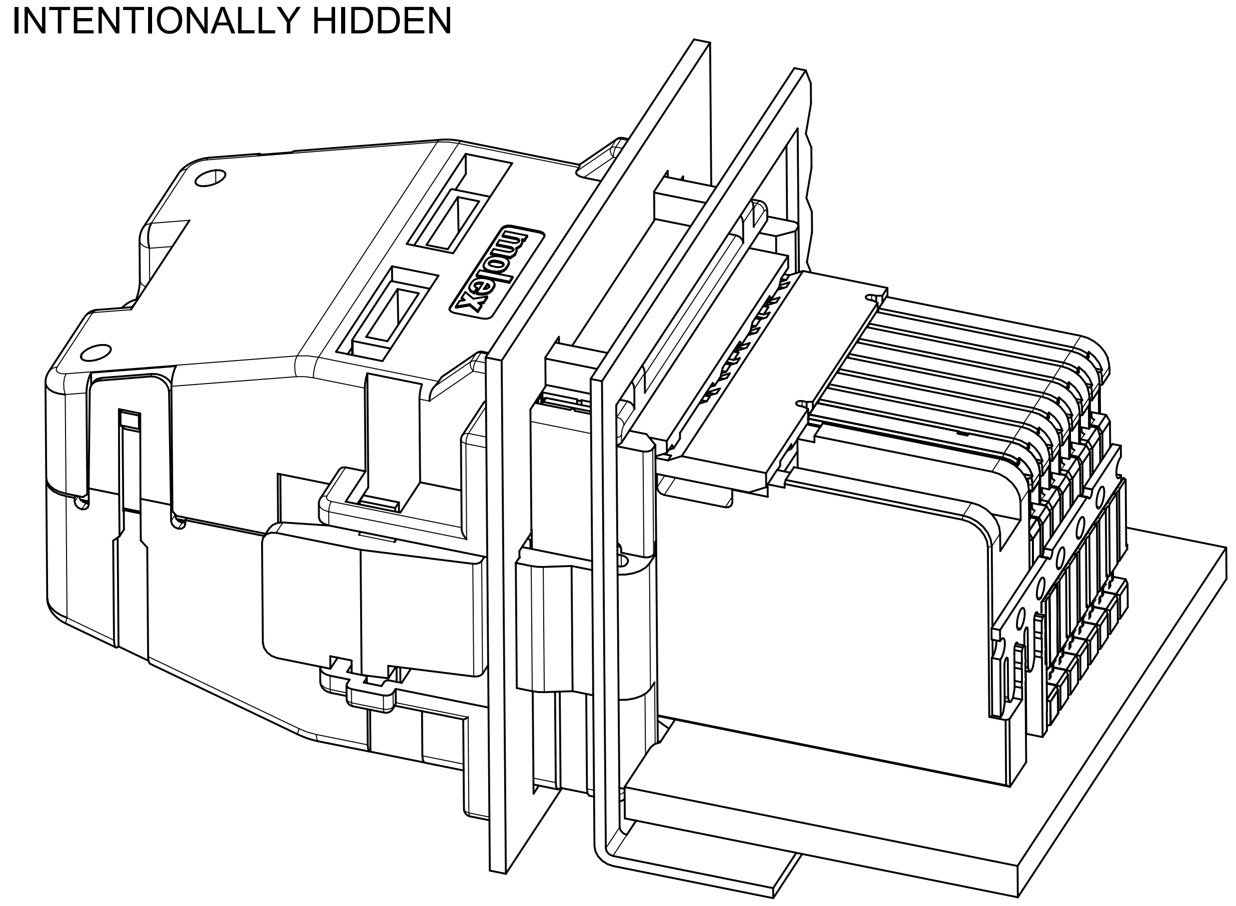
IMPEL PLUS DC MATING PROFILE WITH RESPECT TO PANEL



SECTION C-C
DATA CABLE IS INTENTIONALLY HIDDEN



VIEW D
RECOMMENDED BRICK PANEL
CUT OUT PROFILE

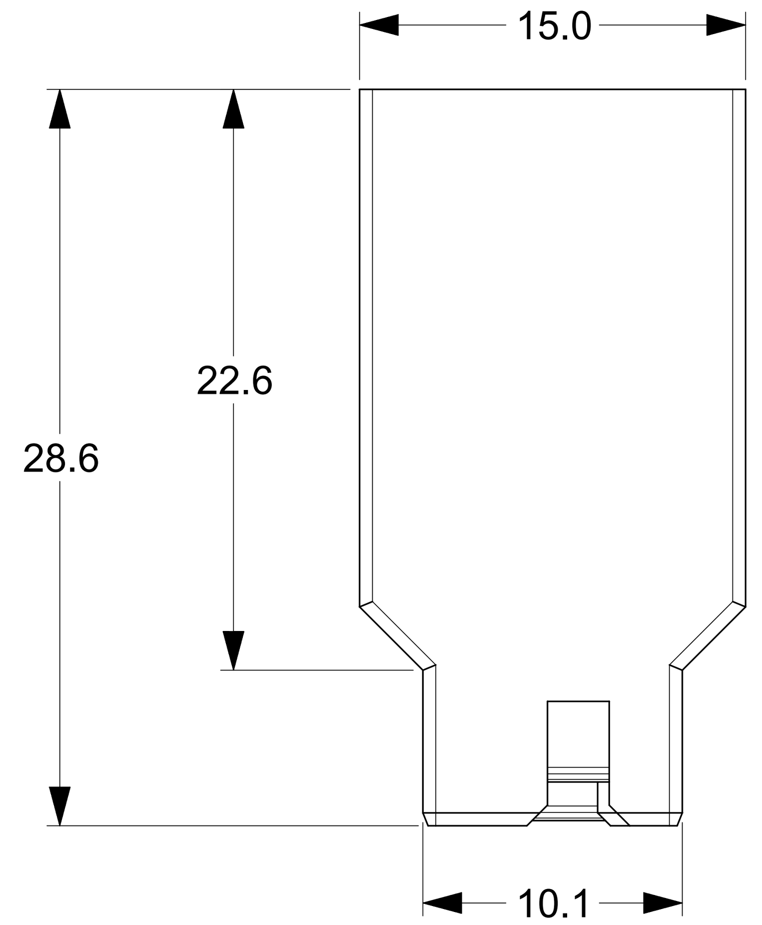


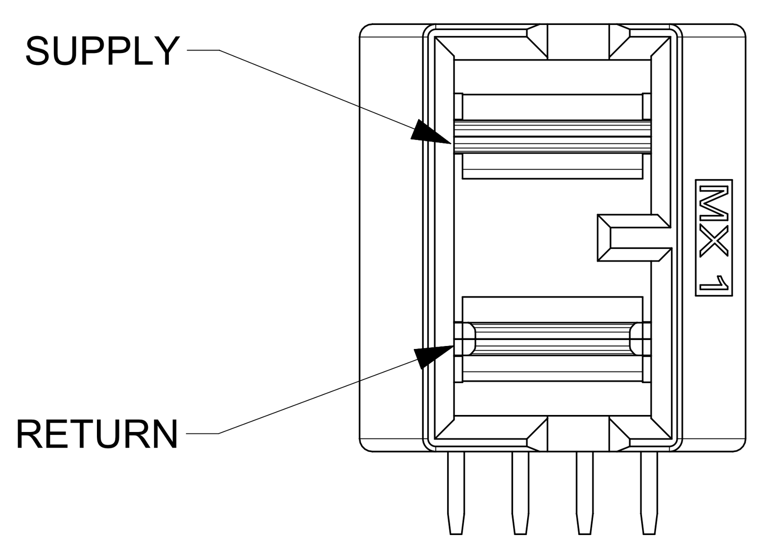
ISOMETRIC VIEW B

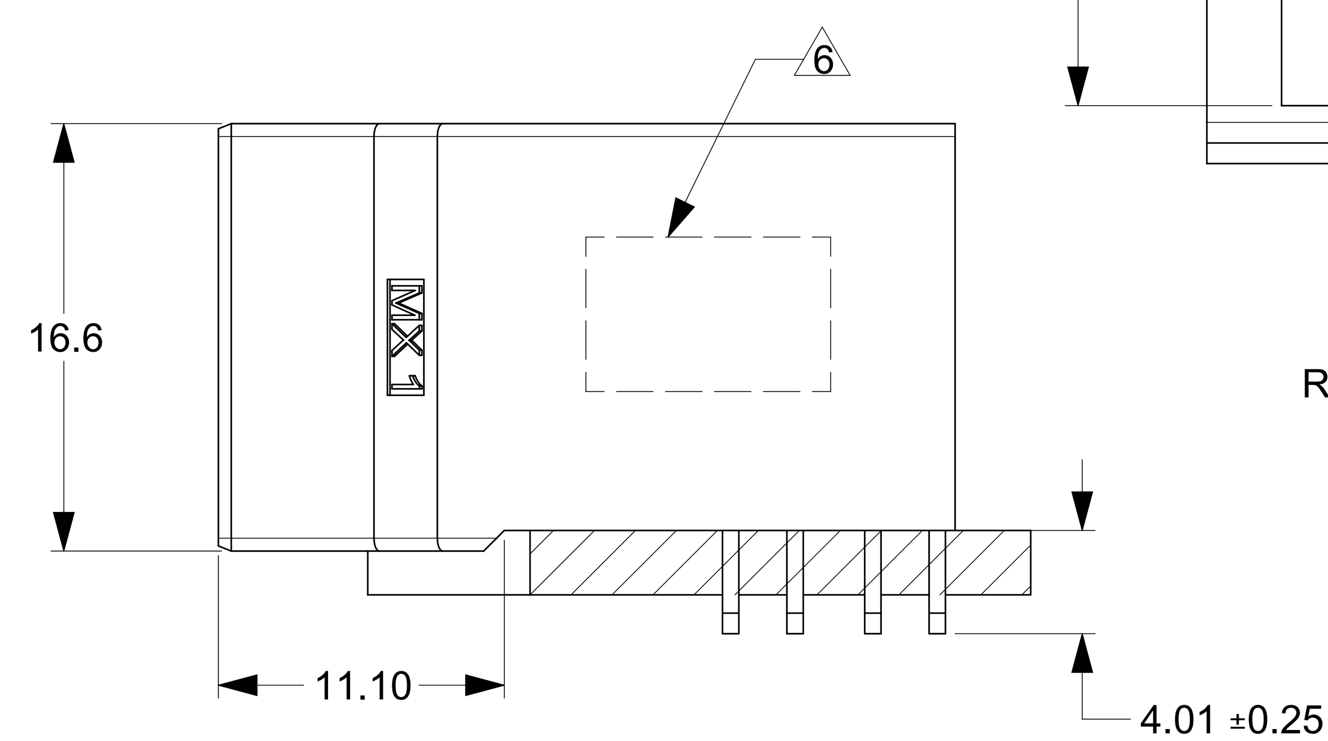
DIMENSION UNITS: mm	SCALE: NTS	CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6	molex
GENERAL TOLERANCES (UNLESS SPECIFIED)		EC NO: 714548	IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)
ANGULAR TOL	± °	DRWN: JLTAN06 2022/07/06	
4 PLACES	±	CHK'D: YLLIM 2022/07/18	
3 PLACES	±	APPR: DWLEE01 2022/07/18	PRODUCT CUSTOMER DRAWING
2 PLACES	± 0.13	INITIAL REVISION:	DOCUMENT NUMBER: 2040669001-SD; DOC TYPE: PSD; DOC PART: 000; REVISION: A1
1 PLACE	± 0.25	DRWN: JLTAN06 2022/04/12	
0 PLACES	±	APPR: DWLEE01 2022/06/27	
DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS	THIRD ANGLE PROJECTION	DRAWING: C-SIZE; SERIES: 204066	MATERIAL NUMBER: SEE TABLE; CUSTOMER: GENERAL MARKET; SHEET NUMBER: 3 OF 3

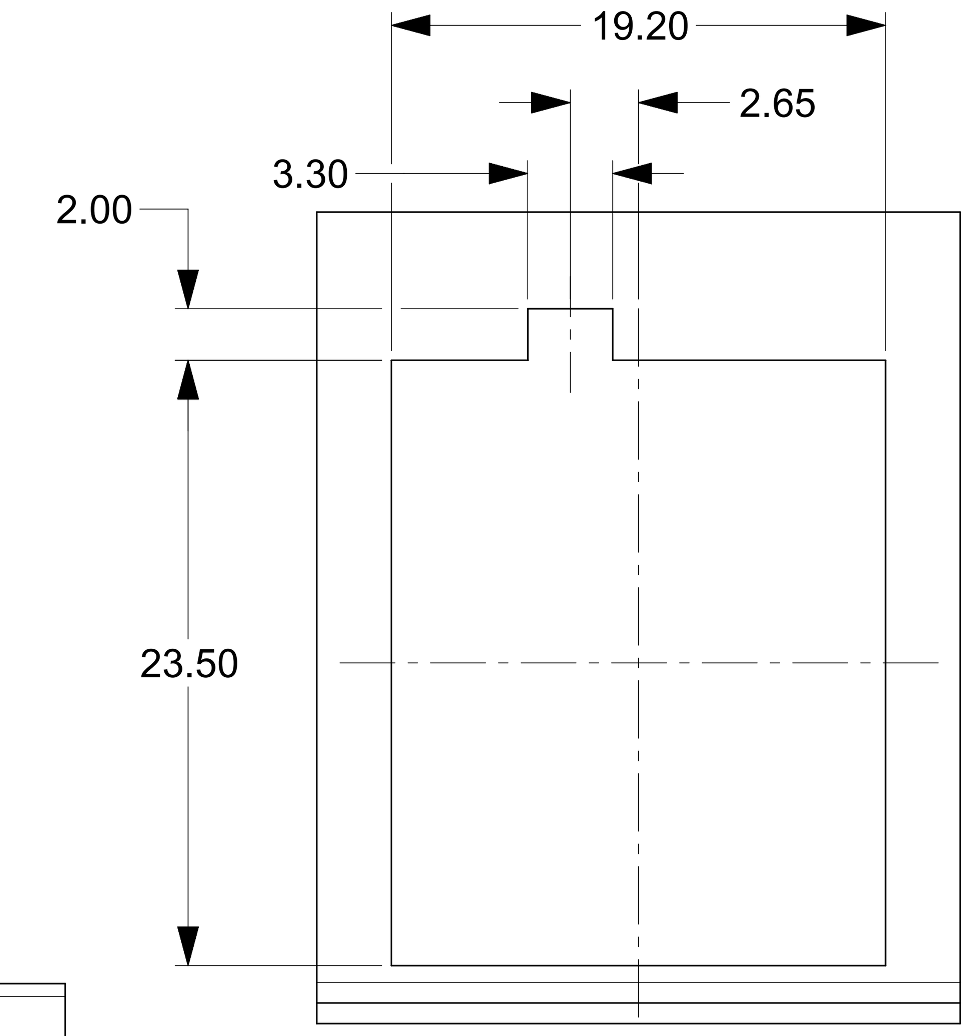
DOCUMENT STATUS	P1	RELEASE DATE	2022/07/18	10:32:07

7.2. Power connector sales drawing

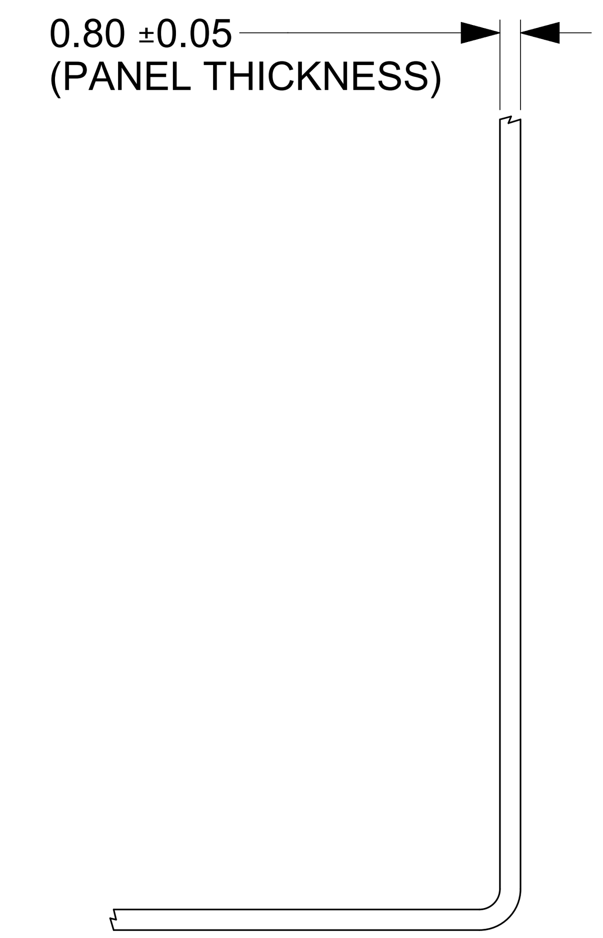


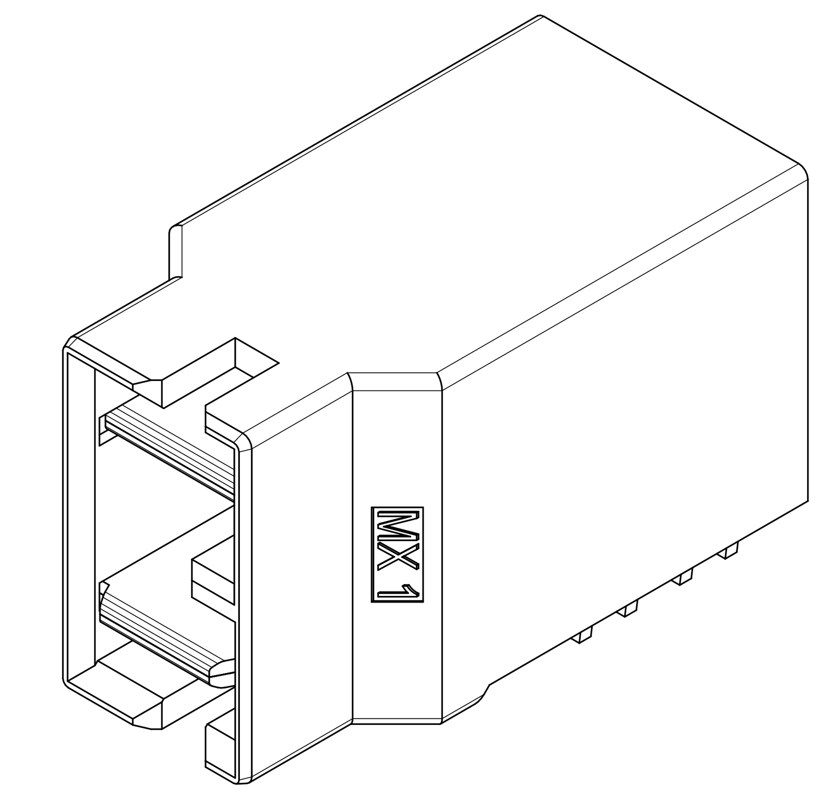


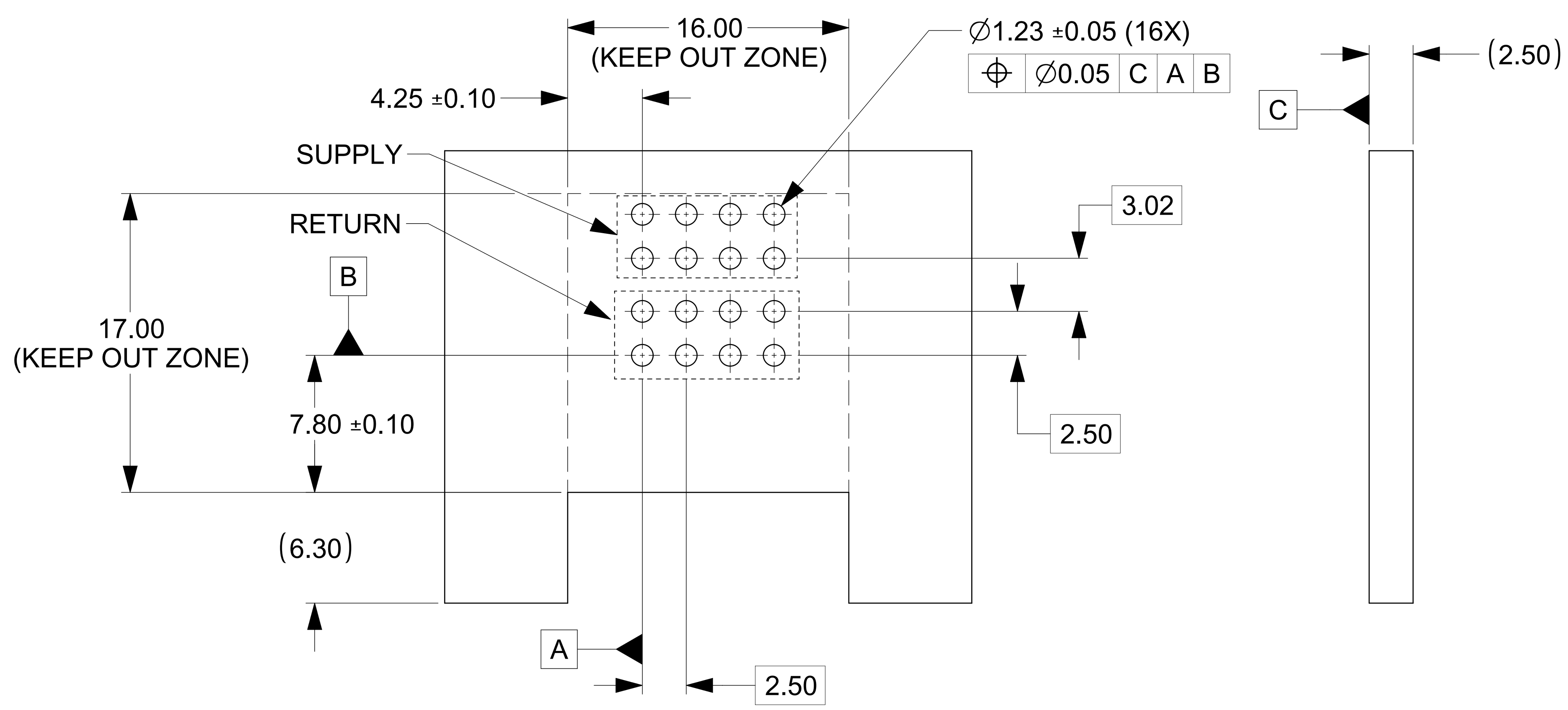




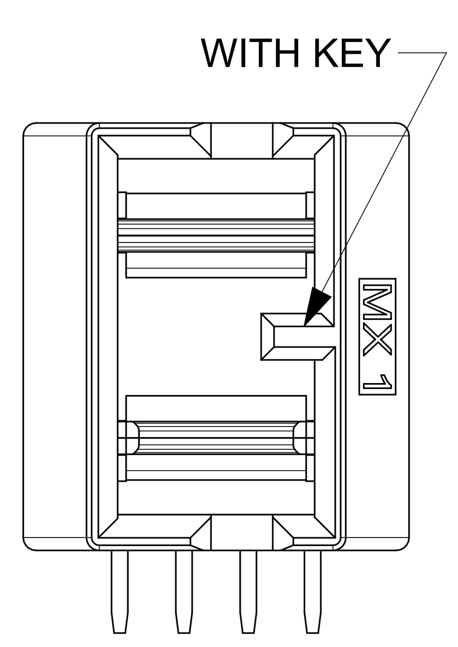
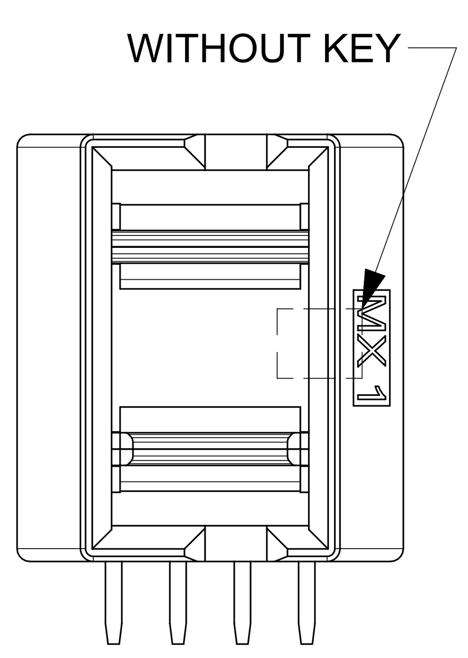
RECOMMENDED PANEL CUTOUT







RECOMMENDED PCB LAYOUT AND KEEP OUT ZONE

WITH KEY	WITHOUT KEY
	
2203700000 (FOR 70 AMPS)	2203700001 (FOR 35 AMPS)

NOTES:
1. MATERIAL:
   HOUSING : LCP, GF, UL94V-0, COLOUR: BLACK.
   TERMINAL: COPPER ALLOY.
2. FINISH:
   CONTACT:
   0.76uM MIN GOLD OVER 1.27um MIN NICKEL UNDERPLATE OVERALL.
   SOLDERTAIL:
   2.54uM MIN TIN OVER 1.27uM MIN NICKEL UNDERPLATING OVERALL
3. PRODUCT SPECIFICATION: TBD
4. PACKAGING SPECIFICATIONS: 2203700000-PK
5. MATE WITH MOLEX PART NUMBER : SEE TABLE
6. DATE CODE & PART NUMBER TO BE LASER MARKED AT APPROXIMATE LOCATION AS SHOWN. (TBA)
7. TERMINALS ARE LUBRICATED.

FOR INFORMATION USE ONLY

PART NUMBER	CURRENT RATING	MATES WITH	NOTES
2203700000	70 AMPS	2203650001	WITH KEY
2203700001	35 AMPS	2203650001 2203650002	WITHOUT KEY

DIMENSION UNITS: mm	SCALE: 4:1
GENERAL TOLERANCES (UNLESS SPECIFIED)	
ANGULAR TOL	± 2.0 °
4 PLACES	±
3 PLACES	±
2 PLACES	± 0.13
1 PLACE	± 0.25
0 PLACES	±
DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS	THIRD ANGLE PROJECTION

CURRENT REV DESC:

EC NO:
DRWN: GSHARMA 2022/07/01
CHK'D:
APPR:

INITIAL REVISION:
DRWN: GSHARMA 2022/02/07
APPR:

molex

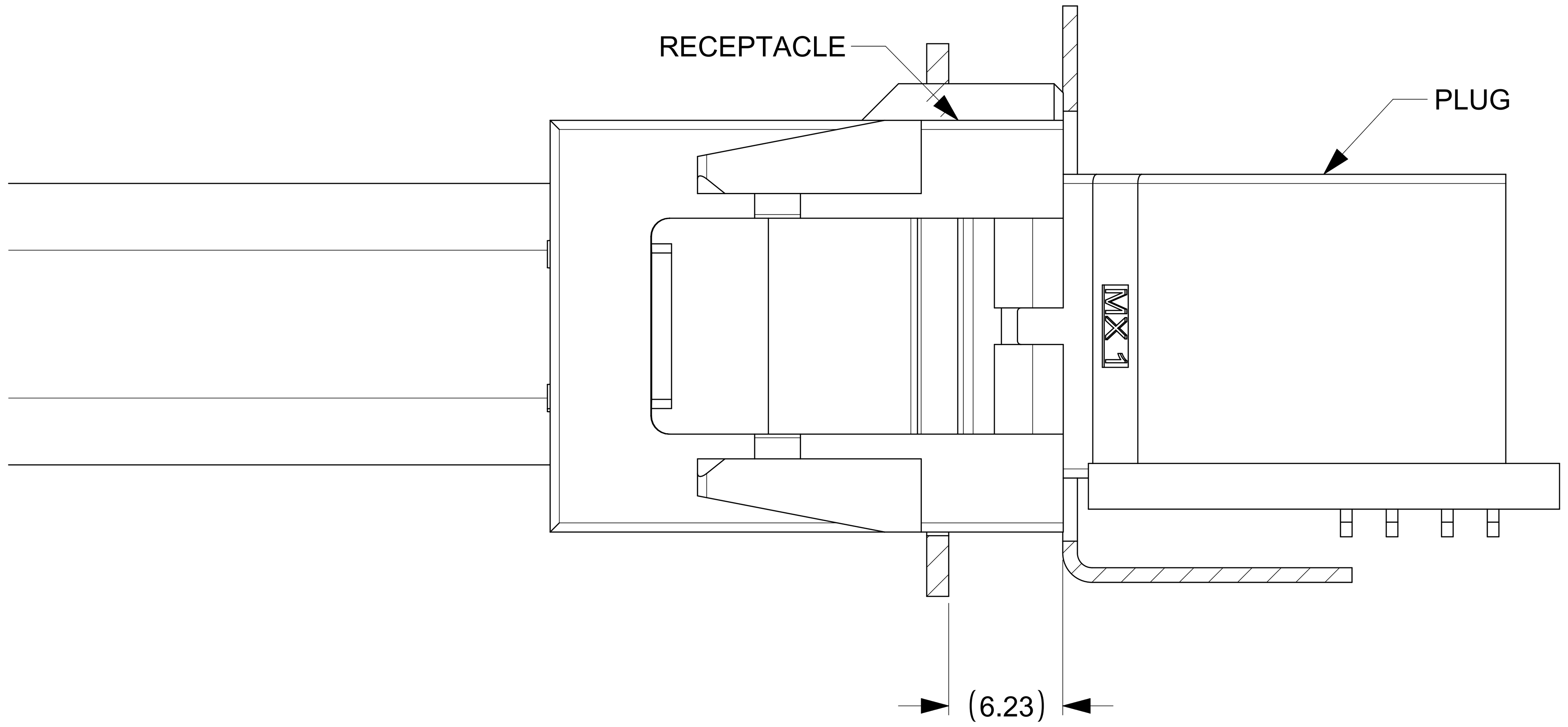
BRICK POWER PLUG CONNECTOR ASSY

PRODUCT CUSTOMER DRAWING

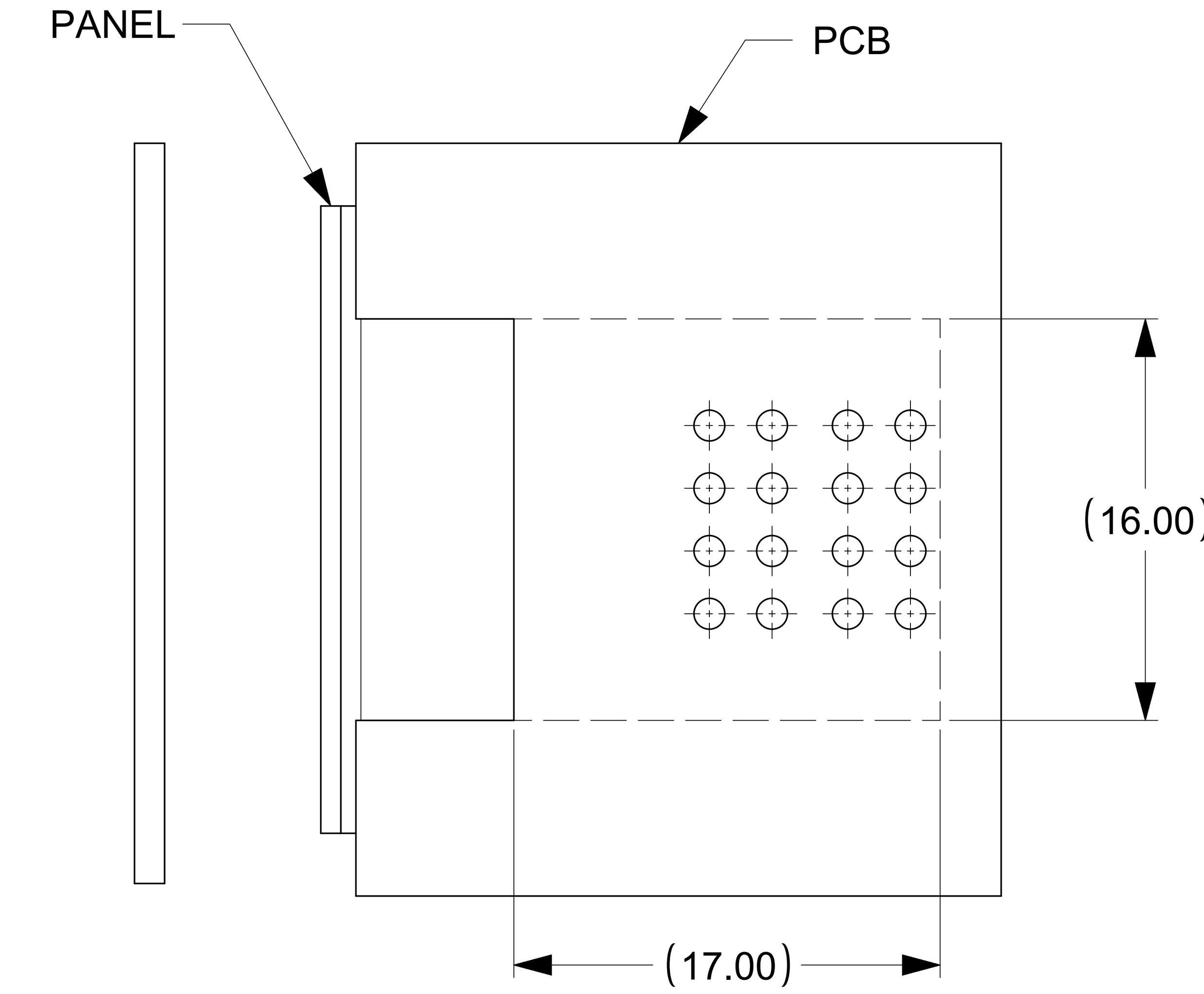
DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2203700000	PSD	000	3

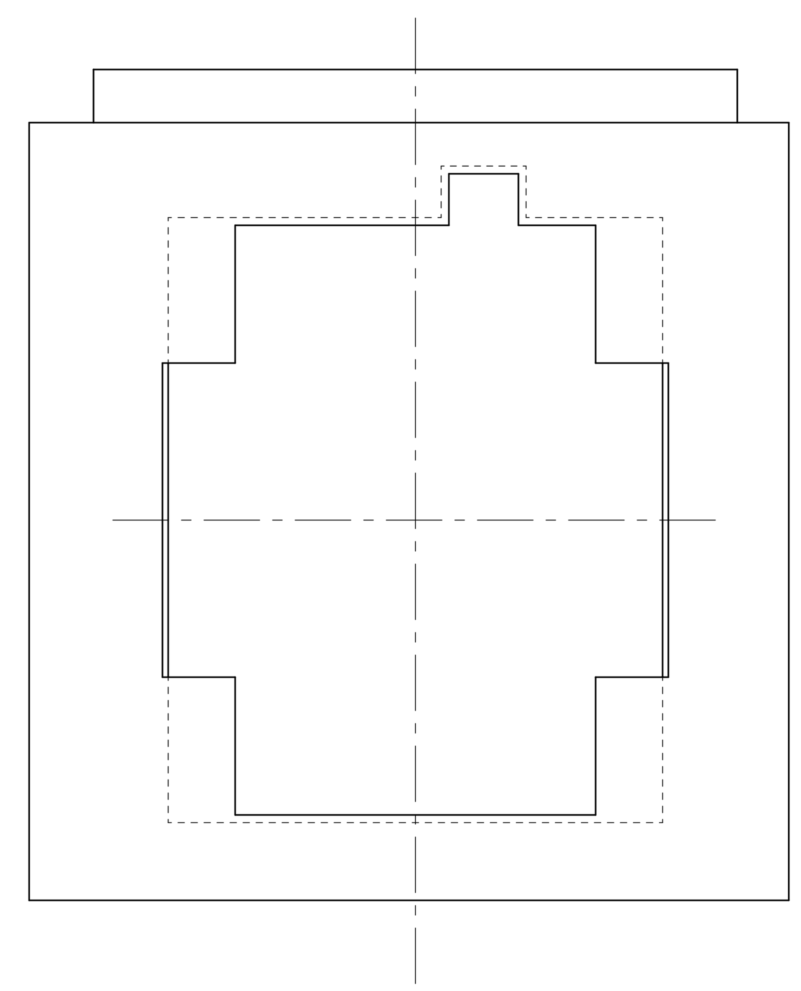
DRAWING	SERIES	MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
D-SIZE	220370	SEE TABLE	GENERAL MARKET	1 OF 2

DOCUMENT STATUS	ER	RELEASE DATE	2022/08/30	18:19:45

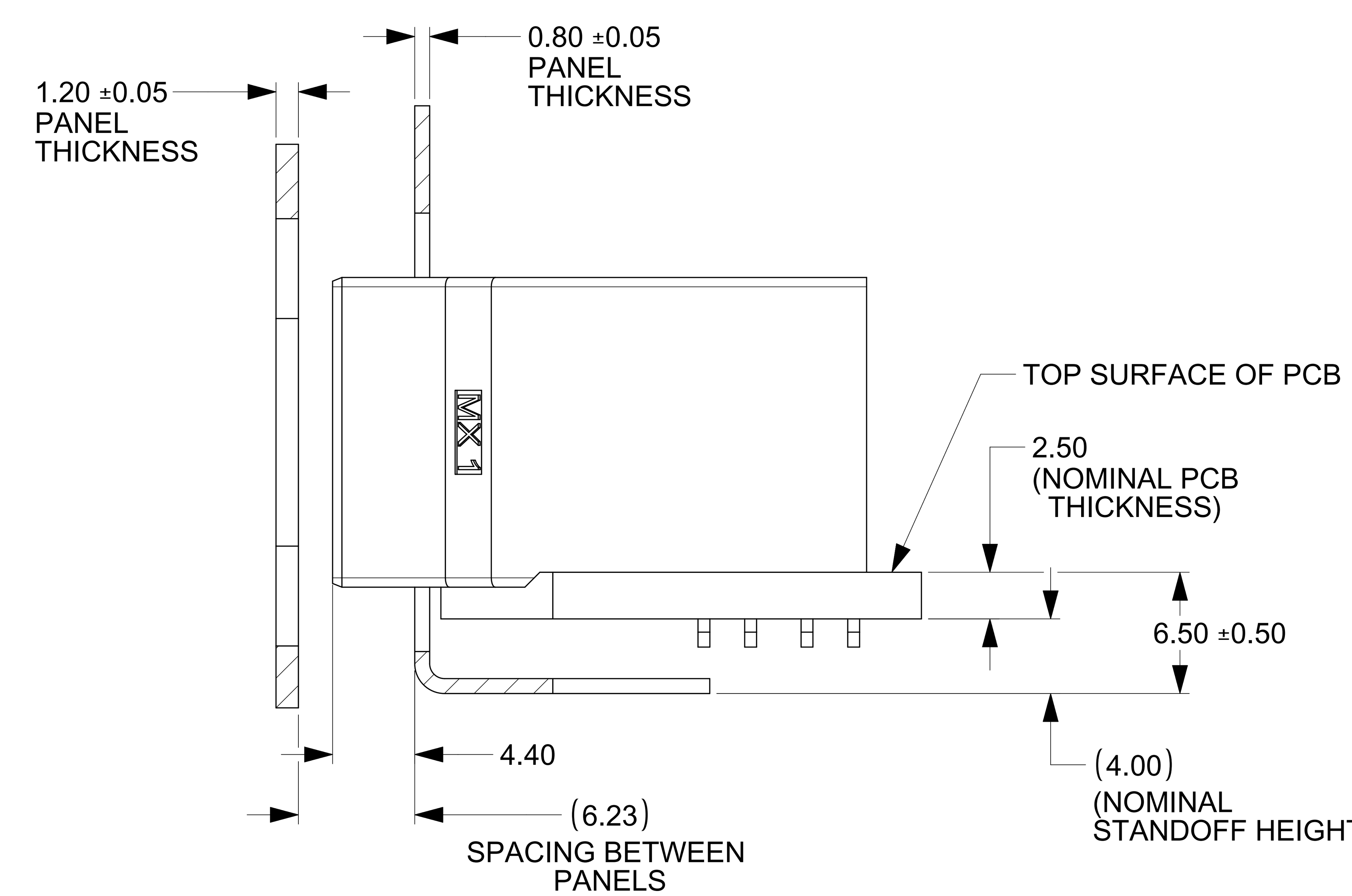


PLUG AND RECEPTACLE MATED PROFILE





BOTH PANELS CUT OUTS ARE CENTRALLY PLACED
ON X-AXIS AS WELL AS Y-AXIS



PLUG MATING PROFILE WITH RESPECT TO PANELS

FOR
INFORMATION
USE ONLY

DIMENSION UNITS	SCALE
mm	4:1

GENERAL TOLERANCES (UNLESS SPECIFIED)

ANGULAR TOL	±	2.0 °
4 PLACES	±	
3 PLACES	±	
2 PLACES	±	0.13
1 PLACE	±	0.25
0 PLACES	±	

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

CURRENT REV DESC:

EC NO:
DRWN: GSHARMA 2022/07/01
CHK'D:
APPR:

INITIAL REVISION:
DRWN: GSHARMA 2022/02/07
APPR:

THIRD ANGLE PROJECTION	DRAWING	SERIES
	D-SIZE	220370

molex

BRICK POWER PLUG CONNECTOR ASSY

PRODUCT CUSTOMER DRAWING

DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2203700000	PSD	000	3

MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
SEE TABLE	GENERAL MARKET	2 OF 2

DOCUMENT STATUS	ER	RELEASE DATE	2022/08/30	18:19:45